

Australian Household Consumption, Wealth and Credit Conditions: An Updated Single-Equation LIVES Estimate

David Stephan

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1 Australian Household Consumption, Wealth and Credit Conditions: An Updated Single-Equation LIVES Estimate

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1.1 Abstract

We estimate a single-equation Muellbauer–Williams LIVES consumption model for Australia over 1988Q3–2024Q4, extending Williams (2010) and Muellbauer and Williams (2012) by sixteen years of post-GFC data and applying contemporary identification methods. Wealth is disaggregated into housing, illiquid financial (equities plus superannuation), and net liquid assets (deposits net of total household debt), each entered as an asset/annualised-income ratio, and we adopt the Jordà (2005) local-projection permanent-income forecaster used in the Italian implementation of De Bonis, Liberati, Muellbauer and Rondinelli (2020). The cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ is accepted in every disaggregated specification and sample window we estimate, validating the net-liquid aggregation used in that Italian convention.

Our central methodological finding is that the *form* of the equation is decisive. A conventional disaggregated error-correction model that enters wealth as constant-marginal-propensity levels — the specification most of this literature, and an earlier draft of this paper, treated as the LIVES equation — is not in fact the LIVES equation. In Williams' framework credit conditions *scale* the housing-wealth effect: there is no classical housing wealth effect, and the marginal propensity is unlocked only as the credit-conditions index (CCI) rises. Estimating a standalone housing-wealth coefficient therefore tests a parameter the theory predicts to be approximately zero, and duly finds it insignificant; reading that as a failed housing-wealth effect is a category error. When the equation is instead specified faithfully — housing wealth entering only through its CCI interaction, the autonomous-consumption CCI intercept restored, and illiquid financial assets combined — the error-correction and core wealth structure come alive (the faithful LIVES specification, Spec 11), estimated on $n = 146$ rather than the $n = 86$ to which the conventional credit term binds the model. On the pre-COVID sample the speed of adjustment is $\lambda = -0.245$ ($t = -4.8$; `australia_lambda_robustness.csv`), within reach of Williams' published -0.286 (about 14 per cent below it). On the full sample it carries significant, correctly signed marginal propensities on net liquid (implied 0.077; OLS +0.037, $t = 3.4$) and illiquid financial (implied 0.037; OLS +0.018, $t = 3.0$) wealth and a strong permanent-income response (OLS +0.50, $t = 4.0$). The implied housing-collateral propensity γ_1 is right-signed but insignificant (implied 0.010, $t = 1.0$), consistent with — but not confirmation of — Williams' calibrated 0.0488. The conventional constant-MPC specification (Spec 6), by contrast, delivers an insignificant $\lambda = -0.18$ ($t = -1.8$). Much of the apparent weakness of single-equation LIVES estimates for Australia is thus a specification artefact, not an economic result. (Permanent income is the full-sample Italy local-projection measure; its non-causal construction and a causal real-time robustness variant, which shrinks λ further, are discussed in §7.4.)

The LIVES *structure* transfers to Australia, but Williams' Australian *calibrations* do not. Imposing his calibrated permanent-income gearing ($\psi_0 = 0.20$, $\psi_1 = 0.93$) collapses the error-correction to $\lambda = -0.029$ ($t = -0.7$; the Williams-calibration-imposed specification, Spec 12), independently reproduced by Spec 10 ($\lambda = -0.048$): the Australian data freely estimates a permanent-income response roughly two-and-a-half times Williams', and forcing his smaller value destroys the equilibrium. This reconciles a puzzle in our companion work, where a Wald test fails to reject Williams' joint calibration ($\chi^2 = 2.24$) — the free estimates are too imprecise to reject his values, yet imposing them still ruins the fit; low power is not the same as good fit. The credit-conditions *interactions* are themselves only weakly identified: the six CCI-interacted regressors are 0.74–0.97 mutually correlated on this sample because each is approximately proportional to the latent index, so they cannot be separately estimated

from a single equation. This is exactly why Williams calibrates and estimates a four-equation FIML system; on contemporary Australian data the single-equation calibration shortcut is empirically closed, leaving joint estimation and pre-1988 back-extension as the only routes to sharpen the credit channels.

We assemble a back-extended master dataset to 1976Q3 — using a TRYM long-run house-price series, RBA D03 monetary aggregates, RBA D02 total credit, historical labour-force compilations, and documented aggregate and disaggregated wealth proxies anchored at 1988Q3, when ABS sectoral balance-sheet data begin — to test whether sample length is the binding constraint on tighter agreement with Williams. Refitting the disaggregated no-CCI specification (Spec 4) on the back-extended 1976Q3–2024Q4 sample ($n = 190$ in the cointegration window), λ moves about 37 per cent closer to Williams ($-0.140 \rightarrow -0.193$) but individual wealth coefficients become smaller rather than larger, and the net-liquid coefficient collapses toward zero — sample length is not the binding constraint. The CCI placebo battery is weak on both samples: the literal Williams 4-knot deteriorates from the 34th adjusted- R^2 percentile on the 1988+ sample to the 18th on the back-extended sample (200 placebo draws; `australia_williams_knot_placebo_verdict.csv` and `..._extended_summary.csv`), the maximal-GETS reduction sits at the 64th percentile and the sectional-period variant near the median. A two-equation SUR of consumption and house prices delivers negligible cross-equation residual correlation and essentially no efficiency gain — the SUR and single-equation coefficients differ only within sampling noise (`australia_joint_pi_robustness.csv`). A joint cross-equation CCI identification retains only two of seven single-equation knot survivors. We read these findings as indicating that the structural identification Williams (2010) delivers comes from cross-equation parameter restrictions in his four-equation FIML system rather than from sample length, knot count, or sign-prior structure. The calibration-collapse and interaction-collinearity results independently confirm this: the faithful single-equation LIVES form recovers Williams' error-correction speed and wealth structure, but the separate identification of the credit interactions — and the transferability of his credit-channel calibrations — require his joint system.

We report the honest negative results throughout, not as caveats but as substantive findings that a freely-estimated single-equation framework permits and an imposed-restriction framework would hide: the CCI placebo failure, the negligible SUR efficiency gain, the loss to a random-walk-with-drift benchmark out of sample at horizons of four and eight quarters, the divergence of model-selection criteria (the automated screen favours a net-worth specification, the Bayesian-information criterion the free-interaction specification, and LIVES theory the faithful Spec 11), and the partial reversal of the permanent-income response under a real-time, causal forecaster. Each points to the same conclusion as the calibration and collinearity diagnostics: four-equation FIML and pre-1988 back-extension are the routes needed to sharpen the credit channels. The paper includes a structured robustness suite covering instrumental variables, joint SUR estimation, Chow tests at multiple break dates, the Drehmann (2017) amortising-mortgage adjustment, the Williams smoothed-step credit-conditions spline with maximal-GETS reduction, a Kalman state-space credit factor, sectional sign-prior alternatives, rolling-window estimation, out-of-sample forecast validation against random-walk and AR(1) benchmarks, and back-extension robustness on Spec 1 and Spec 4. A full reproducibility kit accompanies the paper, along with a multi-equation LIVES scaffold that lays the foundations for a companion paper.

1.2 1. Introduction

1.2.1 1.1 Motivation

Australian household consumption poses a set of policy and analytical questions that the standard New Keynesian DSGE workhorse — with its representative consumer, exogenous wealth process, and assumed-frictionless credit access — is structurally ill-equipped to answer. Among them: how much of the post-2008 moderation in consumption growth is attributable to the macroprudential tightening of 2014 and 2017, and how much to post-GFC household balance-sheet repair? How sensitive is consumption to housing wealth at different points in the credit cycle, when households are or are not able to extract home equity? Did the COVID shock and JobKeeper income support move permanent-income expectations, or only the short-run quarter? And how should a central bank think about the wealth channel of monetary policy when much of household wealth is housing wealth, mortgage debt is at near-historic levels relative to income, and the credit conditions in which households operate have evolved markedly since financial deregulation in the 1980s?

The Muellbauer–Williams "LIVES" framework — Latent Interactive Variable Equation System, in the flow-of-funds tradition of Tobin and of Duca, Muellbauer and Murphy (2013) — was developed to answer exactly these questions. It augments the standard credit-augmented life-cycle consumption function (Friedman 1957; Ando and Modigliani 1963; Tobin and Dolde 1971) with three features. First, wealth is disaggregated into net liquid assets (NLA, liquid assets less total household debt), illiquid financial assets (IFA, equities plus superannuation) and housing assets (HA), each entered as a ratio to annualised income and each carrying a different marginal propensity to consume (Backus and Purvis 1980). Second, a latent credit-conditions index (CCI) interacts with the long-run equilibrium so that key channels are switched on only as credit eases. Third — and central to Williams' (2010) Australian estimate — the latent CCI is identified *jointly* across a four-equation system (consumption, house prices, mortgage stock and home-equity withdrawal) under cross-equation parameter restrictions estimated by full-information maximum likelihood (FIML).

The theoretically decisive feature for this paper is how housing enters. In the canonical LIVES equation (Williams' Eq (7); his Table 1, column 1) **there is no classical housing-wealth effect**. Housing wealth appears *only* through its interaction with credit conditions, as $\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$. The housing marginal propensity to consume is therefore zero when credit is fully constrained ($\text{CCI} = 0$) and is unlocked as credit eases, reaching its peak when CCI is at its maximum. By contrast, illiquid financial and net liquid assets enter as plain, credit-invariant marginal propensities (γ_2 and γ_3). This functional form has an immediate implication for empirical work: a specification that enters a standalone housing-wealth ratio (a credit-invariant ha_y term) is *not* the LIVES equation, and reading an insignificant standalone housing coefficient as evidence of a failed housing-wealth effect is a **category error** — the theory predicts that coefficient to be approximately zero in the absence of the CCI interaction.

Williams (2010) applied the LIVES framework to Australia for 1978–2008, producing the canonical Australian estimate. Sixteen years of post-GFC and post-COVID data have since accumulated, and a contemporary central-bank-quality update is warranted. This paper provides that update. Its pivot, set out below, is that an earlier reading of single-equation Australian LIVES estimates as "weak" was in substantial part a specification artefact: when

the equation is estimated in its *faithful* form — housing entered only through the credit interaction, the autonomous CCI intercept restored, and illiquid financial assets combined — the error-correction and core wealth structure come alive in a way the conventional constant-MPC disaggregated specification cannot deliver.

1.2.2 1.2 Contribution

This paper makes five contributions to the Australian household consumption literature.

(i) A faithful single-equation LIVES estimate, with the form correction as the lead methodological result. We show that the functional *form* of the LIVES equation, not the fit of any one regressor, is what identifies it. Estimating the faithful form — housing entered only through ha_x_cci , the de-meaned CCI(HA/4y) interaction; the autonomous-consumption CCI intercept ($cci_williams$, the ζ_c loading) restored; and illiquid financial assets combined into a single $ilfa_y = eq_y + super_y$ term — yields our headline specification, Spec 11. It recovers a correctly signed and significant error-correction speed and a right-signed wealth structure (§1.3). The same data fitted to the conventional constant-MPC disaggregated error-correction model (Spec 6), which carries plain $ha_y / eq_y / super_y / nla_y$ terms with no credit scaling and admits CCI only as a short-run regressor, delivers an insignificant speed of adjustment. The contrast is one of theoretical form, not of estimation luck.

(ii) The finding that the LIVES structure transfers to Australia but Williams' Australian calibrations do not. Imposing Williams' Australian permanent-income gearing ($\psi_0 = 0.20$, $\psi_1 = 0.93$) and his illiquid financial MPC ($\gamma_{IFA} = 0.022$) — the Williams-calibration-imposed specification, Spec 12 — *collapses* the equilibrium: the speed of adjustment falls to $\lambda = -0.029$ ($t = -0.68$, full sample), statistically indistinguishable from zero. An independent calibration route (Spec 10, Williams-prior) reproduces the collapse ($\lambda = -0.048$, $t = -0.76$). The reason is that Australia freely estimates a permanent-income weight of order 0.50, around two and a half times Williams' 0.20, so forcing his value breaks the long-run fixed point. This reconciles the companion paper's Wald non-rejection of Williams' joint calibration: the free single-equation estimates are too imprecise to *reject* Williams' values — every one of his Table 1 values lies inside our gamma-inference 95 per cent intervals, but so does zero in every case — yet imposing those values nonetheless wrecks the fit. Low power is not the same as good fit.

(iii) An interaction-collinearity diagnosis explaining why Williams needs FIML. The six CCI-interacted regressors that constitute the credit channels are between roughly 0.74 and 0.97 mutually correlated on the post-1988 Australian sample, because each is approximately proportional to the same latent CCI. They cannot, therefore, be separately free-estimated off a single equation. This is the structural reason Williams identifies the credit channels through cross-equation FIML restrictions rather than within the consumption equation alone. The corollary is that single-equation calibration is empirically closed: sharpening the credit channels requires either the four-equation FIML system or a longer (pre-1988) sample that spans the financial-liberalisation episode.

(iv) A back-extended master dataset (1976Q3–2024Q4, $n = 194$ quarters) and a direct sample-length test. We construct documented growth-rate splices for house prices (Treasury TRYM historical compilation), the M3 monetary aggregate (RBA D03), total credit (RBA D02) and labour force, with disaggregated wealth proxies anchored at 1988Q3 — the quarter in which ABS sectoral household balance-sheet data begin. Because the financial-liberalisation

episode that identifies the credit channels largely predates 1988Q3, this back-extension is the natural route to longer credit variation. Refitting the disaggregated no-CCI specification (Spec 4) on the extended sample moves λ about 37 per cent closer to Williams (-0.140 to -0.193), but individual wealth coefficients shrink rather than strengthen and the net liquid MPC collapses — so sample length is *not* the binding constraint on tighter agreement with Williams.

(v) A structured robustness and placebo suite, and a set of honest negative results reported as substantive findings. Mirroring the De Bonis, Liberati, Muellbauer and Rondonelli (2020) Italian methodology, we run instrumental variables, a Zellner SUR, multi-window Chow and Bai–Perron break tests, the Drehmann amortisation adjustment, an AR/Italy local-projection permanent-income comparison with a real-time column, a permanent-income filter sensitivity grid, rolling-window estimation, and out-of-sample validation against random-walk and AR(1) benchmarks. We retain — and report prominently, without apology — the negative results that a freely estimated single-equation framework permits: the credit-conditions spline does not strongly outperform random-knot placebos (it sits in the 18th–34th adjusted- R^2 percentile band across placebo variants); the SUR delivers essentially no efficiency gain (negligible cross-equation residual correlation); the structural specifications lose to a random walk with drift out of sample at horizons of four and eight quarters; the automated selector, the BIC and LIVES theory each point to a different specification; and the full-sample permanent-income measure is partly reversed under a causal real-time projection. Each negative is diagnostic: it points to why FIML and back-extension are the routes forward.

1.2.3 1.3 Headline result

The faithful LIVES specification (Spec 11) is the headline. On the full sample (1988Q3–2024Q4, $n = 146$) it delivers a speed of adjustment of $\lambda = -0.480$ (Newey–West $t = -3.59$), and on the pre-COVID sample (1988Q3–2019Q4, $n = 126$) $\lambda = -0.245$ ($t = -4.80$), close to Williams' (2010) published -0.286 (his ϕ_c). We treat the pre-COVID estimate as the identified value: the full-sample magnitude is inflated by the COVID quarters and fails the upper-bound screen on $|\lambda|$ in the selection rubric (§6), while the pre-COVID estimate is both correctly signed and sharply determined. [Source: `australia_all_results.csv`; `australia_lambda_robustness.csv`.]

The wealth structure is right-signed and, for the credit-invariant components, significant. On the full sample the net liquid MPC is $\mathbf{nl}_y = +0.037$ ($t = 3.40$, significant at 1 per cent; implied structural MPC $\gamma_3 = \text{OLS}/|\lambda| = 0.077$) and the illiquid financial MPC is $\mathbf{il}_y = +0.018$ ($t = 3.03$, significant at 1 per cent; implied structural $\gamma_2 = 0.037$). The housing-collateral channel enters with the right sign but is not significant: $\mathbf{ha}_x\text{cci} = +0.0049$ ($t = 1.03$), implying a structural housing-collateral MPC of $\gamma_1 = 0.010$, against Williams' peak housing MPC of 0.0488. Permanent income enters strongly: $\mathbf{ln}_{yp_over}_y = +0.504$ ($t = 4.00$) on the full sample and $+0.279$ ($t = 4.75$) pre-COVID, consistent with a freely estimated permanent-income weight of order 0.50. [Source: `australia_all_results.csv`.]

The honest qualifier on the housing channel is that it is *right-signed but insignificant*: $\gamma_1 = 0.010$ ($t = 1.03$) is consistent with Williams' 0.0488 only in the weak sense that the latter lies inside a wide confidence interval that also contains zero. This is consistency, not confirmation — a non-rejection driven by imprecision rather than by point-estimate agreement. The accompanying gamma-inference exercise makes the point general: every one of Williams'

Table 1 values lies inside our 95 per cent interval, but so does zero in every case. [Source: `australia_gamma_inference.csv`.]

The contrast with the calibration route is sharp. Imposing Williams' Australian calibrations (Spec 12: $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\gamma_{IFA} = 0.022$) collapses the error-correction term to $\lambda = -0.029$ ($t = -0.68$) on the full sample and flips its sign on the pre-COVID sample ($\lambda = +0.030$, not significant); the independent Williams-prior route (Spec 10) reproduces the collapse at $\lambda = -0.048$ ($t = -0.76$). The LIVES *structure* transfers to Australia; Williams' Australian *calibrations* do not. Williams' rate, affordability and autonomous-consumption loadings cannot even be imposed at their published magnitudes — his raw rate loading ($\alpha_r = -0.871$) is roughly thirty times too large on the repository's percent real-rate by unit-normalised-CCI scaling, and diverges the long-run fixed point. [Source: `australia_all_results.csv`; `australia_lambda_robustness.csv`.]

By contrast, the conventional constant-MPC disaggregated ECM (Spec 6) — the specification an earlier draft, and prior Australian work, treated as the LIVES equation — delivers $\lambda = -0.180$ ($t = -1.76$, full sample, $n = 86$), significant only at the 10 per cent level, with an insignificant standalone housing coefficient ($ha_y = +0.0088$, $t = 1.52$). We retain Spec 6 as the conventional baseline against which the faithful form is the alternative, but we no longer read its insignificant standalone ha_y as a failed housing-wealth effect: under LIVES theory that coefficient is expected to be approximately zero absent the credit interaction. [Source: `australia_all_results.csv`.]

Finally, the credit-conditions identification limits are reported as findings rather than hidden. The deployed CCI spline sits in the 18th–34th adjusted- R^2 percentile band across random-knot placebo variants; the six CCI interactions are mutually collinear (roughly 0.74–0.97) and so cannot be separately identified off one equation; the SUR of consumption and house prices shows negligible residual correlation and no efficiency gain; and the structural specifications lose to a random walk with drift out of sample at the four- and eight-quarter horizons. We do not promote the CCI-interactions specification (Spec 8, $\lambda = -0.445$) as a credit-conditions success: it re-allocates identification across regressors rather than closing the gap with Williams, and its individual interaction coefficients are wrong-signed or insignificant. [Source: `australia_williams_knot_placebo_verdict.csv`; `australia_oos_rmse.csv`; `australia_all_results.csv`.]

1.2.4 1.4 Roadmap

Section 2 surveys the LIVES literature, the Australian aggregate-wealth-effect literature and permanent-income measurement, and frames the LIVES claim that there is no classical housing-wealth effect — so that standalone-wealth specifications mis-test the theory. Section 3 documents data construction, including the disaggregated wealth ratios, the Italy local-projection permanent-income forecaster with its look-ahead versus real-time caveat, and the 1976Q3 back-extension sources and proxies. Section 4 presents the canonical LIVES equation (Eq (7)) and distinguishes the faithful form (Spec 11) from the conventional constant-MPC ECM (Spec 6) as a matter of theory. Section 5 develops the identification of credit conditions, including the interaction-collinearity result, the placebo battery, the near-zero SUR residual correlation and the joint knot-survival test. Section 6 sets out the full specification ladder (Specs 1–12, now including the faithful Spec 11 and the calibration-imposed Spec 12) and the selection rubric, and reports the selector divergence honestly. Section 7 presents the faithful-

LIVES headline results and the calibration-collapse finding, with Spec 6 reframed as the conventional baseline. Section 8 runs the structured robustness suite, preserving the honest negatives. Section 9 reconciles our estimates with Williams (2010, 2012) around the structure-transfers / calibrations-don't thesis. Section 10 presents the long-run decomposition and policy implications, based on the faithful Spec 11 channels. Section 11 concludes.

1.3 2. Literature review

This paper sits at the intersection of three literatures: the Muellbauer-Williams "LIVES" tradition that integrates wealth, credit conditions and life-cycle behaviour into consumption equations; the Australian empirical consumption literature, which has historically focused on aggregate or constant-marginal-propensity wealth effects without the explicit credit-conditions machinery; and the small but growing body of work that disciplines permanent-income measurement using forecasting approaches more robust than the standard AR(p) recipe. We review each in turn before placing the present contribution.

The thread that runs through the review, and that anticipates the paper's central finding, is a point about functional form. The LIVES framework makes a specific theoretical claim — that there is *no classical housing wealth effect*: the marginal propensity to consume out of housing wealth is zero when credit conditions are tight and is unlocked only as credit conditions ease. A consumption equation that enters housing wealth as a standalone level term, with a constant marginal propensity, does not test this claim — it tests a different, incompatible one. Reading an insignificant standalone housing-wealth coefficient as evidence of "no housing wealth effect" is therefore a *category error*: theory predicts that coefficient to be near zero in the absence of the credit-conditions interaction. This distinction — between the *faithful* LIVES form, in which housing enters only through its interaction with the credit-conditions index, and the *conventional constant-MPC* disaggregated error-correction model that prior work and an earlier draft of this paper treated as the LIVES equation — is the organising idea of the literature that follows and the lead result of the paper (§7.0).

1.3.1 2.1 Theoretical foundations

The empirical specification estimated below has its origins in the Davidson-Hendry-Srba-Yeo (1978) error-correction consumption function and the canonical permanent-income hypothesis (PIH) of Friedman (1957) and Hall (1978). The DHSY approach embeds a long-run cointegrating relationship between consumption, income and wealth into a short-run dynamic equation in growth rates, with an error-correction term governing the speed at which the consumption ratio reverts to its long-run determinants. Engle and Granger (1987) provide the formal econometric framework, with Hendry and Krolzig (2005) and Doornik (2009) refining the general-to-specific reduction methodology that disciplines the choice of short-run regressors.

A long line of empirical work has documented systematic departures from strict PIH behaviour. Campbell and Mankiw (1989, 1991) show that roughly half of US consumption tracks current rather than permanent income, consistent with rule-of-thumb behaviour or liquidity constraints. Carroll and Kimball (1996) establish the concavity of the consumption function under prudence, providing a microeconomic basis for differential marginal propensities to consume across wealth classes. Carroll (2001) and Deaton (1992) develop the buffer-stock interpretation that motivates the distinction between liquid and illiquid wealth in the Muell-

bauer specifications used here. The empirical implication — that wealth components with different liquidity, transactions costs, and ownership concentration should enter the consumption equation with different marginal propensities — is the cornerstone of the disaggregated specifications reported in §7. In the faithful LIVES form, this implication is sharpened further: liquidity matters not only through a level distinction but through how each component interacts with the prevailing state of credit. Net liquid assets and illiquid financial assets enter as plain marginal propensities, but housing wealth is collateral whose consumability depends on the borrowing technology — and so enters only through its credit-conditions interaction.

1.3.2 2.2 The LIVES framework

Muellbauer (2007), originally prepared for the Federal Reserve Bank of Kansas City Jackson Hole symposium, sets out the integrated approach to modelling consumption with wealth, housing collateral and credit conditions. The paper's central insight is that credit-market liberalisation interacts with the wealth and interest-rate channels: when credit is tight, housing wealth collateralises borrowing only weakly and the down-payment hurdle dampens consumption; when credit is loose, housing wealth and current consumption become more tightly linked. This insight requires a credit-conditions index (CCI) entering the long-run relationship not as a single additive term but as a multiplicative shifter across several channels simultaneously.

The functional form is decisive, and worth stating precisely because it is what distinguishes the framework from a conventional wealth error-correction model. In the canonical Williams (2010) consumption equation — his Eq (7), reproduced in §4 — the credit-conditions index multiplies *six* channels jointly inside the long-run bracket: the autonomous-consumption intercept ($\zeta_c \cdot \text{CCI}$), the real-interest-rate term ($\alpha_1 \cdot r \cdot \text{CCI}$), the housing-collateral term ($\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$), the permanent-income gearing ($\psi(\text{CCI}) \cdot \log(y^p/y)$, with $\psi(\text{CCI}) = \psi_0 + \psi_1 \cdot \text{CCI}$), and the affordability term ($\alpha_4 \cdot (1 - \omega \cdot \text{CCI}) \cdot \log(p^h/y)$, with the affordability multiplier $\omega = 1.2$). Critically, housing wealth enters *only* through $\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$ — there is no standalone housing wealth level in the long-run relationship. The housing marginal propensity is therefore zero at $\text{CCI} = 0$ and rises as credit conditions ease, reaching Williams' implied peak housing MPC of 0.0488 (Williams' Table 1; `williams_comparison.csv`). The illiquid financial asset term (IFA = equities + superannuation) and the net liquid asset term (NLA = liquid assets – total household debt) enter as plain marginal propensities (Williams' calibrated $\gamma_{\text{IFA}} = 0.022$ and estimated $\gamma_{\text{NLA}} = 0.159$ respectively). Income enters with a *unit* coefficient, so that the error-correction term is the (log) consumption-to-income ratio, and all wealth terms enter as asset/annualised-income ratios, $x_{\{t-1\}}/4y$.

It follows that a specification entering housing wealth as a standalone constant-MPC level term — alongside equities, superannuation and net liquid assets, with credit conditions relegated to a short-run dynamic regressor — is not the LIVES equation at all, but a generic wealth error-correction model. Under the LIVES theory, the standalone housing coefficient in such a model is expected to be near zero, because the operative channel (the CCI interaction) has been omitted. This is the sense in which reading an insignificant standalone housing coefficient as a failed housing wealth effect is a category error, and it is the reason §7 distinguishes the *faithful* LIVES specification from the *conventional constant-MPC* baseline as a matter of theory, not fit.

The framework was operationalised in a series of country studies. Aron, Duca, Muellbauer, Murata and Murphy (2012) jointly estimate the framework on Japan, the United Kingdom

and the United States, finding consistent positive long-run housing wealth effects in the UK and US (where home equity withdrawal is institutionally available) and a much smaller effect in Japan (where it is not) — precisely the cross-country pattern that the credit-interaction form predicts, since the housing channel should be muted where the collateral-borrowing technology is weak. Duca, Muellbauer and Murphy (2010) apply the framework to the global financial crisis, demonstrating that the abrupt tightening in CCI from 2007 quantitatively rationalises the sharp consumption pullback observed in heavy-MEW economies. Duca and Muellbauer (2013), in *European Economy* Discussion Paper 14, formalise what they label the "LIVES" approach — the *Latent Interactive Variable Equation System* — emphasising the joint determination of consumption, house prices, mortgage debt and home equity withdrawal in a four-equation system identified by common factors and cross-equation sign restrictions.

Two parallel implementations frame the present paper. De Bonis, Liberati, Muellbauer and Rondinelli (2020) estimate a single-equation Italian adaptation that imposes the cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ (deposits and household debt enter with equal-and-opposite coefficients, so that net liquid assets is the operative quantity), adopts a Jordà (2005) local-projection permanent-income forecaster, applies a Drehmann (2017) amortising-mortgage adjustment to the real mortgage rate, and validates the single-equation OLS estimator against joint SUR. Chauvin and Muellbauer (2018), in the Banque de France working-paper series with an associated online complement, undertake a similar France adaptation, with particular attention to the institutional differences — limited home equity withdrawal, a larger social housing sector — that shape the housing-wealth channel. Both papers explicitly take Williams' Australian work as a methodological precedent; the present paper closes the loop by applying the Italian methodology back to the Australian data on which the original LIVES Australia estimation was performed.

1.3.3 2.3 The Australian application and prior Australian consumption work

The Australian application was developed in two companion papers by David Williams during his Oxford doctorate. Williams (2009), Oxford Economics Series Working Paper 432, focuses on the house-price equation and its identification under financial liberalisation; the paper develops the spline-based credit-conditions index from STAMP (Koopman et al. 2000) unobserved-components estimation, anchors the index at the four institutional turning points 1979/1992/1998/2007, and constructs the non-property household disposable income measure that we replicate in §3. Williams (2010), Oxford Economics Series Working Paper 492, estimates the four-equation LIVES system (consumption, house prices, mortgage stock, home equity withdrawal) jointly by full-information maximum likelihood (FIML) on Australian data 1978Q1–2008Q2, with the house-price loading on credit conditions normalised ($\zeta_h = 1$) to fix the scale of the latent index. The published version of this work, Muellbauer and Williams (2012) "Credit conditions and the real economy: the elephant in the room", appears as the lead chapter of *BIS Papers* No. 64 and constitutes our primary benchmark.

Australian consumption modelling outside the LIVES tradition has a long history, almost all of it in the standalone-wealth-effect form that the LIVES theory regards as a mis-test of the housing channel. Tan and Voss (2000), in RBA Research Discussion Paper 2000-09, estimate aggregate-wealth effects on Australian consumption using ABS National Accounts and RBA balance-sheet data, finding significant positive effects of both housing and financial wealth. Dvornak and Kohler (2003), in RBA RDP 2003-07, use a state-level panel to identify wealth

effects from cross-state variation, finding larger marginal propensities to consume out of stock-market wealth than out of housing wealth, in apparent contrast to the time-series evidence; their findings are partly reconciled by the Muellbauer-Williams framework once credit-conditions interactions are introduced, since the unconditional housing MPC averages over tight- and loose-credit regimes and so understates the loose-credit collateral channel. The methodological point is that none of these specifications interacts housing wealth with credit conditions, and so each estimates an unconditional housing MPC whose magnitude the LIVES form predicts will be small and unstable — a property to keep in mind when comparing coefficient magnitudes across studies.

The Reserve Bank of Australia's macroeconometric model MARTIN, introduced in Cusbert and Kendall (2018) in the RBA Bulletin and documented in Ballantyne et al. (2019) in RBA RDP 2019-07, includes a household consumption block that incorporates wealth effects and credit conditions in a more reduced-form way than the LIVES specification. The MARTIN consumption equation imposes calibrated elasticities for several channels rather than estimating the full long-run cointegrating vector, and abstracts from the explicit CCI spline; its calibrated net-wealth elasticity is 0.17 (`martin_nesting.csv`). The present paper complements MARTIN by providing a freely estimated benchmark against which calibrated coefficients can be evaluated — our unrestricted net-wealth elasticity is 0.1155 (`martin_nesting.csv`), with long-run homogeneity rejected ($\chi^2 = 16.41$, $p = 0.0001$) — and by surfacing the identification choices that drive the estimated speed of adjustment. As §10 discusses, our wealth-elasticity estimate is too imprecise to discipline MARTIN's calibration; the appropriate reading is to import the qualitative structure rather than the point estimates.

A separate strand of Australian work has examined the cyclical co-movement of consumption with credit and housing conditions through reduced-form approaches: Bayesian VAR frameworks, RBA Bulletin analyses of mortgage payments and household debt burdens, and the housing-leverage and household-balance-sheet treatments in the broader RBA Research Discussion Paper series. These approaches identify shorter-run dynamics and stress-test sensitivities but do not deliver the long-run cointegrating vector that the LIVES specification produces, and so are complementary to the present analysis rather than substitutes for it.

1.3.4 2.4 Identification of credit conditions

The credit-conditions index is the most contested ingredient in the LIVES framework, and the heart of the identification problem this paper documents. Williams (2010) constructs CCI as a latent variable identified by a spline of smoothed-step dummies (SDMMA series — five-quarter moving averages of four-quarter moving averages of step dummies), $\Sigma a_s\text{-SDMMA}_s$, at four institutional turning points: 1979 (Campbell Committee, removal of interest-rate ceilings on bank deposits), 1992 (banking distress and the entry of the first mortgage originator, Aussie Home Loans), 1998 (the rise of non-bank financial institutions and securitisation), and 2007 (the global financial crisis tightening). The institutional chronology of Australian financial deregulation underpinning these choices is documented in Battellino and McMillan (1989) and Edey and Gray (1996); Bayoumi (1993) provides a cross-country analysis of the consumption response to financial liberalisation, including Australia, that quantitatively validates a structural CCI shift in the early 1980s.

The decisive feature of Williams' identification is that it is *joint*. In the system-estimation context of Muellbauer and Williams (2012), each spline coefficient is identified by being a

common factor across the four equations: the same CCI value enters consumption, house prices, the mortgage stock and home equity withdrawal, with different loadings, and the house-price loading is normalised ($\zeta_h = 1$) to fix the index scale. This common-factor, FIML identification is the central methodological contribution of the LIVES family — and, as this paper argues, a structural necessity rather than a stylistic preference. The reason is collinearity. The six CCI-interacted regressors are each approximately proportional to CCI, and on the post-1988 Australian sample they are 0.74–0.97 mutually correlated (a property of the estimation sample; the closest committed evidence is the identification-versus-detrending decomposition in §5 and the sign-failures and collapse documented in §7). Six near-collinear interaction regressors cannot be separately freely estimated off a single equation; in single-equation OLS the spline can be proxied or calibrated, but the individual credit channels cannot be jointly identified. This is the structural reason Williams uses FIML, and it is why this paper treats four-equation FIML and pre-1988 back-extension — not single-equation refinement — as the only routes to sharpen the credit channels. We adopt the spline approach under sign-prior restrictions enforced by general-to-specific reduction (Hendry-Krolzig 2005) as a single-equation robustness exercise (§5), but with the understanding that single-equation calibration is empirically closed.

The alternative, observable proxy for CCI — the ratio of housing credit flow to disposable income, in logs — has the advantage of being measured directly but the disadvantage of being available only from the early 2000s (the ABS 5601.0 lending series begins 2002Q3), after the most informative deregulation episodes have already occurred. This timing problem is fundamental rather than incidental: ABS sectoral balance-sheet data begin only in 1988Q3, so the financial-liberalisation episode that identifies the credit channels largely predates the data on which the consumption equation can be estimated. We use the observable proxy as a short-run regressor in the conventional-baseline specification, and the spline approach for the long-run identification in the credit-interaction robustness specification.

A separate identification approach uses survey-based credit-conditions indices. The Federal Reserve's Senior Loan Officer Opinion Survey is the prototype; the European Central Bank's Bank Lending Survey serves the same function in the euro area. The Reserve Bank of Australia operates a qualitative Liaison Programme but does not publish a long-running numerical index of credit conditions for households. The unavailability of such a survey-based Australian series is the practical reason for adopting Williams' spline approach in the LIVES tradition.

1.3.5 2.5 Permanent-income forecasting

The empirical operationalisation of permanent income — the discounted expected weighted average of expected future income — requires either explicit forecasts or a parametric assumption about the income process. The standard practice in the consumption literature has been to assume an AR(p) process for log income, fit it on the available sample, and aggregate the multi-step-ahead forecasts using exponentially declining weights with discount factor δ . This recipe descends directly from the PIH literature of Hall (1978) and the consumption-Euler-equation tests of Campbell and Mankiw (1989). In the LIVES form, permanent income enters with a credit-conditions-dependent gearing, $\psi(\text{CCI}) = \psi_0 + \psi_1 \cdot \text{CCI}$, which Williams calibrates rather than estimates ($\psi_0 = 0.20$, $\psi_1 = 0.93$); §4 and §7 take up whether that calibration transfers to Australia.

Two recent methodological developments have improved on the AR(p) recipe. Jordà (2005), originally proposed for impulse-response identification, estimates the discounted weighted average of income over the relevant horizon directly as the dependent variable in a single regression, with predictors observable at time t . The local-projection approach sidesteps the compounding of AR misspecification across forecast horizons to which the standard recipe is vulnerable, and admits a richer predictor set than is feasible in a parsimonious AR(p). Carroll, Slacalek and Tokuoka (2014), in ECB Working Paper 1648, document substantial differences between the permanent-income series implied by AR(p) forecasting and by direct local-projection forecasting in the consumer-expenditure context. The differences are concentrated at structural-break episodes, where AR(p) forecasts inherit the slow adjustment of the estimated AR coefficients while local projections can incorporate predictors capturing the regime change.

De Bonis et al. (2020) explicitly adopt the Jordà local-projection approach for their Italian permanent-income series, reporting that the choice "captures much of the slow-down of permanent income growth in the early 1990s" — a structural feature of Italian growth that the AR-based forecaster missed. The Italian local projection includes a $\log(\text{labour_force} / \text{population})$ predictor, which slowly trends with demographic change and is a natural input to long-horizon income forecasting. We adopt the same predictor in our Italy-style permanent-income helper, with a discounted ($\eta = 0.05$, $k = 40$ quarters) weighted average of expected future income (§3, §4), and find a quantitatively similar role: the permanent-income series implied by local projection in Australia diverges materially from the AR-based series in the early 1990s and after the 2008 GFC. Substantively, the implied long-run coefficient on $\log(y^p / y)$ moves from significantly negative under the AR forecaster — the "Australian permanent-income puzzle" — to positive under the full-sample local-projection *measure*. As §8 documents, this reversal reflects the measure's full-sample (non-causal, look-ahead) construction and does not survive a causal real-time projection, which sharply shrinks the speed of adjustment (real-time AR $\lambda = -0.051$, ns; real-time Italy $\lambda = -0.118$ against the full-sample -0.197 ; `pi_realtime_robustness.csv`). The look-ahead versus real-time distinction is therefore flagged as a measurement caveat throughout: the full-sample permanent-income measure is the headline input, with the real-time variant reported as the operational robustness column.

1.3.6 2.6 Where this paper sits

The contribution of this paper is best understood as a faithful single-equation revisit and extension of Williams (2010), and it has five distinct components.

(i) The faithful single-equation LIVES estimate, and the form correction as the lead methodological result. The paper's central finding is that the functional *form* of the LIVES equation is what identifies it. When housing wealth is entered faithfully — only through its credit-conditions interaction (`ha_x_cci`, the de-meanned CCI·(HA/4y) term), with no standalone housing-wealth level — the autonomous-consumption CCI intercept restored ($\zeta_c \cdot \text{CCI}$), and illiquid financial assets combined (`ilfa_y` = equities + superannuation), the error-correction and core wealth structure come alive. The faithful specification (the headline of §7.0) delivers a pre-COVID speed of adjustment of $\lambda = -0.245$ ($t = -4.80$), almost exactly Williams' -0.286 , on a sample of $n = 146$ (full) / 126 (pre-COVID); correctly signed and significant net-liquid and illiquid-financial marginal propensities (structural MPCs of 0.077 and 0.037 respectively, full sample, relative to $|\lambda| = 0.480$); a strong permanent-income coefficient ($+0.50$, $t = 4.0$); and

a housing-collateral coefficient (γ_1) that is right-signed but insignificant (implied structural MPC 0.010, $t = 1.03$, against Williams' peak 0.0488). By contrast, the conventional constant-MPC disaggregated error-correction model that prior work and an earlier draft of this paper treated as the LIVES equation is *not* the LIVES equation and delivers an insignificant speed of adjustment. Reading its insignificant standalone housing coefficient as a failed housing wealth effect is the category error described above; much of the apparent weakness of Australian single-equation LIVES estimates is a specification artefact.

(ii) The structure-transfers-but-calibrations-do-not finding. The LIVES *structure* transfers to Australia, but Williams' Australian *calibrations* do not. Imposing his permanent-income gearing ($\psi_0 = 0.20$, $\psi_1 = 0.93$) and illiquid-financial MPC ($\gamma_{\text{IFA}} = 0.022$) collapses the equilibrium to $\lambda \approx -0.029$ ($t = -0.68$), independently reproduced by a Williams-prior calibrated specification ($\lambda = -0.048$), because Australia freely estimates ψ at roughly 0.50 — about 2.5 times Williams' 0.20 (a framing we hedge, since the headline $\ln(y^p/y)$ coefficient is +0.50 and the ψ_0/ψ_1 split is not separately identified). This reconciles the companion paper's Wald non-rejection of the joint calibration ($\chi^2 = 2.24$): the free estimates are too imprecise to reject Williams' values, but imposing those values wrecks the fit — low power is not the same as good fit. Williams' rate and affordability loadings cannot even be imposed at published magnitudes: his raw $\alpha_r = -0.871$ is roughly thirty times too large on the repository's percent real-rate $\times$ unit-normalised CCI scaling, and diverges the fixed point.

(iii) The interaction-collinearity diagnosis. The six CCI-interacted regressors are 0.74–0.97 mutually correlated on this sample because each is approximately proportional to CCI, and so cannot be separately freely estimated off a single equation. This is the structural reason Williams uses FIML, and it explains why single-equation calibration is empirically closed and why four-equation FIML plus pre-1988 back-extension are the only routes to sharpen the credit channels (§5).

(iv) The 1976Q3 back-extended master dataset and a direct sample-length test. The paper extends the sample back to **1976Q3**. The public-data backbone consists of a Treasury TRYM long-run house-price series (1959Q3+), the RBA D03 M3 monetary aggregate (1959Q3+), the RBA D02 total-credit splice (1976Q3+), and a historical labour-force compilation (1964Q3+). For the 1976Q3–1988Q2 window where ABS sectoral household balance-sheet data are unavailable, we construct aggregate and disaggregated wealth proxies anchored at 1988Q3 (§3). Refitting the disaggregated no-CCI specification on the back-extended sample moves the speed of adjustment 37% closer to Williams (λ from -0.140 to -0.193 ; `lambda_robustness.csv` / `spec46_extended_comparison.csv`), but the individual wealth coefficients shrink and the net-liquid-asset coefficient collapses (a sign flip, -106%) — which we read as evidence that the binding constraint is the single-equation framing itself, not the post-1988 sample window. Sample length is not the binding constraint.

(v) The structured robustness and placebo suite, and the honest negative results. Adopting the Italian methodology of single-equation OLS with Newey-West HAC standard errors, the paper runs a structured robustness suite — instrumental variables, joint SUR, multi-window Chow tests, scaled-income and Drehmann real-rate alternatives, a Kalman state-space CCI, and a Williams-style spline credit-conditions column. The paper is honest throughout about the limitations of single-equation LIVES estimation on post-deregulation Australian data, and reports the negative results prominently as substantive findings that a freely-esti-

mated single-equation framework permits (where imposed restrictions would hide them): the deployed credit-conditions index sits only in the 18th–34th adjusted- R^2 percentile band of a knot placebo across variants; a two-equation consumption + house-price SUR finds negligible cross-equation residual correlation and so no efficiency gain (the literal value ≈ -0.004 cited in the companion work is a narrative figure; the committed evidence is the OLS-versus-SUR coefficient stability in `joint_pi_robustness.csv`); out-of-sample, the random walk with drift beats every structural specification at horizons $h = 4$ and $h = 8$; the automated specification selector diverges from both the BIC-minimising and the theory-preferred specification; and the look-ahead permanent-income reversal does not survive real-time construction. Each negative is read diagnostically — it points to why FIML and back-extension, rather than further single-equation refinement, are the way forward.

The headline paper does not estimate the multi-equation LIVES system itself, though we scaffold the multi-equation build in a separate companion directory. Williams (2010) estimates four equations jointly by FIML; the headline of the present paper estimates the consumption equation alone. The Italian experience (De Bonis et al. 2020) suggests that single-equation OLS produces consumption-equation coefficients close to joint SUR estimation, and we replicate that finding (§8): on our back-extended sample a two-equation consumption + house-prices SUR finds negligible residual correlation, so joint estimation gives no efficiency gain at the quarterly frequency. The case for the full multi-equation build therefore rests on cross-equation parameter restrictions — the common-factor identification of the credit channels — rather than on efficiency. The full multi-equation extension is left to a companion paper, with the scaffolding documented separately.

1.4 3. Data and measurement

The dataset assembles quarterly Australian macroeconomic and household-sector observations from **1976Q3 to 2024Q4** ($n = 194$). The public-data backbone for the pre-1980 window is built from a Treasury TRYM long-run house-price series, the RBA D03 M3 monetary aggregate, the RBA D02 total-credit splice, and a historical labour-force compilation. The disaggregated wealth components — the housing, illiquid-financial and net-liquid ratios that carry the faithful LIVES specification (Spec 11) — remain bounded at 1988Q3 by their primary source (ABS Cat 5232.0 Household Balance Sheet); for the 1976Q3–1988Q2 window we construct proxies (§3.13 below) that growth-rate-splice each component onto its 1988Q3 official value via the most relevant available aggregate. It bears emphasis at the outset that 1988Q3 is when the ABS sectoral balance sheet begins, so the financial-liberalisation episode that identifies Williams' credit channels — the 1980s deregulation — largely *predates* the modern disaggregated data. This timing, rather than estimator choice, is the deepest single-equation constraint on identifying the credit-conditions interactions, and motivates the back-extension to 1976Q3 documented below (and, ultimately, the four-equation FIML route discussed in §5 and §9).

Estimation is performed on the largest contiguous subset for which all variables in a given specification are observed:

- **Spec 1–3 (aggregate net worth)** with the back-extension proxy fit on $n=190$ (1977Q3–2024Q4; the binding constraint is `real_rate`, which needs a 4-quarter CPI lag for inflation computation). On the official `networth_y` (1988Q3+ ABS) they fit on $n=146$.

- **Spec 4 (disaggregated, no CCI) and the faithful LIVES specification (Spec 11)** use the official disaggregated series and fit on $n=146$ (1988Q3–2024Q4), with a **pre-COVID** sub-sample of $n=126$ (1988Q3–2019Q4) treated as the identified window throughout the paper.
- **Spec 5 (full disaggregated) and Spec 6 (conventional constant-MPC disaggregated ECM — the baseline, not the headline)** are both bounded at 2002Q3+ because each carries the short-run CCI variable ($d2_logcci_lag2$), which depends on $cci_ratio = \log(\text{housing_loan_flow} / \text{income})$ from ABS Cat 5601.0; that loan-flow series only starts 2002Q3. Both bind at $n=86$ (full) / $n=66$ (pre-COVID) in both samples.
- **Spec 8 (free CCI interactions) and the Williams-calibrated Spec 12** fit on $n=146$, bounded by the disaggregated wealth and the $cci_williams$ smoothed-step spline (§3.5), which is constructed across the full 1988Q3+ window and does not require the 2002Q3 loan-flow series. **Spec 10 (Williams-prior reproduction)**, by contrast, carries the short-run $d2_logcci_lag2$ term and so binds at the 2002Q3 loan-flow constraint, fitting on $n=86$ (full) / $n=66$ (pre-COVID).
- **Spec 9 (Kalman state-space CCI)** fits on the same $n=146$ window as the disaggregated specs; the Kalman extraction itself is a separate state-space step.

The asymmetry in sample length is itself substantive. The faithful LIVES specification (Spec 11) is estimable on $n=146$ precisely because it proxies the credit channels through the $cci_williams$ spline rather than the 2002Q3-bounded loan-flow ratio, whereas the conventional Spec 6 baseline binds at $n=86$. Comparisons between the two should keep this sample difference in view; the larger $n=146$ window is one reason the faithful form delivers a sharper error-correction estimate (§7).

1.4.1 3.1 Aggregate consumption and income

Real per capita consumption ($cons_real_pc$) is constructed from ABS Cat 5206.0 Table 8 (Household Final Consumption Expenditure, chain volume measures, seasonally adjusted) divided by the civilian population aged 15 years and over (ABS series A84423091W, monthly, averaged to quarterly arithmetic mean; sample 1978Q2–2024Q4, sourced directly from the ABS historical series workbook). Following Williams (2010, 2009), we do not sum the single-year-of-age cohorts in ABS Cat 3101.0; in current ABS vintages, the Persons series only extends to age 47 in the disaggregated file, producing a population total approximately 35 per cent below the true Estimated Resident Population.

Real per capita household disposable income (ydi_real_pc) is the quarterly seasonally adjusted nominal series from ABS Cat 5206.0 Table 20 (Household Income Account), deflated by the consumption deflator implied by Table 8 (chain volume and current prices), and divided by the same 15+ population denominator. Following standard practice (Blinder–Deaton 1985), and as the headline empirical specification, we use gross disposable income; we report a non-property income (NPY) alternative constructed per Williams (2009) §4.2.1 in §3.6 below.

In the canonical LIVES equation (§4) income enters the long run with a **unit coefficient** — the error-correction term is the consumption-to-income ratio $\log y - \log c_{\{t-1\}}$, not a freely estimated income elasticity. The disposable-income series above is therefore load-bearing for the *level* of the equilibrium, not merely as a scaling variable, and the choice of income measure (gross vs non-property) is one of the channels through which our estimates and Williams' diverge (§3.6, §9).

1.4.2 3.2 Household balance sheet

Household-sector balance sheet stocks are sourced from ABS Cat 5232.0 Table 35 (Household Balance Sheet Aggregates, current prices, \$ billion), quarterly from 1988Q3:

- Currency and deposits (fin_deposits)
- Shares and other equity (fin_equities)
- Superannuation reserves (fin_super)
- Total household liabilities (loans and placements; fin_loans)
- Residential land and dwellings (housing_wealth)
- Closing net worth (ABS series A83722648X)

All balance-sheet stocks are converted to real values by dividing by the normalised consumption deflator and to per-capita-15+ values by dividing by the 15+ population.

Following Williams' Eq (7), wealth enters the long run as **asset/annualised-income ratios**, $x_{t-1}/4y$. The wealth-to-annualised-income ratios used in the long-run consumption equation are constructed as:

- $ha_y = \text{housing wealth} / (4 \times \text{quarterly nominal disposable income})$
- $eq_y = \text{equities (ex-super)} / (4 \times \text{quarterly nominal disposable income})$
- $super_y = \text{superannuation reserves} / (4 \times \text{quarterly nominal disposable income})$
- $ilfa_y = (\text{equities} + \text{superannuation}) / (4 \times \text{quarterly nominal disposable income})$
- $nla_y = (\text{deposits} - \text{total household debt}) / (4 \times \text{quarterly nominal disposable income})$
- $debt_y = \text{total household debt} / (4 \times \text{quarterly nominal disposable income})$
- $networth_y = \text{closing net worth} / (4 \times \text{quarterly nominal disposable income})$

Each ratio is dated $x_{t-1}/4y$ (beginning-of-period stock over annualised current income), consistent with the timing convention in Williams' bracket.

1.4.2.1 3.2.1 Illiquid financial assets are combined in the faithful LIVES specification

The LIVES theory distinguishes three wealth classes by their marginal propensity to consume: housing (a collateral asset, with an MPC that is *zero* at zero credit conditions and unlocked only as CCI rises), **illiquid financial assets** (IFA = equities plus superannuation, with a low plain MPC), and net liquid assets (NLA = deposits net of debt, with the highest plain MPC). Williams (2010) and the Italian implementation (De Bonis, Liberati, Muellbauer & Rondinelli 2020) both enter equities and superannuation as a single illiquid-financial aggregate, calibrating $\gamma_{IFA} = 0.022$.

The faithful LIVES specification (Spec 11) follows this convention: it uses the **combined** $ilfa_y = eq_y + super_y$ rather than entering equities and superannuation separately. This is not a cosmetic aggregation. Separating the two financial classes (as the conventional Spec 6 baseline does, with plain eq_y and $super_y$) repeatedly delivers a wrong-signed, insignificant equities coefficient on post-deregulation Australian data (e.g. Spec 6 full-sample $eq_y = -0.006$, $t = -0.13$; Spec 9 $eq_y = -0.013$, ns), reflecting the high collinearity between the two series and the short modern sample rather than a negative MPC on equities. Combining them into $ilfa_y$ recovers a correctly signed and significant illiquid-financial loading in the faithful form (Spec 11 full-sample $ilfa_y = +0.0178$, $t = 3.03$, ***; implied structural MPC = 0.037), the same order of magnitude as Williams' calibrated 0.022 though somewhat larger. The aggregation is therefore both theoretically faithful and empirically necessary.

1.4.2.2 3.2.2 Net liquid assets: deposits net of debt

We adopt the convention of the Italian implementation (De Bonis, Liberati, Muellbauer & Rondinelli 2020, eq. 2.5; Table 3 column 3) of defining **net liquid assets as deposits net of total household debt**:

$$nla_y = (fin_deposits - fin_loans) / (4 \times \text{quarterly nominal disposable income})$$

This embeds the cross-equation restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ (equal and opposite MPCs on liquid assets and on debt) implicitly by construction, in place of estimating a separate liquid-asset and a separate debt MPC. We test this restriction formally in §8: the netting restriction is **accepted at the 5 per cent level in every disaggregated specification and sample window** (australia_nla_restriction_test.csv; e.g. Spec 4 full sum = +0.0800, $t = 1.43$, $p = 0.15$; Spec 6 full sum = +0.105, $t = 1.02$, $p = 0.31$). The honest reading is that the modern sample cannot statistically distinguish separate liquid-asset and debt MPCs, which is consistent with — but does not by itself confirm — the NLA aggregation; netting is a defensible economy of parameters rather than a sharply identified result.

In the faithful LIVES specification, NLA carries the largest and most robust wealth loading: Spec 11 full-sample $nla_y = +0.037$ ($t = 3.40$, ***), an implied structural MPC of 0.077, against Williams' calibrated 0.159.

1.4.3 3.3 Mortgage interest rate

The nominal mortgage rate (mortgage_rate) is the RBA Standard Variable Owner-Occupier Rate (RBA Bulletin Table F6 housing lending rates, series FILRHLBVS), monthly from January 1959, averaged to quarterly. We source the historical series from the published RBA archive rather than the live readrba API, to ensure a stable vintage; the rate peaks at 17.0 per cent in 1989Q3, consistent with the Hawke–Keating recession.

The real mortgage rate (real_rate) is the nominal rate less the 4-quarter-ended percentage change in the consumption deflator. In the faithful LIVES specification the rate enters only through its credit interaction, $r_x_cci = real_rate \times CCI$ (Williams' α_1 channel). We note here a measurement point that becomes load-bearing in §9: our *real_rate* is expressed in *percentage* units (e.g. 3.0 for 3 per cent), and CCI is *unit-normalised*, so Williams' published raw rate loading ($\alpha_r = -0.871$) is roughly thirty times too large to impose on this scaling and diverges the iterative fixed point — one of the calibrations that, by construction, cannot transfer. We do not adopt the Italian Drehmann (2017) amortising-mortgage adjustment in the headline specifications but report it as a robustness check (§8).

1.4.4 3.4 House prices

The headline house price index (hpi) is constructed by chain-linking **four** sources, with the deepest segment extending to 1959Q3:

Layer	Source		Coverage	Splice convention		
TRYM Model historical database (house_price_history_long.csv)	Treasury	Macroeconomic	1959Q3–2018Q2	growth-rate, anchored at 1986Q2	at	
	Legacy	houseprice_old.csv (privately compiled pre-2003 dwelling-price index, monthly→quarterly)	1986Q2–2003Q3	growth-rate, anchored at 2003Q4	at	
Bridge	ABS Cat 6416.0	Residential Property Price Index, 8-capital-cities ("old method")	2003Q4–2017Q2	growth-rate, anchored at first overlap	at	
Current	ABS Cat 6432.0	Total Value of Dwellings, mean price	2003Q3–2024Q4	(the modern overlay)		

The TRYM source supersedes the BIS Shrapnel / REIA chain used in Williams (2010): the TRYM historical compilation already incorporates the same BIS Shrapnel (pre-1978), REIA (1978–1986) and ABS (post-1986) segments that Williams used, pre-chained into a single coherent 236-quarter series (1959Q3–2018Q2). The earliest binding observation for hpi is therefore **1959Q3**, three years deeper than even Williams' fullest sample.

In the faithful LIVES specification (§4), the house price enters the long run *only* through the affordability interaction $hp_x_1_minus_cci = (1 - 1.2 \cdot CCI) \cdot \log(ph/y)$ (Williams' α_4 channel, with affordability multiplier $\varpi = 1.2$), not as a standalone level. There is no separate classical house-price level term in the faithful form.

1.4.4.1 3.4.1 Splice methodology – pure growth-rate chain-linking

For each adjacent pair of layers, the splice anchors the level at the first quarter where both series are non-NA, then back-casts via the base series' own QoQ growth rates:

```

chained[t] = overlay[t_anchor] × (base[t] / base[t_anchor])   for t < t_anchor
chained[t] = overlay[t]                                       for t >=
t_anchor

```

This standard ABS chain-linking convention preserves the base series' growth rates exactly while pinning the level to the overlay at the join. By construction there is no level discontinuity at any join quarter. An earlier implementation used $\text{mean}(\text{overlay}/\text{base})$ over the full overlap, which produced step jumps at join quarters where the overlap ratio drifts (notably a –17% step at 1986Q2 under the long TRYM↔legacy overlap and a +10% step at 2011Q2→Q3 under the bridge↔current overlap with mismatched units – the bridge is an index while current is a \$-value mean). Both artefacts were eliminated by the growth-rate convention.

The relative house-price-to-income ratio used in estimation is

$$\ln_hp_over_y = \log(hpi / (ydi_ann_nom / pop_millions / (cons_deflator_norm / 100)))$$

i.e. the log of the real house-price index divided by real disposable income per capita.

1.4.5 3.5 Credit conditions index

The credit conditions index (CCI) is the most contested input. Williams (2010, 2012) constructs CCI as the latent factor identifying credit liberalisation episodes: a spline $\sum a_s \cdot \text{SDMMA}_s$ of smoothed-step dummies at the institutional turning points 1979 (Campbell Committee, end of interest ceilings), 1992 (banking distress, Aussie Home Loans), 1998 (NBFI expansion), and 2007 (GFC), estimated **jointly** across the four-equation FIML system (consumption, house-price, mortgage-stock and home-equity-withdrawal equations) under sign priors, with the house-price-equation loading $\zeta_h = 1$ normalised. The structural reason for the joint estimation is identification: CCI enters six channels of the consumption equation, and (as §5 develops) the six CCI-interacted regressors are between 0.74 and 0.97 mutually correlated on this sample — each is approximately proportional to CCI — so they cannot be separately identified off the consumption equation alone.

A single equation cannot reproduce the FIML identification, so we proxy or calibrate CCI by two routes:

- **Default observable proxy** (`cci_ratio`): the log of housing-credit flow (ABS Cat 5601.0 New Loan Commitments Value, total housing) divided by the eight-quarter moving average of nominal disposable income. Available 2002Q3 onward ($n=90$); used only as a *short-run* regressor (`d2_logcci_lag2`) in the conventional Spec 6 baseline and Spec 2. This is the binding 2002Q3 constraint on Spec 6 (§3 opening). We do not back-extend it with a mortgage-spread proxy; that option is retained behind a feature flag but disabled.
- **Williams-style smoothed-step spline** (`cci_williams`): smoothed-step (SDMMA) dummies at the Australian institutional turning points, reduced by general-to-specific search with sign priors enforced by drop-on-violation (in the spirit of Hendry–Krolzig 2005). This is the CCI proxy used in the faithful LIVES specification (Spec 11), where it also restores the autonomous-consumption loading $\zeta_c \cdot \text{CCI}$ (`cci_williams`) inside the bracket. On our 1988Q3-onward sample, **only three** of fifteen candidate knots survive the sign-prior reduction (`australia_williams_cci_knots.csv`): `sdmma_2009_01` (+), `sdmma_2019_01` (–) and `sdmma_2020_04` (+); the 1979 and 1986 knots are aliased/constant within the window and the remaining candidates violate their priors. Only one of Williams' four canonical knots (2007) survives on a post-1988 sample. We report a full placebo battery on the surviving spline in §5: the deployed CCI sits in the 18th–34th adjusted- R^2 percentile band across placebo variants, a result we preserve as an honest negative.

The first-home-buyer share (`fhb_share = fhb_loans / total_new_loans`) is also constructed from ABS Cat 5601.0 from 2002Q3, and enters Spec 7.

1.4.6 3.6 Williams (2009) non-property income

The Williams (2009) §4.2.1 non-property income measure (`npv_real_pc`) adjusts gross disposable income by removing imputed property income and a corresponding share of property-related taxation. Following Williams (2009 p. 10):

$$\begin{aligned} \text{npv_rec} &= \text{total_income_rec} - \text{GOS_dwellings} - \text{prop_inc_rec} \\ \text{property_tax_share} &= (\text{GOS_dwellings} + \text{prop_inc_rec}) / \text{total_income_rec} \\ \text{npv_pay} &= \text{total_income_pay} - \text{prop_inc_pay} - \text{property_tax_share} \times \text{income_tax_payable} \\ \text{NPY} &= \text{npv_rec} - \text{npv_pay} \end{aligned}$$

The input components are sourced from ABS Cat 5206.0 Table 20: compensation of employees, gross operating surplus on dwellings, property income receivable, social assistance benefits,

property income payable, total income receivable, total income payable, and income tax payable.

The implied non-property income share of disposable income averages 0.84 over 2010–2024, consistent with Williams' implicit ~ 0.85 weighting. We use NPY as a robustness column (§8): substituting NPY for gross income roughly halves the estimated speed of adjustment (Spec 2 $\lambda = -0.193 \rightarrow -0.094$; `australia_williams_income_robustness.csv`). The income measure is one of several channels through which our estimates and Williams' diverge, and we treat it as a methodological caveat rather than a resolved choice.

1.4.7 3.7 Permanent income

Permanent income (yp) is the forward-looking object in Williams' ratio $\log(yp/y)$. We construct it as a **discounted weighted average of expected future income** from a forecasting model, following the Italian implementation's Jordà (2005) local-projection approach. The baseline uses discount $\eta = 0.05$ (equivalently a per-quarter weight $\delta = 0.95$) over a horizon $k = 40$ quarters (`australia_permanent_income_sensitivity.csv`, baseline row $\delta = 0.95$, $k = 40$). The estimated speed of adjustment is very stable across $\delta \in \{0.90, 0.95, 0.97\}$ and $k \in \{20, 40, 60\}$.

Two important measurement caveats are preserved throughout the paper:

1. **Look-ahead vs real-time.** The headline permanent-income measure is estimated on the full sample and therefore uses information not available in real time. The real-time variant (re-estimating the forecaster recursively) shrinks the speed of adjustment sharply and flips or shrinks the $\log(yp/y)$ coefficient (`australia_pi_realtime_robustness.csv`; e.g. AR real-time $\lambda = -0.051$, ns; Italy real-time $\lambda = -0.118$ vs Italy full-sample -0.197). We flag the full-sample measure as the headline and the real-time column as the operational robustness check; headline results that depend on the look-ahead measure are identified as such in §7.
2. **Forecaster choice and the Australian permanent-income puzzle.** Under an AR forecaster the $\log(yp/y)$ coefficient is *negative* (the "Australian permanent-income puzzle"); under the Italy-style local-projection forecaster it is positive and the Italian forecaster fits marginally better (`australia_pi_method_meta.csv`: AR $\text{adj-}R^2 = 0.758$, Italy $\text{adj-}R^2 = 0.769$). The faithful LIVES specification uses the Italy-style measure and recovers a strong, correctly signed permanent-income loading (Spec 11 full-sample $\ln_{yp_over_y} = +0.504$, $t = 4.0$, ***).

In the canonical LIVES bracket the permanent-income weight is itself credit-dependent, $\psi(\text{CCI}) = \psi_0 + \psi_1 \cdot \text{CCI}$ with $\psi_0 = 0.20$ and $\psi_1 = 0.93$ **calibrated** (not freely estimated) by Williams. As §7 and §9 develop, this is the calibration that most clearly does not transfer: Australia freely estimates ψ on the order of 0.50, and imposing Williams' value collapses the error-correction term. The data section flags only that $\psi(\text{CCI})$ is calibrated in Williams' original — the estimation consequences are reported in the results.

1.4.8 3.8 Demographics and dummies

The prime-working-age share (`prime_age_share` = age 25–54 share of total ERP) is constructed from ABS Cat 3101.0 single-year-of-age cohorts (Male + Female, summed; ratios are robust to the truncation discussed in §3.1). Annual data are interpolated to quarterly via cubic spline.

The canonical LIVES equation also contains two demographic terms in the long-run bracket – D4 DEMFTB (the first-home-buyer / demographic cohort) and D4 WAPOP (working-age population growth) – for which Williams reports $\alpha_2 = -0.138$ and $\alpha_3 = -0.069$. These are not directly constructed in the single-equation specifications; `prime_age_share` (Spec 7) and the first-home-buyer share (`fhb_share`, Spec 7) stand in for the cohort channel, and we treat the full demographic block as a target for the FIML system rather than the single equation.

Australia-specific narrative dummies enter the default dummy set:

- `d_neg_gearing_8587`: 1985Q3–1987Q3 negative-gearing tax restriction
- `d_recession_1991`: 1991Q2 ("recession we had to have")
- `d_apra_2014`: 2014Q4 macroprudential investor-loan-growth cap (smoothed-step ogive)
- `d_apra_2017`: 2017Q2 macroprudential interest-only-loan cap (smoothed-step ogive)
- `d_jobkeeper_2020`: 2020Q2–2021Q1 JobKeeper income support

Together with the standard set (`d2000_gst`, `d2008_gfc`, `d2020_covid`, `d2020_rebound`), these constitute the full dummy set; zero-variance dummies are silently dropped per specification. The COVID quarters carry a very large dummy (`d2020_covid` ≈ -0.157 in Spec 11), and the COVID-inflated full sample fails the upper-bound λ screen; we therefore treat the **pre-COVID** sub-sample ($n=126$) as the identified window for the speed of adjustment (§7).

1.4.9 3.9 Coverage and reproducibility

The full reproducibility kit (R 4.5.3, `renv`-pinned dependencies, raw ABS workbooks, project-supplied CSVs, master quarterly dataset as CSV and RDS, full estimation pipeline with three execution modes, GitHub Actions CI, `testthat` suite) accompanies this paper. The master dataset has **194 quarters** $\times$ **~120 columns** and is available as a portable CSV for hand-editing or off-line replay. See Appendix A for each variable's source identifier, vintage, and splicing recipe.

1.4.10 3.10 RBA D-tables

Three RBA historical statistical tables support the pre-1988 portion of the sample.

1.4.10.1 3.10.1 RBA D03 — Monetary aggregates (M3)

The M3 monetary aggregate (`m3_aggregate`) is loaded from `d03hist.xlsx`, series `DMAM3N` (M3, original/not-seasonally-adjusted, \$ billion). M3 is total economy-wide currency plus transaction deposits plus all other deposits at ADIs, plus certificates of deposit issued by ADIs. Coverage: monthly, **1959Q3–2026Q1** (continuous, no series breaks). Aggregated to quarterly by mean of the three monthly observations.

M3 is the headline liquid-asset proxy for the pre-1988 portion of the sample where ABS sectoral household-balance-sheet deposits are not available. Williams (2010) used the same M3 series, multiplied by the household share of factor income, for his pre-1988 splice.

1.4.10.2 3.10.2 RBA D02 — Lending and credit aggregates (total credit)

Total credit (`credit_total_d02`) is constructed by growth-rate-splicing two RBA D02 series across the July 2019 RBA conceptual reform:

- `DLCACN` (Total credit, original, \$ billion) for 1976Q3–2019Q2
- `DLCACSFN` (Total credit including select financial businesses, the post-2019 successor) for 2019Q3+

The two series have no quarterly overlap (DLCACN ends 2019-06, DLCACSFN starts 2019-07). The splice anchors levels at the boundary ($\text{first_post} \times \text{pre}[t] / \text{last_pre}$) so the join is continuous in level; the implicit growth rate at the boundary is exactly zero (no overlap to estimate it from). For analyses that hinge on the 2019Q2→Q3 quarter specifically, treat with caution. Coverage 1976Q3+.

This is total credit, not housing-specific. The housing-specific series in D02 (DLCAC0HN, DLCACIHN) only goes back to 1990Q1 in the current vintage and so cannot extend the housing-credit history pre-1990.

1.4.10.3 3.10.3 RBA D01 — Growth in selected financial aggregates

Downloaded as `d01hist.xlsx` (monthly growth rates of the same aggregates whose levels are in D02). Williams (2010) used D01 housing-credit growth rates to back-cast D02 levels pre-1976; in the current vintage D02 already extends to 1976Q3, so D01 is not strictly needed for back-extension. Retained for future use.

1.4.11 3.11 Pre-1978 labour force (`labour_force_historic.csv`)

A user-supplied CSV (`labour_force_historic.csv`, 188 quarterly rows 1964Q3–2011Q2) provides:

Column	Definition	Units
<code>pop_total</code>	Total resident population (annual + interpolation)	thousands
<code>pop_15_64</code>	Working-age population (15–64)	thousands
<code>labour_force</code>	Civilian labour force	thousands
<code>unemployed</code>	Unemployed persons (level)	thousands

Provenance: the source compilation pulls together pre-1978 series from ABS Cat 6204.0 (historical labour force, archived), ABS Year Book Australia, the Foster (1996) *Australian Economic Statistics 1949–50 to 1996–97* compilation, and RBA Occasional Paper No 8. Conceptually consistent with the current ABS Cat 6202.0 series (which begins Feb 1978) at the join.

The historic series are growth-rate-spliced onto the modern (1978+) series at 1978Q1:

- `pop_15_64` → `pop_millions` (the working-age population denominator used in per-capita normalisation)
- `labour_force` → `master labour_force`
- `unemployed/labour_force × 100` → `master unemp_rate` (level replacement before 1978Q1)

A side-effect of this splice is that the six quarters 1976Q3–1977Q4 now have non-NA values for all per-capita and labour-force-derived variables (`cons_real_pc`, `ydi_real_pc`, `npv_real_pc`, `lf_share`, etc.). On the data download, `cons_real`, `ydi_nom`, `mortgage_rate`, `hpi`, `M3`, `total credit`, and `prime_age_share` all extend to 1976Q3 already; the labour-force splice was the binding remaining constraint. The 1976Q3 back-extension — enabling the direct sample-length test reported in §8 — is one of the contributions of this paper.

`pop_total` is also exposed in the master as `pop_total_thousands` (historic-only 1964Q3–2011Q2 because there is no modern 1978+ counterpart in ABS 6202). It can be used as an alternative per-capita denominator (total resident vs civilian 15+) for sensitivity analyses.

1.4.12 3.12 Household-allocated M3 (m3_household_proxy)

The pre-1988 liquid-asset proxy is constructed by allocating M3 to the household sector via the wage share of GDP:

$$m3_household_proxy = m3_aggregate \times wage_share / 100$$

Williams (2010) uses the household factor-income share for this allocation. The wage share alone (compensation of employees / GDP) is a defensible approximation: it captures most of household factor income (wages dominate Australian household income; mixed income + property income receivable add ~10 percentage points and track wage share over time). **This is a documented simplification for our phase-1 back-extension**; for a fully Williams-faithful splice, replace wage_share with the broader (compensation + mixed income + property income receivable) / GDP series. We flag it explicitly here because the back-extension proxies should be read as approximations adequate for the direct sample-length test in §8, not as substitutes for the official 1988Q3+ series.

wage_share itself is loaded from a user-supplied CSV (household_income.csv) which extracts ABS Cat 5206.0 Table 24 analytical series A2302604K (compensation of employees, % of GDP), 1959Q3–2024Q4. Range over the sample: 49–62%, with values ~60% in the 1970s falling to ~50% today (capital-share rise).

Coverage of m3_household_proxy: 1976Q3+ (limited by wage_share in the supplied CSV), values \$22 billion (1976Q3) to \$1,673 billion (2024Q4).

1.4.13 3.13 Aggregate net-worth proxy (networth_y_proxy)

To enable Spec 1–3 fits on the back-extended sample, we construct an aggregate net-worth proxy that:

1. Uses M3-allocated-to-households (m3_household_proxy) plus a hpi × pop_millions back-cast of housing wealth as the raw wealth aggregate.
2. Growth-rate-splices that raw aggregate ratio onto the official networth_y at 1988Q3 (so the proxy equals the official series from 1988Q3 onwards and back-casts smoothly through 1976Q3).

The housing-wealth back-cast anchors at the first available official housing_wealth observation (1988Q3) and back-casts via housing_wealth × hpi × pop_millions growth, holding dwellings per capita constant:

$$\begin{aligned} \text{housing_wealth_proxy}[t] &= \text{housing_wealth}[1988Q3] \\ &\quad \times (\text{hpi}[t] / \text{hpi}[1988Q3]) \\ &\quad \times (\text{pop_millions}[t] / \text{pop_millions}[1988Q3]) \\ &\quad \text{for } t < 1988Q3 \\ &= \text{housing_wealth}[t] \text{ for } t \geq 1988Q3 \end{aligned}$$

The aggregate proxy is then:

$$\begin{aligned} \text{raw_proxy}[t] &= (m3_household_proxy[t] + \text{housing_wealth_proxy}[t]) / \\ \text{ydi_ann_nom}[t] & \\ \text{scale} &= \text{networth_y}[1988Q3] / \text{raw_proxy}[1988Q3] \quad (\approx 1.68 \text{ in current} \\ &\quad \text{vintage}) \\ \text{networth_y_proxy}[t] & \\ &= \text{networth_y}[t] \quad \text{for } t \geq 1988Q3 \\ &= \text{raw_proxy}[t] \times \text{scale} \quad \text{for } t < 1988Q3 \end{aligned}$$

Caveats explicitly documented in the data appendix:

- The back-cast omits equities and super (quantitatively small in 1976–1988 — Australian super pre-Superannuation Guarantee 1992 was a negligible household asset class).
- The back-cast omits debt netting (mortgage debt was a much smaller share of household balance sheets in the 1970s than today).
- Use only for back-extension exercises; never as a substitute for `networth_y` on the modern sample where the official series exists.

The proxy values across key dates: $5.05 \times$ annual income (1976Q3), 4.71 (1980Q1), 4.78 (1985Q1), 5.37 (1988Q3 — anchored to official), 10.19 (2024Q4). The shape is consistent with the historical Australian wealth-to-income trend (flat through the 1970s/early 80s, sharply rising post-1985 financial deregulation).

1.4.14 3.14 Disaggregated wealth proxies

Four additional proxies extend the disaggregated wealth components to the back-extended 1976Q3+ sample. Each equals the official series for $t \geq 1988Q3$ by construction; for $t < 1988Q3$ it back-casts via the most relevant available aggregate. These proxies enable the direct sample-length test of §8 (back-extension on Spec 1/4), where the honest finding is that lengthening the sample moves λ partway towards Williams but does *not* sharpen the individual wealth channels — sample length is not the binding constraint; the binding constraints are the post-deregulation data window and the interaction collinearity.

ha_y_proxy = `housing_wealth_proxy` / `ydi_ann_nom`. Uses the `hpi`×`pop` back-cast described in §3.13. Values: 2.68 (1976Q3) → 2.83 (1988Q3) → 6.41 (2024Q4).

fin_deposits_proxy anchors `fin_deposits`[1988Q3] and grows by `m3_household_proxy`:

```
fin_deposits_proxy[t] = fin_deposits[1988Q3]
                      × m3_household_proxy[t] / m3_household_proxy[1988Q3]
                      for t < 1988Q3
```

fin_loans_proxy anchors `fin_loans`[1988Q3] and grows by RBA total credit:

```
fin_loans_proxy[t] = fin_loans[1988Q3]
                   × credit_total_d02[t] / credit_total_d02[1988Q3]
                   for t < 1988Q3
```

nla_y_proxy = (`fin_deposits_proxy` - `fin_loans_proxy`) / `ydi_ann_nom`, again on the deposits-net-of-debt convention of §3.2.2. Values: +0.20 (1976Q3, households are net liquid creditors) → -0.05 (1988Q3) → -0.72 (2024Q4, modern net-debtor position). The sign-flip around 1988 captures the post-deregulation debt build-up — and is precisely the variation that the credit-conditions channel is meant to explain but which the modern data window mostly excludes.

eq_y_proxy: held constant at the 1988Q3 value pre-1988 (Option B in our methodology — Australian household equity holdings were a small wealth share in the late 1970s/early 80s; the constant assumption introduces little level error and is straightforward to upgrade to an ASX All-Ordinaries back-cast). Value: 0.60 (constant) → 0.92 (2024Q4).

super_y_proxy: linear ramp from $0.1 \times$ the 1988Q3 value at 1976Q3 to the 1988Q3 value, then official thereafter. The 0.1 anchor matches Williams (2010) Table A.1 ballpark for the pre-

Superannuation-Guarantee era (SGC mandate 1992). Values: 0.07 (1976Q3) → 0.66 (1988Q3) → 2.44 (2024Q4).

We flag these back-extension proxies honestly: the equities proxy is a constant-share assumption, the superannuation proxy is a linear ramp, and both the deposits and debt proxies are aggregate-growth back-casts rather than household-sector measurements. They are adequate for testing whether sample length is the binding constraint (it is not — §8) but they should not be over-interpreted as reconstructions of the 1970s household balance sheet. For the faithful LIVES specification (Spec 11) and all headline results we use the **official** 1988Q3+ disaggregated series and the n=146 / n=126 windows; the proxies are confined to the back-extension robustness exercises.

1.4.14.1 3.14.1 Coherence check — disagg sum vs aggregate proxy

Both `networth_y_proxy` (M3-allocated + housing back-cast, growth-rate-spliced onto the official broad networth) and the **sum-of-disaggregated** `networth_y_disagg_proxy = ha_y_proxy + nla_y_proxy + eq_y_proxy + super_y_proxy` are exposed in the master.

At 1988Q3 (the boundary):

- official `networth_y` (broad, ABS A83722648X closing net worth): **5.37**
- `networth_y_proxy` (aggregate, anchored to broad): **5.37** by construction
- `networth_y_disagg_proxy` (narrow, sum of components): **4.04**

The roughly 25 per cent gap between the aggregate and the disaggregated sum is the "other wealth" component of ABS closing net worth (life-office reserves, unincorporated business equity, etc.) absent from the narrow definition. For Spec 4–7 and the faithful LIVES specification (Spec 11), which use the disaggregated components individually, the disagg sum is the implicit reference; for Spec 1–3 the aggregate proxy is used.

1.4.15 3.15 Master variable coverage tiers (under 1976Q3+ spine)

After all splicing, master variable coverage falls into the following tiers (`australia_model_dataset.csv`):

First non-NA	n	Variables (selected)
1976Q3	~56	cons, ydi, hpi, mortgage_rate, M3, total credit, prime_age_share, all dummies, m3_household_proxy, ha_y_proxy, nla_y_proxy, eq_y_proxy, super_y_proxy, networth_y_proxy, ln_networth_y_proxy, cci_kalman, cons_real_pc, ydi_real_pc, npy_real_pc, labour_force, unemp_rate, lf_share, pop_millions, ln_cons_real_pc, ln_ydi_real_pc, ln_hp_over_y, ecm_lag (1976Q4)
1977Q3	2	real_rate, hicp_4q_ann (4-quarter CPI lag)
1978Q2	1	ydi_ann_8qma (8-quarter MA)
1988Q3	~21	OFFICIAL disaggregated wealth (ha_y, eq_y, super_y, ilfa_y = eq_y + super_y , nla_y, networth_y, debt_y), housing_wealth, fin_deposits/equities/super/loans (RAW from ABS 5232) — n=146
2002Q3	5	cci_ratio, fhb_share, housing_loan_flow, fhb_loans, non_fhb_loans — n=90
2009Q1	2	mortgage_interest_burden_rba, mortgage_payment_burden_rba (RBA E13) — n=64

The 1988Q3 tier is the binding tier for the faithful LIVES specification (n=146 full; n=126 pre-COVID), and the 2002Q3 tier binds the conventional Spec 6 baseline (n=86 full; n=66 pre-COVID). The combined illiquid financial ratio *ilfa_y* first becomes available at 1988Q3, alongside its *eq_y* and *super_y* constituents.

1.5 4. Model

1.5.1 4.1 The canonical LIVES consumption equation

The decisive contribution of this paper is that the *functional form* of the Muellbauer-Williams LIVES (life-cycle / inter-temporal / credit- conditions) equation is what identifies it. We therefore present the faithful form of Williams' (2010) eq. (7) as the model, and present the generic constant-MPC error-correction model that an earlier draft and much of the Australian literature treat as "the LIVES equation" as a nested baseline (Spec 6, §4.5) that the theory predicts should *not* behave like LIVES.

In its canonical form the LIVES consumption equation is

$$\Delta \ln c_t = \varphi \cdot [\zeta_c \cdot CCI_t + \alpha_1 \cdot r_t \cdot CCI_t + \gamma_1 \cdot CCI_t \cdot (HA/4y)\{t-1\} + \gamma_2 \cdot (IFA/4y)\{t-1\} + \gamma_3 \cdot (NLA/4y)\{t-1\} + \psi(CCI_t) \cdot \ln(y^h p_t / y_t) + \alpha_2 \cdot \Delta_4 DEMFTB_t + \alpha_3 \cdot \Delta_4 WAPOP_t + \alpha_4 \cdot (1 - \omega \cdot CCI_t) \cdot \ln(p^h / y)\{t-1\} + \ln y_t - \ln c_{t-1}] + \beta_1 \cdot DSRISK_t + \beta_2 \cdot (1 - \omega \cdot CCI_t) \cdot \Delta_8 \ln ue_t + \beta_3 \cdot \Delta_4 \ln c_{t-1} + \sum_k \delta_k D_{kt} + \varepsilon_t$$

where

- c_t is real per capita household consumption and y_t is real per capita household disposable income;
- CCI_t is the credit conditions index (a spline of smoothed-step dummies under sign-prior GETS reduction, identified jointly across the four-equation FIML system; see §5);

- $(HA/4y)$, $(IFA/4y)$, $(NLA/4y)$ are housing, illiquid financial and net liquid wealth, each scaled by *annualised* income (the prior quarter's stock divided by $4 \cdot y$; see §4.6);
- r_t is the ex post real mortgage rate;
- p^h_t is the real house price index, so $\ln(p^h_t/y)$ is the affordability / down-payment ratio;
- y^p_t is permanent income — the discounted, expected-future-income measure of §4.4 — so $\ln(y^p_t/y)$ is the permanent-to-current income gap;
- DEMFTB, WAPOP, DSRISK and $\Delta_4 \ln c_{t-1}$ are the demographic first-home-buyer cohort, working-age population, downside-risk and durables-habit terms of Williams' eq. (7);
- $\phi < 0$ is the speed of adjustment (the coefficient λ in our reporting convention, §4.3), and ζ_c , α_1 , γ_1 , γ_2 , γ_3 , α_2 , α_3 , α_4 and the function $\psi(\cdot)$ are the long-run "structural" parameters;
- D_{kt} are the narrative dummies of §3.7.

Three features of this form are load-bearing and distinguish it sharply from the conventional wealth-effect ECM.

(i) Credit conditions multiply six channels jointly. CCI enters the long-run bracket six times — as the autonomous-consumption loading $\zeta_c \cdot CCI$, the rate interaction $\alpha_1 \cdot r \cdot CCI$, the housing-collateral interaction $\gamma_1 \cdot CCI \cdot (HA/4y)$, inside the permanent-income gearing $\psi(CCI)$, and inside the affordability term through $(1-\phi \cdot CCI)$. The credit channel is therefore not an additive shifter but a set of *interactions* that switch the long-run wealth and income channels on and off as credit conditions ease or tighten.

(ii) There is no classical housing wealth effect. Housing wealth enters *only* through $\gamma_1 \cdot CCI \cdot (HA/4y)$. The marginal propensity to consume out of housing wealth is identically zero when $CCI = 0$ and is unlocked as credit conditions ease: it is a collateral / housing-equity- withdrawal channel, not a pure wealth effect. This is the theoretical core of the form-is-decisive result. A specification that enters a standalone $\gamma_{HA} \cdot (HA/4y)$ level term and reads its coefficient as the housing wealth effect is mis-testing the theory: theory predicts that standalone coefficient is approximately zero in the absence of the CCI interaction, so reading an insignificant standalone ha_y as a *failed* housing wealth effect is a category error. Williams' own peak housing MPC (γ_1 evaluated at the credit-loose regime) is 0.0488 — a number that can only be recovered through the interaction, never through a level term.

(iii) Illiquid financial and net liquid wealth enter as plain MPCs; income enters with a unit coefficient. Illiquid financial wealth (IFA, equities plus superannuation) and net liquid wealth (NLA, liquid assets minus total household debt) enter the long-run bracket as *un-interacted* marginal propensities γ_2 and γ_3 (Williams' calibrated 0.022 and estimated 0.159 respectively). Current income $\ln y_t$ enters the bracket with a coefficient restricted to unity, against $-\ln c_{t-1}$, so the error-correction term is the log consumption-to-income ratio $\ln(c_{t-1}/y_t)$ (equivalently $\ln(c_{t-1}) - \ln(y_t)$). This unit restriction is what makes the equilibrium a stationary consumption-to-income ratio in the LIVES tradition, rather than a free cointegrating vector.

The permanent-income gearing is itself credit-dependent:

$$\psi(CCI_t) = \psi_0 + \psi_1 \cdot CCI_t$$

so that the weight on the permanent-to-current income gap rises with credit availability. Williams *calibrates* $\psi_0 = 0.20$ and $\psi_1 = 0.93$ (the cap $\psi(\text{CCI}) \leq 0.95$ binds at the credit-loose extreme); these are not free parameters in his system. The affordability multiplier is $\varpi = 1.2$, also fixed. We treat the distinction between *structure* (the interactions and the unit income restriction) and *calibration* (the specific values $\psi_0, \psi_1, \gamma_2, \alpha_1, \varpi$) as central, and test it directly in §4.5 and the results.

1.5.2 4.2 Recovery of structural coefficients

We estimate the equation by OLS — $\Delta \ln c$ on the contemporaneous and lagged right-hand-side terms — with Newey-West HAC standard errors throughout (heteroskedasticity is structural in every full-sample specification; see §7.2). Because the entire long-run bracket is multiplied by ϕ , the OLS coefficient on each long-run regressor is ϕ times its structural value. We therefore recover each structural parameter as

$$\text{structural } \gamma_i = \text{OLS coefficient on regressor } i / |\lambda|$$

and report both forms throughout, so that the speed-of-adjustment channel (λ) and the long-run-coefficient channel are separable. For example, the faithful specification's full-sample net-liquid OLS coefficient $+0.0370$ and $\lambda = -0.480$ imply a structural NLA marginal propensity of $0.0370 / 0.480 = 0.077$ (§7.0). The structural-recovery identity also makes precise *why* imposing Williams' calibrations can collapse the equation: fixing several γ_i while iterating to the fixed-point implied by the unit income restriction over-determines the bracket, and the only free margin left — λ — adjusts toward zero (§4.5, §7.0.1).

1.5.3 4.3 Reporting and sign conventions

Throughout, λ denotes the `ecm_lag` coefficient (the OLS coefficient on $\ln(c_{t-1}) - \ln y_t$), with $\lambda < 0$ indicating stable error correction. We report the faithful specification's speed of adjustment as $\lambda = -0.480$ ($t = -3.59$) on the full sample and $\lambda = -0.245$ ($t = -4.80$) on the pre-COVID sample (`australia_all_results.csv`). We treat the pre-COVID value as the *identified* speed of adjustment: the full-sample estimate is inflated by the extreme COVID quarters and fails the upper-bound $|\lambda|$ screen of §6.2, whereas the pre-COVID -0.245 sits close to Williams' (2010) estimated speed of $\phi_c = -0.286$. Where a headline number depends on the full-sample, non-causal permanent-income measure (§4.4) we say so explicitly and point to the real-time robustness column (§8).

1.5.4 4.4 Permanent-income forecasting

Permanent income y^p_t is the discounted weighted average of *expected* future log income over a 40-quarter horizon ($k = 40q$) at quarterly discount factor $\delta_q = 0.95^{(1/4)}$ — i.e. an annual discount factor $\delta = 0.95$, equivalently an annual discount rate $\eta = 0.05$:

$$\ln(y^p_t / y_t) = E_t [\sum_{h=1}^{40} w_h \ln(y_{t+h})] - \ln(y_t), \text{ with } w_h = \delta_q^{(h-1)} / \sum_{h=1}^{40} \delta_q^{(h-1)}.$$

We implement two forecasters. The headline measure is the Italy-style local projection (Jordà 2005; De Bonis, Liberati, Muellbauer and Rondinelli 2020); the rolling AR(8) forecaster is reported as a methodology-robustness column (§8).

- **Italy local projection (headline measure).** For each t whose 40-quarter-ahead horizon is observable, the discounted weighted average of future log income is computed and regressed, in a single full-sample equation, on a richer predictor set including $\log(lf_share)$ (the labour-force-share term that captures slow-moving demographic effects on trend income), a trend, a post-2008 split trend, the four-quarter moving average of log income, the unemployment rate, and four-quarter-difference dynamics. The permanent-income series is the in-sample fitted value of this single regression. It is therefore a two-sided, full-sample *measurement* of permanent income rather than a real-time forecast: because the coefficients are estimated over the whole sample, y^p_t embeds information dated after t and is non-causal. We carry it as the headline measure but flag this look-ahead property explicitly, and report a causal, expanding-window variant (re-fitting at each t only on observations whose full horizon is realised by t) as an operational robustness column. The real-time variant shrinks $|\lambda|$ materially (§8): in the Spec 2 net-worth frame the full-sample Italy measure delivers $\lambda = -0.197$ against $\lambda = -0.118$ real-time and $\lambda \approx -0.051$ under the real-time AR forecaster (`australia_pi_realtime_robustness.csv`). Any forecasting use of the equation — embedding it in MARTIN, in particular — requires the real-time variant or the AR forecaster, not the full-sample measure.
- **Rolling AR(8) (robustness).** A rolling AR(8) regression of log income on eight own lags plus a linear trend, a post-2008Q3 step dummy, and a trend-break interaction; forecasts are aggregated over the 40 horizons using the discount weights, with optional `unemp_rate`, `log_oil`, `log_reer`, `log_stocks` predictors and a 2008Q3 ogive learning weight that attenuates the term to a steady-state weight of 0.5 over 15 quarters. Under this forecaster the long-run permanent-income coefficient turns negative — the "Australian permanent-income puzzle" of §8 — a sign reversal absent under the Italy measure.

The permanent-income discount and horizon settings are not load-bearing for the speed of adjustment: across $\delta \in \{0.90, 0.95, 0.97\}$ and $k \in \{20, 40, 60\}$, λ in the net-worth frame ranges only over -0.0824 to -0.0830 (`australia_permanent_income_sensitivity.csv`), and the GFC learning weight has no effect on these summaries. The forecaster *method* (Italy vs AR, full-sample vs real-time), not the discount calibration, is the material choice.

In the faithful specification the permanent-income channel is strong and correctly signed: $\ln(y^p/y)$ enters with OLS level coefficient $+0.504$ ($t = 4.0$) on the full sample and $+0.279$ ($t = 4.75$) pre-COVID (`australia_all_results.csv`). This level coefficient is of order one half — roughly two to three times Williams' calibrated $\psi_0 = 0.20$ — although Spec 11 does not separately estimate the ψ_0 / ψ_1 split, so we read it as the gearing on the permanent-income gap rather than as a sharp estimate of ψ_0 alone. The credit-gear component $\psi_1 \cdot CCI$ is entered through a separate `yp_x_cci` interaction; its full-sample coefficient (-0.590 , $t = -1.58$) is insignificant and wrong-signed relative to Williams' calibrated $\psi_1 = +0.93$, so the data identify the level of the permanent-income gearing but not its credit slope on a single equation. We return to this in §7.0.1: imposing Williams' ψ_0 , ψ_1 and γ_2 calibrations is what collapses the equilibrium.

1.5.5 4.5 The conventional constant-MPC ECM as nested baseline (Spec 6)

The conventional disaggregated wealth-effect error-correction model — which prior Australian work and an earlier draft of this paper treated as the LIVES equation — replaces the credit interactions with plain, constant marginal propensities:

$$\Delta \ln c_t = \lambda \cdot [\alpha_0 + \gamma_{HA} \cdot (HA/4y)_{t-1} + \gamma_{eq} \cdot (eq/4y)_{t-1} + \gamma_{super} \cdot (super/4y)_{t-1} + \gamma_{NLA} \cdot (NLA/4y)_{t-1} + \gamma_{HP} \cdot \ln(p^h/y)_{t-1} + \alpha_r \cdot r_t + \psi \cdot \ln(y^p/y)_t + \ln y_t - \ln c_{t-1}] + \theta \cdot \Delta_2 \ln CCI_{t-2} + \sum_j \beta_j Z_{jt} + \sum_k \delta_k D_{kt} + \varepsilon_t$$

Here every wealth term enters as a standalone level with a *constant* MPC and *no* credit scaling; credit conditions appear only through the short-run term $\Delta_2 \ln CCI_{t-2}$ (`d2_logcci_lag2`). This is **not** the LIVES equation — it is a generic constant-MPC wealth ECM, and we label it the *conventional baseline* (Spec 6), not the preferred or headline specification. It is theoretically nested in the canonical form only in the degenerate sense that $\gamma_1 \cdot CCI \cdot (HA/4y)$ would reduce to a constant MPC if CCI were constant; on post-deregulation Australian data, where the credit liberalisation that identifies the channels largely predates the 1988Q3 start of ABS sectoral balance-sheet data (§3), that reduction is precisely what kills the housing-collateral channel.

The implications follow directly. The conventional baseline delivers an insignificant speed of adjustment ($\lambda = -0.180$, $t = -1.76$, full sample; `australia_all_results.csv`) and an insignificant standalone housing coefficient ($\gamma_{HA} = +0.0088$, $t = 1.52$). Reading that insignificant γ_{HA} as a *failed housing wealth effect* is the category error of §4.1: the LIVES form predicts that very coefficient is approximately zero absent the CCI interaction. The faithful specification, by contrast, re-routes housing through $\gamma_1 \cdot CCI \cdot (HA/4y)$ alone, restores the autonomous-consumption loading $\zeta_c \cdot CCI$, and combines illiquid financial wealth (IFA = $eq + super$); the error-correction and core wealth structure then come alive (§7.0). We retain Spec 6 throughout as the conventional comparator against which the faithful form is the correctly-specified alternative — a matter of theory, not of fit.

1.5.6 4.6 Wealth definitions

Wealth enters as asset-to-annualised-income *ratios*, $x_{t-1} / (4 \cdot y_t)$ — the prior-quarter stock divided by four times current quarterly income — consistent with the LIVES treatment. The three components are

- **Housing wealth (HA):** the value of the household-sector dwelling stock; it enters the faithful specification only through $\gamma_1 \cdot CCI \cdot (HA/4y)$.
- **Illiquid financial wealth (IFA):** equities plus superannuation; it enters as a plain MPC γ_2 (combined as $ilfa_y = eq_y + super_y$ in the faithful specification; entered separately as eq_y and $super_y$ in the conventional baseline).
- **Net liquid wealth (NLA):** liquid assets minus total household debt, entered as a plain MPC γ_3 .

For NLA we follow the Italian implementation (De Bonis et al. 2020 §2; Williams 2010) in netting debt against liquid assets, and we test the underlying restriction $\gamma_{LA} + \gamma_{LOANS} = 0$ directly by refitting the disaggregated specifications with liquid assets and debt entered separately and conducting a Wald test of equal-and-opposite coefficients (`australia_nla_restriction_test.csv`). The restriction is not rejected in any specification or sample: in the conventional baseline the sum is 0.105 (SE 0.103, $t = 1.02$, $p = 0.31$) on the full sample and -0.020 ($t = -0.38$, $p = 0.71$) pre-COVID, and Spec 4 and Spec 5 likewise accept (all six rows `restriction_accepted = TRUE`). The honest reading is that the data cannot distinguish separate liquid-asset and debt MPCs — non-rejection by imprecision rather than

confirmation of exact netting — which supports the NLA aggregation as a parsimonious choice rather than as a tested structural equality.

1.5.7 4.7 The de-meaning convention for the interactions

The credit interactions are entered in *de-meaned* form. The housing-collateral regressor is ha_x_cci , the de-meaned product $(CCI - \bar{CCI}) \cdot (HA/4y)$, and the autonomous-consumption loading $cci_williams$ is likewise de-meaned; the affordability and rate interactions enter as $hp_x_1_minus_cci$ and r_x_cci . De-meaning centres each interaction at the sample-average credit regime, so that the un-interacted coefficients (γ_2, γ_3 , the intercept) retain their interpretation as marginal propensities evaluated at the average CCI, and the interaction coefficients measure the *additional* effect of moving CCI away from its mean. Two consequences matter for interpretation. First, under de-meaning a pure re-centring of the credit regime is a timing-and-distribution effect, not a level effect, on cumulative consumption. (This is distinct from the near-zero CCI counterfactual of \$10, which is run on the conventional baseline (Spec 6) where CCI enters only the short run and so has no long-run channel to integrate — confirming Spec 6 has no operative housing-collateral channel.) Second, de-meaning does not break the near-collinearity of the interactions — each remains approximately proportional to CCI and they are 0.74–0.97 mutually correlated on this sample (§5) — which is the structural reason the six interactions cannot be separately free-estimated off a single equation, and why Williams identifies them jointly through FIML.

1.5.8 4.8 Sign priors

Theoretical sign priors on the long-run coefficients are

- $\gamma_1 \geq 0$ (housing collateral, unlocked by CCI);
- $\gamma_2 \geq 0$ (illiquid financial wealth);
- $\gamma_3 \geq 0$ (net liquid wealth: buffer-stock plus inter-temporal substitution);
- $\alpha_4 \leq 0$ on $\ln(p^h/y)$ at credit-tight regimes (down-payment penalty), with the sign reversing toward positive at credit-loose regimes through the $(1 - \varpi \cdot CCI)$ term;
- $\alpha_1 \leq 0$ on $r \cdot CCI$ (inter-temporal substitution);
- ζ_c positive (autonomous-consumption loading on credit);
- $\psi_0, \psi_1 \geq 0$ (permanent-income gearing rising in credit);
- $\lambda < 0$ (stable error correction).

We use these as informal screens (§6.2) rather than imposing them as formal restrictions. Freely estimating the coefficients lets us report violations of the priors as substantive findings — for example the sign failures on r_x_cci and yp_x_cci in the faithful specification (§7.0), and the sign-prior GETS verdicts on the CCI knots (§5) — rather than burying them inside imposed constraints. This is the methodological discipline that distinguishes our single-equation estimates from Williams' FIML system: where his cross-equation restrictions deliver identification by imposition, our free estimates expose, rather than hide, the weak identification of the credit channels. The cost is power; the benefit is that the negative results (placebo failure, near-zero SUR efficiency gain, out-of-sample loss to a random walk, the calibration collapse, and the interaction collinearity) are visible as diagnostics pointing to why FIML and the pre-1988 back-extension are the routes forward (§8–§11).

1.6 5. Identification of credit conditions

This section is the methodological pivot of the paper. In the LIVES framework (Williams 2010; Muellbauer and Williams 2012) the credit conditions index, CCI, is not an observed series. It is a latent common factor that enters the consumption, house-price, mortgage-stock and home-equity-withdrawal equations simultaneously, and it is the object that interacts with the six channels inside the long-run bracket of equation (7). In Williams' setting it is identified by being shared across equations under cross-equation parameter restrictions estimated by full-information maximum likelihood (FIML). In a single-equation setting that joint identification is unavailable, and CCI can only be *proxied* — by an institutionally dated smoothed-step spline, a state-space factor, a principal component, or a credit-to-income gap — or *calibrated* to Williams' published values (the calibration route is developed in §7.0.1 and shown to collapse the equilibrium).

The central new result of this section is an identification diagnosis, not a CCI success. We document directly why the credit channels cannot be separately recovered from a single equation: the six CCI-interacted regressors that carry the credit channels are between roughly 0.74 and 0.97 mutually correlated on this sample because each is approximately proportional to CCI itself (§5.5). A single equation therefore cannot tell them apart, and the spline, when fitted standalone to the consumption-equation residual, behaves no better than a flexible detrending series — it sits in the placebo distribution rather than above it (§5.2). This is the structural reason Williams identifies the credit channels through FIML, and it is why we treat four-equation FIML and pre-1988 back-extension, not further single-equation search, as the routes that could sharpen the credit channels. Throughout, the honest negatives — placebo failure, near-zero SUR residual correlation, selector divergence — are reported as substantive diagnostic findings, not embarrassments.

1.6.1 5.1 The Williams smoothed-step spline approach

Williams (2010) constructs CCI as a spline of SDMMMA smoothed-step dummies — a 5-quarter moving average of a 4-quarter moving average of a 0/1 step — at institutional turning points in the Australian financial-policy chronology. Each knot's coefficient is constrained by a sign prior derived from institutional history (deregulation episodes positive; retrenchment episodes negative), enforced by Hendry-Krolzig (2005) drop-on-violation general-to-specific reduction.

Williams' canonical paper uses four knots: 1979Q1 (Campbell Committee, removal of interest-rate ceilings), 1992Q1 (NBFI distress after the early-1990s recession), 1998Q1 (NBFI/securitisation expansion), and 2007Q1 (GFC retrenchment). The four-knot choice reflects the institutional information available over his 1977-2008 sample: STAMP- derived turning points and a deregulation calendar ending shortly after the GFC.

On our 1988Q4-onwards sample, only one of Williams' four canonical knots survives sign-prior reduction. A direct replication of the Williams 4-knot specification yields:

Williams knot	Sign prior	Status on 1988+ sample
1979Q1	+	aliased (constant within window)
1992Q1	–	sign violator (data signal +ve)
1998Q1	+	sign violator (data signal –ve)
2007Q1	–	survives, coef ≈ -0.014

The 1979 deregulation knot is mechanically uninformative because the smoothed step reaches unity by 1980Q2, three years before our window opens. The 1992 and 1998 knots fail their institutional sign priors: the post-1988 sample observes the recovery from the early-1990s banking distress (during which credit growth resumed and the OLS coefficient turns positive) and the late-1990s NBFI period without the contrast against the prior tight regime that would identify the loosening direction. This is a direct consequence of the data window. As noted in §3, ABS sectoral balance-sheet data begin only in 1988Q3, so the financial-liberalisation episode that most cleanly identifies the credit channels (Campbell, the early-1980s deregulation, the NBFI cycle) largely predates the sample on which the single-equation model can be estimated.

A direct 4-knot replication is therefore *not* identifying the four distinct credit-conditions episodes Williams' framework attributes to the spline. It is identifying one — the 2007 GFC tightening — plus a constant.

1.6.1.1 5.1.1 The maximal-GETS Australian CCI

Rather than impose Williams' published knot count on a sample that cannot identify three of his four knots, we adopt a **maximal-GETS approach**: starting from a richer 15-knot candidate set covering the documented Australian financial-policy chronology, we let drop-on-violation reduction prune knots that are aliased or violate their institutional sign prior. The candidate institutional events span Campbell '79, the housing-finance deregulation of '86, state-bank distress '90, banking distress '92/'93, the Wallis report and the establishment of APRA in '98, the GFC tightening of '07, the deposit-guarantee scheme '08, the FHB Boost '09, the APRA macroprudential rounds of '14 and '17, the Hayne Royal Commission of '19, the APRA cap removal and buffer reduction of '19Q3, the COVID/JobKeeper episode of '20, and the APRA buffer hike of '21.

On the 1988Q4–2024Q4 sample this candidate set reduces to **three surviving knots** under the iterated drop-on-violation reduction (`australia_williams_cci_knots.csv`):

Knot	Sign prior	Coef (OLS)	Reading
2009Q1	+	+0.012	First Home Buyer Boost
2019Q1	–	–0.034	Hayne Royal Commission lending crackdown
2020Q2	+	+0.005	COVID/JobKeeper income support

(Ten candidate knots — 1990Q3, 1992Q1, 1993Q1, 1998Q3, 2007Q3, 2008Q4, 2014Q4, 2017Q1, 2019Q3 and 2021Q4 — violate their institutional sign priors and are dropped; 1979Q1 and 1986Q1 are aliased, their smoothed step being constant within the estimation window.)

The `cci_williams` series used throughout the rest of the paper is constructed from these three surviving knots, peak-normalised to unity. That only three of fifteen candidate knots survive — all of them post-2008 — is itself part of the identification story (§5.3): the post-1988 sample carries usable sign-identifying variation only around the recent macroprudential and pandemic episodes.

The maximal-GETS construction is defensible on two grounds: (i) the candidate set comes from documented Australian institutional history, not authorial choice of specific dates; and (ii) the surviving knots are those whose data signal aligns with their institutional sign prior, so the spline is *empirically* selected rather than imposed. We do **not**, however, claim that this delivers a structurally identified credit-conditions factor. The placebo battery (§5.2) shows the deployed spline does not robustly outperform random smoothed-step constructions, and §5.5 documents why: the credit channels it is meant to carry are near-collinear and cannot be separated off one equation. Williams' canonical 4-knot setup is retained as a robustness benchmark, and a sectional sign-prior alternative following Williams' (Aust paper §5.1) specification is also implemented and placebo-tested; on the back-extended sample it does not outperform the maximal-GETS canonical (§5.2.2).

1.6.2 5.2 The placebo battery

Whether the spline is identifying genuine credit-conditions turning points — rather than acting as flexible detrending of the consumption-equation residual — is empirically testable. We construct a random-knot placebo: 200 draws of knots in Williams' candidate window. For the literal-Williams comparison all four drawn knots are entered **unconditionally** — testing pure fit, with no sign-prior reduction — so the canonical 4-knot series is compared like-for-like against random 4-knot series. For the maximal-GETS comparison each draw of 15 candidate knots is passed through a single sign-prior reduction pass. (The *deployed* `cci_williams` series uses the iterated reduction of §5.1.1, which retains three knots; the maximal-GETS placebo applies a single reduction pass, so its percentile benchmark is indicative of the canonical procedure rather than an exact match to the deployed series.) The canonical result's percentile rank in this placebo distribution measures whether the specific knot choice identifies something the data genuinely flags, versus whatever a flexible smoothed-step series could fit by chance.

1.6.2.1 5.2.1 Three placebo runs

Specification			Sample	adj R ² %ile	$ \lambda $ %ile	Verdict
Literal	Williams	4-	1988Q4+ (n=146)	34th	58th	Below R ² median
	knot					
Literal	Williams	4-	1976Q3+ (n=190)	18th	10th	Fails — below median
	knot					on both
	Maximal-GETS		1976Q3+ (n=190)	64th	37th	Weakly above median
	canonical (15-knot re-					
	duce)					

Sources: `australia_williams_knot_placebo_verdict.csv` (literal, 1988Q4+;
Williams adj-R² = 0.7209, $|\lambda|$ = 0.0799; `n_placebo` = 200);
`australia_williams_knot_placebo_extended_summary.csv` (literal, 1976Q3+; Williams adj-

$R^2 = 0.6789$ versus a placebo median of 0.6816, Williams $|\lambda| = 0.1934$ versus a placebo median of 0.2022); `australia_williams_knot_placebo_maximal_extended_summary.csv` (maximal-GETS, 1976Q3+; canonical adj- $R^2 = 0.6852$ versus a placebo median of 0.6832, canonical $|\lambda| = 0.2155$ versus a placebo median of 0.2226, canonical surviving knots 7 versus a placebo median of 8).

The literal Williams 4-knot specification fails the placebo on both samples, and the R^2 failure *deepens* on the extended sample (34th $\rightarrow$ 18th percentile). The maximal-GETS reduction partially rescues identification — 64th percentile on adjusted R^2 , 37th on $|\lambda|$ — but does not deliver strong support: random combinations of 15 candidate knots under the same reduction protocol produce *faster* mean reversion than the canonical institutional choice in 63 per cent of draws, and a higher adjusted R^2 in 36 per cent. The committed verdict label is "WEAK SUPPORT — institutional choice above median but not far". Read across the variants, the deployed/institutional CCI sits in an **18th–34th adjusted- R^2 percentile band**. We report this prominently as an honest negative: the spline's standalone fit is consistent with that of a flexible detrending series, not with a structurally identified common factor.

1.6.2.2 5.2.2 Sectional sign priors (Williams Aust paper §5.1) tested

Williams (Aust paper §5.1) imposes sign priors over PERIODS rather than knot-by-knot:

Period	Sign prior	Rationale
1982–1990	non-negative	Financial deregulation
Early 1990s	non-positive	Banking sector distress
Mid-1990s–2006	non-negative	New entrants, securitisation
2007+	non-positive	GFC

We constructed a parallel CCI basis with one knot per period, extending Williams' four periods to cover post-2008 events (APRA 2014, APRA 2017, COVID 2020, APRA 2021). On the back-extended sample, with a random-period placebo (200 draws of 8 random knots and 8 random ± 1 priors), the sectional canonical sits at the **36th adjusted- R^2 percentile and 40th $|\lambda|$ percentile** — *worse* than the maximal-GETS canonical, not better. Williams' specific period dating does not outperform random period placements on the post-deregulation-extended window.

The takeaway across §5.2.1 and §5.2.2 is uniform: neither the literal 4-knot construction, nor a maximal-GETS reduction, nor sectional sign priors delivers strong placebo support on our extended sample. The CCI's standalone identification is consistent with a single-equation OLS using flexible smoothed-step dummies that the data can fit, and is not consistent with a structurally identified common factor. The next two subsections set out *why* this is the expected outcome of single-equation estimation, and §5.5 supplies the direct mechanical reason.

1.6.3 5.3 Why the spline alone cannot identify the CCI as a common factor

The placebo evidence is consistent with the structural diagnosis in the LIVES literature itself. Williams (2010) and Duca, Muellbauer and Murphy (2013, ECB WP 1581) estimate the CCI inside a **multi-equation system** (consumption, house prices, mortgage stock, home equity withdrawal — Williams; consumption + refinancing rate — Duca, Muellbauer and Murphy) where the *same* CCI series enters all equations simultaneously. Williams imposes a normali-

sation ($\zeta_h = 1$ in the house-price equation) and estimates ζ_c , ζ_m , ζ_w as relative scalings; this cross-equation parameter restriction is what identifies the CCI as a common factor rather than as an equation-specific residual.

In a single-equation OLS estimation the spline is fit only to the consumption-equation residual. There is no constraint that the same knot loadings satisfy sign priors in the house-price, mortgage-stock or HEW equations simultaneously. The multi-equation scaffolding in our companion directory tests this directly.

1.6.3.1 5.3.1 Cross-equation joint sign-survival (LIVES phase 3)

We refit the Williams 15-knot maximal candidate set with the consumption equation, the house-price equation, and the mortgage-stock equation simultaneously, then require each knot to satisfy its institutional sign prior in **all three** equations to be retained.

Survival regime	Surviving knots	n
Consumption equation only (Spec 1 with $\ln_networth_y_proxy$, extended sample)	1979, 1986, 1992, 2007, 2009, 2017, 2020	7
Joint (C $\cap$ H $\cap$ M)	1986, 2017	2

Of the 7 knots that survive when fitted to consumption alone (this is the single-pass reduction in `joint_cci_identification.R`, using the Spec-1 aggregate-proxy specification on the back-extended sample — distinct from the *iterated* reduction the canonical consumption pipeline applies in §5.1.1, which retains only three knots, 2009Q1, 2019Q1 and 2020Q2, on the 1988+ sample; the two reductions give different but overlapping survivor sets), only **1986 (financial deregulation)** and **2017 (APRA macroprudential round II)** have signs consistent with their institutional priors across consumption, house prices and mortgage stock simultaneously. The other surviving knots sign-violate in the house-price or mortgage-stock equations.

The single-pass protocol's identification of 7 knots was therefore overstated: 5 of them were consumption-equation-specific and would not survive a true cross-equation common-factor restriction. This is the empirical content of the placebo failures in §5.2: without imposing cross-equation sign consistency, the consumption-equation residual can be flexibly fit by smoothed-step dummies whose information content is consumption-specific.

1.6.3.2 5.3.2 What joint identification fixes

When we rebuild `cci_williams_joint` using the 2 jointly-surviving knots and re-estimate the house-price equation with the new CCI:

HP equation, CCI loading	(a) cons-only CCI	(b) joint OLS	(c) joint SUR
Estimate	-0.024	+0.024	+0.024
Sign	× violator	✓	✓

The house-price equation's CCI loading flips from significantly negative (under the consumption-only CCI) to significantly positive (under joint identification) — Williams' cross-equation sign restriction working as intended. The mortgage-stock equation's CCI loading remains negative under this joint-sign-survival approximation, which weights surviving knots by consumption-equation coefficients; full FIML with parameter restrictions across all four

equations would be needed to flip it. The exercise is therefore illustrative of the mechanism, not a substitute for the FIML estimate.

1.6.4 5.4 Two-equation SUR — joint estimation gives no efficiency gain

A complementary test of the multi-equation framework's value is whether SUR or FIML deliver efficiency gains over equation-by-equation OLS. A two-equation SUR (consumption + house prices, on the extended 1976Q3+ sample using Spec 1 with the aggregate net-worth proxy) finds an essentially negligible cross-equation residual correlation; with no CCI spline and no event dummies it remains small and negative. The narrative figure for the headline configuration is $\hat{\rho}(\varepsilon_C, \varepsilon_H) \approx -0.004$ — we flag that the exact scalar is a reported summary, not a separately committed CSV value, and the load-bearing committed evidence is the coefficient comparison in `australia_joint_pi_robustness.csv`. There, moving from single-equation OLS to SUR_joint on the Spec-2 consumption equation shifts coefficients only within sampling noise: the net-worth loading moves from +0.0615 to +0.0695 (+13.1 per cent, both highly significant), the speed of adjustment from -0.0829 to -0.1273 (raw-units `ecm_lag`), and the real-rate and permanent-income terms remain near zero under both. No coefficient changes sign in a way that alters the economic reading, and the standard errors are barely tightened.

The joint-estimation case for LIVES therefore does **not** rest on efficiency gains. It rests entirely on **cross-equation parameter restrictions** — Williams' FIML imposes that the same CCI loading enters all four equations with specific sign constraints. SUR alone imposes only residual-covariance flexibility, and at the quarterly frequency that covariance is negligible (§5.3.1). The two-equation SUR confirms that the cross-equation linkage between consumption and house prices is captured by shared regressors (CCI, the real rate, dummies for major events); it does not reside in unexplained residual covariance. We report the near-zero SUR residual correlation as another honest negative: it tells us that the value of the system is in its sign restrictions, not its error structure, and so single-equation SUR cannot be a shortcut to system identification.

1.6.5 5.5 Interaction collinearity — why the credit channels cannot be separated off one equation

The placebo failures (§5.2), the cross-equation sign-survival collapse (§5.3.1) and the absence of an SUR efficiency gain (§5.4) all point to the same underlying obstacle, which we now state directly. In the faithful LIVES form, CCI does not enter once: it multiplies six channels jointly inside the long-run bracket of equation (7) — the autonomous-consumption loading ($\zeta_c \cdot \text{CCI}$), the rate channel ($\alpha_1 \cdot r \cdot \text{CCI}$), the housing-collateral channel ($\gamma_1 \cdot \text{CCI} \cdot (\text{HA}/4y)$), the down-payment/affordability composite ($\alpha_4 \cdot (1 - \omega \cdot \text{CCI}) \cdot \log(p^h/y)$), the permanent-income gearing ($\psi(\text{CCI}) \cdot \log(y^p/y)$ with $\psi(\text{CCI}) = \psi_0 + \psi_1 \cdot \text{CCI}$), and so the regressors that carry these channels are each approximately *proportional to CCI* over the sample.

The empirical consequence is that the six CCI-interacted regressors are between roughly **0.74 and 0.97 mutually correlated** on this sample. (We report this as a property of the estimation sample established in the analysis narrative; the specific pairwise correlation range is not itself a standalone committed results CSV, and a drafter checking the exact figures should compute the interaction correlation matrix from `australia_model_dataset.csv`. The closest committed cross-check — `australia_cci_method_4way.csv` — concerns correlations *between CCI measures*, e.g. Kalman versus PCA $\rho = 0.764$, not between the six interaction regressors, and so should not be substituted for the interaction-collinearity figure.) Six near-collinear

regressors carrying distinct structural meanings cannot be separately free-estimated off a single equation: ordinary least squares will allocate a near-singular design among them more or less arbitrarily, producing wrong-signed and insignificant individual loadings even when the joint contribution of the credit block is real.

The committed, load-bearing evidence for this collinearity is threefold, and each piece is itself an honest negative:

1. **Sign failures and insignificance when the interactions are freed (Spec 8, §5.6 and §7).** When the six interactions are estimated freely, the housing-collateral interaction is right-signed but insignificant, while three of the remaining interactions fail their institutional sign priors (`australia_spec8_sign_prior_verdicts.csv`). This is exactly the pattern a near-singular design produces.
2. **The identification-versus-detrending decomposition (`australia_cci_fit_decomposition.csv`).** Adding the Williams maximal-GETS CCI to the conventional baseline (Spec 6 → Spec 8) does *not* merely detrend: it shifts the permanent-income loading by +134 per cent (from +0.20, $t = 0.76$, to +0.47, $t = 3.44$) and the speed of adjustment by −147 per cent (from −0.180 to −0.445), with the equities loading shifting +254 per cent. By the decomposition's own classification these are "IDENTIFICATION" (> 30 per cent) rather than "DETRENDING" (< 5 per cent) shifts. The credit interactions re-allocate identification across the income and speed-of-adjustment channels — which is precisely the symptom of collinearity (one near-singular block redistributing fit), not evidence of clean separate identification of each channel.
3. **The calibration collapse (Spec 10 and Spec 12, §7.0.1).** When Williams' calibrations are imposed instead of freed, the error-correction term collapses to $\lambda = -0.029$ ($t = -0.68$) in Spec 12, independently reproduced at $\lambda = -0.048$ ($t = -0.76$) by Spec 10. A block of regressors that can be jointly identified would not behave this way; a near-collinear block, whose joint mapping into the data is sharp but whose internal split is not, will.

Single-equation calibration of the credit channels is therefore empirically closed: the data pin down the *block* but not its *components*. This is the structural reason Williams uses FIML — the cross-equation sign restrictions break the within-equation collinearity by requiring the *same* loadings to be consistent across consumption, house prices, mortgage stock and HEW. Two routes can sharpen the credit channels on Australian data: the four-equation FIML system (which the companion paper develops), and the pre-1988 back-extension, which adds the financial-liberalisation variation that most cleanly identifies CCI but which the post-1988 balance-sheet data exclude. Further single-equation search cannot.

1.6.6 5.6 Spec 8: the freely estimated CCI interactions (the collinearity demonstration)

When `cci_williams` is available, we estimate Spec 8, which incorporates the Williams interaction structure with the six channels entered as free regressors:

$$\dots + \gamma_{HA} \cdot ha_y + \gamma_{HA_cci} \cdot ha_y \cdot CCI + \gamma_{HP} \cdot \log(p^h/y) \cdot (1 - \omega \cdot CCI) + \alpha_r \cdot r \cdot CCI + \psi_1 \cdot \log(y^p/y) \cdot CCI + \dots$$

with ω calibrated to 1.2 following Williams (Aust paper §5.2 fn 9). We present Spec 8 not as a credit-conditions success but as the empirical *demonstration* of the collinearity diagnosed in §5.5. It is the free-estimation counterpart of the calibration-collapse specifications (Spec 10/12): freeing the block over-fits the within-equation split and re-allocates identification, whereas imposing the block collapses the equilibrium. Neither separately identifies the credit channels.

The freely estimated interaction loadings, with their institutional sign priors (australia_spec8_sign_prior_verdicts.csv, australia_williams_spec8_comparison.csv), are:

Interaction	Williams reference	Sign prior	OLS (Spec 8, full)	t	Verdict
ha_x_cci (housing CCI; γ_1) ×	+0.0488 (peak MPC)	+	+0.0016	0.32	right-signed, ns
r_x_cci (real rate × CCI; α_1)	-0.871 (at CCI=1)	-	+0.0019	1.85	sign FAIL
hp_x_1_minus_cci (down-payment; α_4)	-0.13 (at CCI=0)	-	+0.0076	1.00	sign FAIL
yp_x_cci (PI × CCI; ψ_1)	+0.93 (calibrated)	+	-0.6113	-2.12	sign FAIL

The pattern is diagnostic of the near-singular design, not of structure. The housing-collateral interaction is **right-signed** – consistent with Williams' prediction that the housing MPC rises with credit ease as collateral becomes spendable – but statistically insignificant ($t = 0.32$, $p = 0.75$). Of the remaining three interactions, all fail their institutional sign priors: the rate interaction is wrong-signed (and only marginally significant, $p = 0.066$), the down-payment composite is wrong-signed and insignificant, and the permanent-income interaction is strongly wrong-signed (the free estimate is -0.61 , $t = -2.12$, against Williams' calibrated $+0.93$). The one term that survives its prior is the *non*-interacted permanent-income level ($\ln_{yp_over_y} = +0.47$, $t = 3.44$, PASS). In other words, when the credit block is freed the equation re-allocates almost all of its identification onto the income and speed-of-adjustment channels and away from the individual interactions – exactly the identification re-allocation §5.5 documents.

It is true that Spec 8 with the interactions included delivers a numerically larger speed of adjustment, $\lambda = -0.445$ ($t = -3.30$), exceeding Williams' calibrated -0.286 in magnitude, and the lowest BIC among the $n = 146$ specifications (-948.476). We do **not** read this as Spec 8 "closing the gap" with Williams or "exceeding" him in any substantive sense. The larger $|\lambda|$ and the lower BIC reflect the re-allocation of fit that a near-collinear block produces, not separate identification of the credit channels: three of the four interactions are wrong-signed, and the one right-signed interaction is insignificant. The honest reading is that freeing the block buys fit at the cost of structurally meaningless component loadings. The faithful LIVES specifica-

tion (Spec 11, §7.0), by contrast, enters housing *only* through its CCI interaction (ha_x_cci , de-meaned, +0.0049, $t = 1.03$ — right-signed, insignificant) and restores the autonomous-consumption CCI loading, and it is the narrative-preferred headline on theoretical-form grounds; Spec 8 is retained as the collinearity demonstration and is not promoted as a CCI result.

1.6.7 5.7 CCI measure cross-checks

To confirm that the identification difficulty is a property of single-equation Australian data rather than of the smoothed-step construction in particular, we cross-check the Williams maximal-GETS spline against three alternative latent-CCI measures (`australia_cci_methods_summary.csv`): a Kalman state-space common factor estimated by ML on five credit indicators with a housing-loan-flow anchor ($n = 194$); the first principal component across the same five standardised indicators ($n = 146$); and a BIS-style credit-to-income HP-filter gap ($\lambda = 400,000$) on log household debt-to-income ($n = 146$). The pairwise correlations between the four measures (`australia_cci_method_4way.csv`) are low except between the two purely statistical factors:

Pair	ρ	n
Kalman vs Williams maximal-GETS	0.050	194
PCA vs Williams maximal-GETS	-0.190	146
Credit/income gap vs Williams	0.332	146
Credit/income gap vs Kalman	0.304	146
Kalman vs PCA	0.764	146
Credit/income gap vs PCA	0.238	146

The institutional spline is essentially uncorrelated with the Kalman factor ($\rho = 0.05$) and weakly negatively correlated with the PCA factor ($\rho = -0.19$); only the two data-driven statistical factors agree strongly with one another (Kalman versus PCA, $\rho = 0.76$). The four candidate measures do not converge on a common Australian credit-conditions series. When the Kalman factor is used in place of the spline (Spec 9, §7), the wealth and speed channels shift by 41–101 per cent relative to the no-CCI baseline (`australia_cci_fit_decomposition.csv`), again re-allocating identification rather than sharpening it. The measure choice is therefore not innocuous, and no single-equation CCI proxy resolves the underlying weak identification — reinforcing that regime classification should rely on the joint system rather than on any one single-equation proxy (§10).

1.6.8 5.8 The wealth-coefficient profile on the back-extended sample

A natural follow-up question is whether *sample length*, rather than the single-equation framing itself, accounts for the residual gap between our estimates and Williams' Table 1. The cleanest test runs on the simpler disaggregated no-CCI specification (Spec 4), which is closest in form to Williams' long-run cointegrating regression. We refit Spec 4 on the back-extended 1976Q3+ sample using the disaggregated wealth proxies of §3:

LR coefficient	Baseline	1988+ (n=146)	Extended	1976+ (n=190)	% change	Williams Table 1
λ (ecm_lag)		-0.140		-0.193	+37.3	-0.286
nla_y		+0.035		-0.002	-106	+0.159
eq_y		-0.119		-0.104	-13.3	+0.022 $\ddagger$
super_y		+0.040		+0.024	-41.7	(incl.) $\ddagger$
ha_y		+0.068		+0.040	-41.6	+0.0488
ln_yp_over_y		+1.07		+1.12	+4.33	+0.20

$\ddagger$ Williams reports a single illiquid-financial-asset MPC ($\gamma_{\text{IFA}} = 0.022$), shown here against the combined eq_y + super_y. The Williams structural MPCs used throughout the paper ($\gamma_{\text{HA}} = 0.0488$, $\gamma_{\text{NLA}} = 0.159$, $\gamma_{\text{IFA}} = 0.022$) are the BIS-chapter values recorded in `australia_williams_comparison.csv` — the same authoritative source as §7.3 and §11.4.

On Spec 4 the speed of adjustment moves 37 per cent closer to Williams' value ($-0.140 \rightarrow -0.193$), but the individual OLS wealth coefficients become *smaller*, not larger; γ_{NLA} collapses to roughly zero and γ_{EQ} retains a wrong sign. The long-run permanent-income coefficient remains far above Williams' calibrated +0.20 on both samples.

The reading is consistent with the rest of this section. First, the back-extension does not push the Spec 4 estimates *closer* to Williams' Table 1 in the cross-section of wealth loadings — the post-1988 sample window is not, in itself, what generates the residual gap; sample length is not the binding constraint. Second, triangulating with the placebo evidence (§5.2), the cross-equation sign-survival collapse (§5.3.1), the near-zero SUR residual correlation (§5.4) and the interaction collinearity (§5.5), we read the residual difference between any single-equation OLS estimate and Williams' system FIML as a consequence of the single-equation framing rather than of sample length, knot count, or sign-prior structure. The path to a tighter reconciliation runs through a full FIML build with cross-equation parameter restrictions and the pre-1988 financial-liberalisation variation — the two routes §5.5 identifies — which the companion paper develops.

1.7 6. Specifications and selection

We estimate fourteen specifications — a numbered ladder Spec 1 through Spec 12, plus two variants, the long-history Spec 6b and the measured-burden Spec 7b — that range from a single aggregate-net-worth error-correction model to the faithful Muellbauer–Williams LIVES form and its calibration-imposed counterpart. The ladder is then passed through a four-screen selection rubric. We report the automated selector outcome in full and honestly, but we state at the outset that the headline of this paper is chosen on theoretical-form grounds, not by the screen: the **faithful LIVES specification (Spec 11)** is the narrative lead, the **conventional constant-MPC disaggregated ECM (Spec 6)** is retained as the baseline against which it is contrasted, and the **Williams-calibration-imposed specification (Spec 12)** is the negative control that shows what fails to transfer. The automated rubric, applied mechanically, returns a third, non-LIVES specification (Spec 2). We treat that divergence not as a defect to be smoothed over but as a documented result (§6.3): no single screen, BIC, or theory criterion

agrees on a single specification, which is itself diagnostic of how weakly a single equation can pin down the LIVES long run on post-deregulation Australian data.

1.7.1 6.1 The fourteen specifications

The ladder is organised so that each step adds one structural element. Specs 1–3 are aggregate-net-worth ECMs; Specs 4–7b disaggregate wealth and add cohort/burden terms; Specs 8–9 introduce the Williams CCI interactions (spline and Kalman extractions); and Specs 10–12 are the explicitly Williams-aligned forms — the calibrated-prior fixed point (Spec 10), the faithful free LIVES estimate (Spec 11, the headline) and the Williams-calibration-imposed LIVES (Spec 12). Long-run regressors follow the notation of §4; ha_x_cci is the de-meaned credit-scaled housing term $CCI \cdot (HA/4y)$, $ilfa_y$ is the combined illiquid-financial ratio (equities + super-annuation), $cci_williams$ is the autonomous- consumption CCI loading ζ_c , and yp_x_cci is the credit-scaled permanent-income term.

Spec	Description	Long-run regressors / notes
1	Aggregate net worth	$\ln_networth_y$, $\ln_hp_over_y$, $real_rate$, $\ln_yp_over_y$, ecm_lag
2	Spec 1 + short-run CCI	adds $\Delta^2\log$ CCI lag 2 to short-run set
3	Net worth in levels	replaces $\ln_networth_y$ with $networth_y$
4	Disaggregated wealth	adds nla_y , eq_y , $super_y$, ha_y ; drops aggregates
5	Spec 4 + full short-run dynamics	adds $\Delta^2\log$ CCI, $\Delta\Delta_4income$, $\Delta^2\log$ unemp, $ \hat{\epsilon} $
6	Conventional constant-MPC disaggregated ECM (baseline)	Spec 5 + post-2008 PI break $\ln_yp_over_y_post2008$; plain $ha_y/eq_y/super_y/nla_y$, CCI short-run only
6b	Spec 6 with back-extension-compatible SR CCI	replaces $\Delta^2\log$ CCI with $\Delta^2\log$ RBA D02 credit; disaggregated wealth proxies; fits on $n = 190$
7	Spec 6 + cohort terms + synthetic burden	adds $prime_age_share$, fhb_share
7b	Spec 7 with RBA E13 measured burden	post-2009 sample only
8	Williams CCI interactions (six, free)	Spec 4 + $r \times CCI$, $\log(HP/y) \times (1 - 1.2 \cdot CCI)$, $\log(y^p/y) \times CCI$, ha_x_cci
9	Spec 8 with Kalman state-space CCI	replaces smoothed-step spline with state-space extraction
10	Williams-prior calibrated	$\gamma_IFA = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\omega = 1.2$; iterative fixed-point OLS
11	Faithful LIVES (free, headline)	housing via ha_x_cci only; ζ_c CCI intercept $cci_williams$ restored; IFA combined ($ilfa_y$); nla_y , $hp_x_1_minus_cci$, r_x_cci , $\ln_yp_over_y$, yp_x_cci
12	Williams-calibration-imposed LIVES	Spec 11 form with $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\gamma_IFA = 0.022$ imposed via iterative fixed-point

The crucial structural difference is between Spec 6 and Spec 11. Spec 6 carries the wealth components as plain, constant marginal propensities (ha_y , eq_y , $super_y$, nla_y) and lets the credit-conditions index enter only through a short-run term, $\Delta^2\log$ CCI at lag 2. It is the form that the broader empirical literature, and an earlier draft of this work, treated as "the" LIVES equation. It is not. As §4 sets out, the LIVES theory predicts no classical housing-wealth effect: the housing marginal propensity is zero at $CCI = 0$ and is unlocked only as credit

conditions ease, so housing must enter through the interaction term `ha_x_cci` and not as a standalone level. Spec 11 imposes exactly that form — housing enters only via `ha_x_cci`, the autonomous- consumption CCI loading ζ_c (`cci_williams`) is restored, and the two illiquid-financial components are combined into `ilfa_y` as the theory specifies. The contrast between the two is therefore a matter of functional form dictated by theory, not of fit; §7 shows that the form is decisive for whether the error-correction and wealth structure come alive.

1.7.2 6.2 The four selection screens

Following the structural-econometrics tradition (Hendry and Krolzig 2005; Doornik 2009), we screen each estimable specification through four formal tests, with BIC as a tiebreak. The screens and their per- specification outcomes are recorded in `australia_spec_selection.csv`; the four-sample λ stability that feeds the stability screen is in `australia_lambda_robustness.csv`; the cointegration statistics are in `australia_cointegration.csv`.

1. **Sign screen** — every long-run coefficient carrying a non-ambiguous theoretical prior (§4.2) has the correct sign.
2. **Cointegration screen** — an Engle–Granger residual test: an ADF (with drift) on the residual of the static long-run regression rejects the no-cointegration null at 5 per cent, evaluated against MacKinnon (1991, 2010) critical values keyed to the number of regressors in the cointegrating regression, *not* the univariate Dickey–Fuller value. Phillips–Ouliaris and single-equation Johansen results are reported alongside.
3. **Speed-of-adjustment screen** — λ has the correct (negative) sign and lies in the interval (0.02, 0.30).
4. **Stability screen** — Chow at 2008Q3 is not rejected at the 1 per cent level, *and* λ is sign-stable across at least 3 of the 4 sample variants (full, pre-COVID, COVID-dropped, COVID rich-dummies).

Two screens warrant comment for the LIVES interaction specifications (Specs 8–12). The cointegration screen is reported as "NA" for these forms in `australia_cointegration.csv`: with up to ten regressors (`coint_n_vars` of 10, 9, 5, 9 and 3 for Specs 8–12 respectively) the static long-run regression that the Engle–Granger test requires is not well posed, and the test is not run. And the upper bound of the speed-of-adjustment screen ($|\lambda| < 0.30$) is binding for the strongest-adjusting forms: Specs 8 and 11 both adjust faster than the 0.30 ceiling on the full sample ($|\lambda| = 0.445$ and 0.480), and so are flagged as failing the $|\lambda|$ screen even though their λ is correctly signed, strongly significant, and sign-stable across all four sample variants. We return to this in §6.3 and §7: the full-sample λ for the faithful LIVES form is inflated by the COVID quarters, and the identified value is the pre-COVID estimate, $\lambda = -0.245$ ($t = -4.80$), which lies comfortably inside the screen interval and within roughly 15 per cent of Williams' -0.286 .

1.7.3 6.3 Selector outcome

Under the canonical `PI_METHOD = 'italy'` setting, the automated four-screen selector returns **Spec 2** (aggregate log net worth plus a short-run CCI term) as the only specification that passes the sign, speed-of-adjustment and stability screens with a finite BIC — and it is not the LIVES form. The full screen card, taken directly from `australia_spec_selection.csv`, is:

Spec	Signs	Coint	λ	Stability	BIC
1	✓	×	✓	×	-919.2
2 (selector-best)	✓	×	✓	✓	-500.8
3	✓	×	✓	×	-919.8
4	×	×	✓	×	-906.8
5	×	×	✓	✓	-493.0
6 (conventional baseline)	×	×	✓	×	-493.8
6b	×	×	✓	×	-1116.3
7	×	×	×	×	-499.8
7b	×	×	×	×	-363.0
8	✓	NA	×	×	-948.5
9	×	NA	✓	×	-895.9
10	✓	NA	✓	×	-493.7
11 (faithful LIVES, headline)	✓	NA	×	×	-945.0
12	×	NA	✓	×	-892.5

Source: `australia_spec_selection.csv`; BIC values are Schwarz from `australia_all_diagnostics.csv`. Spec 6b carries the lowest (best) BIC only because it is fitted on the longest, $n = 190$ back-extended sample and so is not directly comparable to the $n = 146$ or $n = 86$ forms.

Several patterns emerge, and we read them as a coherent statement about the limits of single-equation identification rather than as a clean verdict.

No specification clears the cointegration screen. Evaluated against MacKinnon critical values keyed to the regressor count (`coint_adf_5pct_cv` ranging from -4.42 to -6.13 across the estimable forms), no single-equation specification rejects the no-cointegration null. The disaggregated forms come closest — Specs 4–6 reach $ADF \approx -3.2$ on the long-run residual (Spec 4/5: -3.23 against -5.23 ; Spec 6: -3.22 against -5.47) and the back-extended Spec 6b reaches -3.77 against -5.47 — but none crosses its critical value. The aggregated Specs 1–3 fall far short ($ADF \approx -0.56$ to -1.13). Phillips–Ouliaris likewise fails for Specs 1–3, while single-equation Johansen returns $r = 1$ for every estimable form. A static single-equation long run between consumption and its wealth/income determinants is therefore not formally established on this sample. This is one of the paper's recurring themes (§5, §7.3, §9): the long-run identification Williams obtains comes from his cross-equation FIML system, not from any single equation. Because the screen no longer discriminates, selection falls to the remaining three screens with the BIC tiebreak.

The sign screen passes for both the faithful LIVES form and Spec 8. Spec 11 passes the sign screen — its long-run coefficients with unambiguous priors (`nla_y`, `ilfa_y`, `ha_x_cci`, `ln_yp_over_y`) are all correctly signed, with `nla_y` = $+0.037$ ($t = 3.40$), `ilfa_y` = $+0.018$ ($t = 3.03$) and `ln_yp_over_y` = $+0.504$ ($t = 4.00$) significant and right-signed on the full sample, and

$ha_x_cci = +0.0049$ ($t = 1.03$) right-signed though insignificant. Spec 8, the six-interaction free form, also passes the sign screen at the aggregate level. By contrast the disaggregated Specs 4–6 *fail* the sign screen under the canonical Italy LP forecaster: because $|\lambda|$ rises markedly relative to the AR forecaster, the implied γ on each wealth component shrinks and small negative coefficients on individual components — eq_y in particular (Spec 6 full: $eq_y = -0.0063$, wrong-signed) — tip the sign screen. We read this as an identification artefact of the constant-MPC disaggregation, not as a substantive reversal of the illiquid-financial channel, and it is precisely the artefact that the faithful LIVES form removes by combining the illiquid components into $ilfa_y$ (which is then correctly signed and significant).

The faithful LIVES form fails the $|\lambda|$ upper-bound screen on the full sample. Spec 11's full-sample $\lambda = -0.480$ ($t = -3.59$) exceeds the 0.30 ceiling, as does Spec 8's $\lambda = -0.445$ ($t = -3.30$). Both are correctly signed, strongly significant and sign-stable across all four sample variants. The mechanical screen failure is therefore not a sign or significance failure; it reflects the full-sample λ being inflated by the COVID quarters. On the pre-COVID sample Spec 11's $\lambda = -0.245$ ($t = -4.80$) lies inside the screen interval and within roughly 15 per cent of Williams' -0.286 — which is why §7 treats the pre-COVID estimate as the identified value.

The calibration-imposed forms pass the $|\lambda|$ screen for the wrong reason. Specs 10 and 12 pass the speed-of-adjustment screen on the full sample, but only because their λ has *collapsed* toward zero (Spec 10: -0.048 , $t = -0.76$; Spec 12: -0.029 , $t = -0.68$), placing it just inside the lower edge of the (0.02, 0.30) interval. Both fail the stability screen — their λ is not sign-stable across samples (`lambda_sign_stable_across_samples = FALSE` in `australia_lambda_robustness.csv`; Spec 12's pre-COVID λ flips to $+0.030$, Spec 10's COVID-dropped λ flips to $+0.005$). A screen that is satisfied by a near-zero, sign-unstable adjustment speed is exactly the case the stability screen exists to catch. The economic content of this collapse — that imposing Williams' Australian calibrations wrecks the equilibrium — is developed in §7.0.1 and §9.

No single criterion selects the headline. The automated screen returns Spec 2, a non-LIVES aggregate-net-worth form; BIC over the comparable $n = 146$ specifications favours Spec 8 (-948.5 , the best finite-sample BIC), with the faithful Spec 11 close behind (-945.0); and LIVES theory favours Spec 11. These three criteria do not agree. We treat this **selector divergence as a documented limitation, not a result to be argued away**. It is the single-equation counterpart of the identification problem that runs through the paper: the six CCI interactions are 0.74–0.97 mutually collinear (§5), the long run does not cointegrate in a single equation, and the data are too imprecise for any mechanical rule to isolate the structural LIVES form. We do *not* promote Spec 8's fast adjustment ($\lambda = -0.445$) as a CCI success; its interaction coefficients are wrong-signed or insignificant (§7) and it re-allocates identification across the interaction block rather than sharpening any single channel.

Accordingly, the body of this paper leads with the **faithful LIVES specification (Spec 11)** on theoretical-form grounds — it is the form Williams (2010) and the LIVES tradition adopt, it passes the sign screen, and (on the identified pre-COVID sample) it recovers Williams' error-correction speed. We retain **Spec 6 as the conventional constant-MPC baseline** — not the headline or the preferred specification — because it is the form prior work treated as LIVES and because it permits the $\gamma_LA + \gamma_LOANS = 0$ net-liquid restriction test (§8). We carry **Spec 12 (and its independent reproduction Spec 10) as the negative control** that shows the

structure transfers but the calibrations do not, and **Spec 2 as the automated selector-best** comparator. The full per-specification coefficient vectors, with Newey–West HAC standard errors and now including the Spec 11 and Spec 12 columns, are in Appendix B.

1.8 7. Results

1.8.1 7.0 The LIVES headline specification (Spec 11)

The specification we lead with, Spec 11, is the faithful Muellbauer–Williams Eq (7) form. It differs from the conventional disaggregated ECM (Spec 6, §7.1 below) in three structural respects, each dictated by the theory rather than by fit: (i) housing wealth enters **only** through its credit-conditions interaction $CCI \cdot (HA/4y)$ — there is no standalone ha_y level, because the Williams housing m.p.c. is zero at $CCI = 0$; (ii) the autonomous-consumption intercept $\zeta_c \cdot CCI$ is restored; and (iii) illiquid financial assets are combined into a single ratio (equities + superannuation) rather than split, which removes a collinear, wrong-signed equities coefficient. The CCI interactions are de-meanned over the post-1979 sample, following Williams.

Full-sample (1988Q4–2024Q4, $n = 146$, $adj-R^2 = 0.81$) long-run coefficients (australia_all_results.csv, Spec 11 rows; diagnostics from australia_all_diagnostics.csv):

Term	OLS coef	t-stat	Implied $\gamma (= OLS/ \lambda)$	Williams
$ha_x_cci (\gamma_1, \text{housing} \times CCI)$	+0.0049	+1.03	0.010	0.049
$nla_y (\gamma_{NLA})$	+0.0370	+3.39	0.077	0.159
$ilfa_y (\gamma_{IFA})$	+0.0178	+3.03	0.037	0.022
$cci_williams (\zeta_c)$	+0.0046	+0.47	—	0.190
$hp_x_l_minus_cci (\alpha_4)$	+0.0142	+2.69	—	−0.130
$r_x_cci (\alpha_1)$	+0.0027	+3.57	—	−0.871
$ln_yp_over_y (\psi_0)$	+0.504	+4.00	—	0.20
$yp_x_cci (\psi_1)$	−0.590	−1.58	—	0.93
$ecm_lag (\lambda)$	−0.480	−3.59	(= 1)	−0.286

The contrast with Spec 6 (§7.1) is stark. **The error-correction and core wealth structure come alive.** The speed of adjustment is $\lambda = -0.480$ ($t = -3.59$) on the full sample and $\lambda = -0.245$ ($t = -4.8$) pre-COVID — almost exactly Williams' **−0.286** — against Spec 6's insignificant -0.180 ($t = -1.76$); the full-sample value is inflated by the COVID quarters and fails the upper-bound $|\lambda|$ screen, so we read the pre-COVID estimate as the identified one. The net-liquid and illiquid-financial m.p.c.s are now individually significant and correctly signed on the full sample (implied $\gamma_{NLA} = 0.077$, $\gamma_{IFA} = 0.037$, the latter close to Williams' 0.022; both become individually insignificant on the smaller pre-COVID sample, Table 11.1), and the permanent-income response is strong ($t = 4.0$). Because the CCI-spline interactions replace Spec 6's 2002Q3-binding cci_ratio short-run term, the model estimates on $n = 146$ rather than $n = 86$ — a near-doubling of the identification window. Reading a standalone, insignificant ha_y in Spec 6 as a failed housing wealth effect was therefore a category error: the theory predicts that coefficient to be ≈ 0 , and the housing effect lives in the credit interaction.

The honest qualifier is that **the credit-conditions interactions themselves remain weakly identified**. The housing-collateral term ha_x_cci (γ_1) is correctly signed but insignificant ($t = 1.03$; implied m.p.c. 0.010 against Williams' 0.049), the ζ_c intercept is correctly signed but insignificant, and the real-rate and affordability interactions carry the wrong sign and significance on the full sample but flip to the correct sign pre-COVID (e.g. ψ_1 on yp_x_cci is -0.59 full-sample but $+0.13$ pre-COVID). This is the signature of two compounding problems documented in §5 and below: the six CCI-interacted regressors are 0.74–0.97 mutually correlated (each $\approx$ proportional to the latent index), and the 1980s–1990s financial-liberalisation episode that identifies the credit channels largely predates the 1988Q3 household-balance-sheet data.

1.8.2 7.0.1 The calibration route does not transfer (Spec 12, Spec 10)

Because the interactions cannot be freely identified, the natural single-equation response is Williams' own: calibrate the credit channels and estimate only what the data can support. Spec 12 imposes Williams' scale-robust calibrations ($\gamma_{IFA} = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$) via an iterative fixed-point offset and frees only the housing-collateral m.p.c., the net-liquid m.p.c. and λ . (His real-rate, affordability and intercept loadings cannot be imposed at their published magnitudes: the repo's percent-scaled real rate against a unit-normalised CCI makes $\alpha_1 = -0.871$ roughly thirty times too large, diverging the fixed point.)

The result is decisive and negative: **imposing Williams' permanent-income calibration collapses the error-correction to $\lambda \approx 0$** (Spec 12: $\lambda = -0.029$, $t = -0.68$; independently reproduced by the pre-existing Spec 10, $\lambda = -0.048$, which keeps the rate and affordability channels free). The mechanism is that the Australian data freely estimates $\psi_0 \approx 0.50$ — about two-and-a-half times Williams' 0.20 — so forcing his value injects a large, mis-signed contribution that destroys the equilibrium. The LIVES *structure* transfers; Williams' Australian *calibrations* do not. This sits with, and explains, the companion paper's Wald result that the joint calibration is *not rejected* ($\chi^2 = 2.24$, $p = 0.90$): the freely estimated coefficients are too imprecise to reject Williams' values, but imposing them still ruins the fit — low power is not the same as good fit. Sharpening the credit channels therefore requires either the four-equation FIML system (whose cross-equation restrictions supply the identifying variation the single equation lacks) or a pre-1988 back-extension that recovers the financial-liberalisation episode — not a single-equation calibration.

1.8.3 7.1 Conventional disaggregated specification (Spec 6) — baseline

Spec 6 is the constant-marginal-propensity disaggregated ECM. We retain it here not as the preferred or headline result — that role belongs to the faithful LIVES form of §7.0 — but as the **conventional baseline** against which Spec 11 is the faithful alternative. It is the specification an earlier draft, and much of the Australian aggregate-wealth-effect literature, would treat as "the LIVES equation": it carries plain ha_y , eq_y , $super_y$ and nla_y with no credit scaling, and CCI enters only as a short-run term ($d2_logcci_lag2$). It is therefore the right object against which to make the form-is-decisive point concrete, not the equation we recommend.

Over the full 1988Q4–2024Q4 sample Spec 6 fits on $n = 86$ after lag truncation. The binding constraint is cci_ratio from ABS Cat 5601.0, which begins 2002Q3; this also prevents Spec 6 from being back-extended to the 1976Q3+ window without either replacing the short-run CCI term with a longer-history credit aggregate (e.g. $\Delta^2 \log$ of $credit_total_d02$) or setting it to

zero pre-2002. We retain the 2002Q3+ binding constraint here and report the back-extension exercise on the simpler Spec 4 in §7.3 and §8.15.

The long-run coefficients of Spec 6 under canonical Italy LP are (australia_all_results.csv, Spec 6 rows):

Term	OLS coef	NW SE	t-stat	Implied γ (= OLS/ $ \lambda $)	Sign OK
ha_y	+0.0088	0.0058	+1.52	+0.049	✓
nla_y	+0.0354	0.0369	+0.96	+0.196	✓
eq_y	-0.0063	0.0471	-0.13	-0.035	×
super_y	+0.0117	0.0079	+1.48	+0.065	✓
ln_hp_over_y	-0.0170	0.0199	-0.85	-0.094	n/a
real_rate	-0.00018	0.0011	-0.17	-0.0010	✓
ln_yp_over_y	+0.1999	0.2633	+0.76	+1.110	n/a
ln_yp_over_y_post2008	+0.2360	0.2030	+1.16	+1.311	n/a
ecm_lag (λ)	-0.1801	0.1025	-1.76	(=1)	✓

(Short-run regressors and event dummies omitted from this table; see Appendix B for the full coefficient vector.)

The picture is the mirror image of Spec 11. In summary:

- **Speed of adjustment.** $\lambda = -0.180$ (NW SE 0.103), $t = -1.76$, $p = 0.084$ — only borderline significant, about 63 per cent of Williams' published -0.286 . The error-correction does not come decisively to life in this form.
- **Housing wealth.** OLS coefficient $+0.0088$, $t = 1.52$ — statistically indistinguishable from zero as a *standalone* level. This is the coefficient the theory predicts to be ≈ 0 absent the CCI interaction; reading it as a failed housing wealth effect is the category error §7.0 identifies. The implied structural $\gamma_{HA} = 0.049$ is numerically close to Williams' 0.0488, but §7.3.1 shows that closeness is a non-rejection driven by imprecision, not confirmation.
- **Net liquid assets.** OLS $+0.0354$ ($t = 0.96$, insignificant), implied $\gamma_{NLA} = 0.196$ — about 23 per cent above Williams' 0.159, in the same direction and order of magnitude. The $\gamma_{LA} + \gamma_{LOANS} = 0$ cross-equation restriction is accepted at the 5 per cent level (§8.5).
- **Illiquid financial wealth.** Decomposed into equities ($\gamma = -0.035$, wrong-signed but $t = -0.13$, statistically indistinguishable from zero) and superannuation ($\gamma = +0.065$, $t = 1.48$); the combined $\gamma_{IFA} = 0.030$ sits above Williams' calibrated 0.022. The negative point estimate on equities is a small-sample identification artefact of the disaggregated split: combining the two illiquid components into a single *ilfa_y* ratio (as Spec 11 does, §7.0) restores a positive, significant coefficient ($+0.018$, $t = 3.03$).
- **House-price affordability.** OLS -0.0170 , implied $\gamma = -0.094$; Spec 6 does not include the affordability $\times (1 - \omega\text{-CCI})$ interaction that Williams' framework uses to identify this channel, so the level coefficient is not a like-for-like comparison.
- **Real mortgage rate.** OLS -0.00018 , insignificant in the level; the credit-conditions-contingent rate effect Williams models enters only through the $r \times \text{CCI}$ interaction, which Spec 6 omits.

- **Permanent income.** Base coefficient +0.200 (SE 0.263, $t = 0.76$), plus a post-2008 break of +0.236 (SE 0.203, $t = 1.16$) — neither individually significant. The base coefficient is close to Williams' calibrated +0.20 in sign and broad magnitude.
- **Diagnostics.** $\text{adj-}R^2 = 0.81$, Durbin–Watson 2.18, $\text{AR}(1)$ $p = 0.26$, $\text{AR}(4)$ $p = 0.14$ (no serial correlation), RESET $p < 0.01$ (functional-form misspecification remains), heteroskedasticity structural (NW HAC SEs throughout). λ is sign-stable across all four sample variants (full –0.180, pre-COVID –0.123).

The key reading is comparative, not absolute: Spec 6 reproduces a *plausible profile of point estimates* but at conventional significance levels its error-correction and wealth channels are not individually identified. Spec 11 recovers the same theory on the same data with significant, correctly signed core channels — the difference is the functional form, not the sample, the data vintage, or the estimator.

1.8.4 7.2 Diagnostics summary

Diagnostic results for all fourteen specifications (full sample) are summarised below; full per-spec output is in `australia_all_diagnostics.csv`.

Spec	n	adj R^2	DW	AR(1)	AR(4)	Het	RESET	BIC
1 (LogNetWorth)	146	0.731	2.34	rej	rej	struct	rej	–919.2
2 (LogNetWorth_CCI)	86	0.769	2.40	rej	OK	struct	rej	–500.8
3 (LevelNetWorth)	146	0.732	2.35	rej	rej	struct	rej	–919.8
4 (Disagg_NoCCI)	146	0.729	2.39	rej	rej	struct	rej	–906.8
5 (FullDisagg)	86	0.798	2.31	rej	rej	struct	rej	–493.0
6 (Preferred/baseline)	86	0.807	2.18	OK	OK	struct	rej	–493.8
6b (LongHistSRCCI)	180	0.707	2.11	OK	OK	struct	rej	–1116.3
7 (CohortBurden)	86	0.833	2.20	OK	rej	struct	rej	–499.8
7b (RBABurden)	64	0.869	2.16	OK	rej	struct	rej	–363.0
8 (CCI_Interactions)	146	0.821	1.87	OK	rej	struct	rej	–948.5
9 (KalmanCCI)	146	0.737	2.20	OK	rej	struct	rej	–895.9
10 (WilliamsPrior)	86	0.778	2.17	OK	OK	struct	OK	–493.7
11 (LIVES_Headline)	146	0.812	1.80	OK	OK	struct	rej	–945.0
12 (LIVES_Calibrated)	146	0.686	2.09	OK	rej	struct	rej	–892.5

(Table regenerated from `australia_all_diagnostics.csv`; "OK"/"rej" are the 5 per cent verdicts of the Breusch–Godfrey AR(1)/AR(4) and RESET tests, and "Het" is the `het_diagnosis` field. AR(4) is borderline for Spec 11, $p = 0.055$.)

Four diagnostic patterns are worth noting. First, **heteroskedasticity is structural in every full-sample specification** — the Breusch–Pagan rejection survives dropping the four event quarters in every case — so the Newey–West HAC standard errors used throughout are necessary rather than a precaution. Second, **RESET rejects in every specification bar the Williams-prior Spec 10** ($p = 0.054$), indicating functional-form misspecification that the

single-equation framing does not fully resolve (consistent with §5.3/§7.3/§9); the faithful Spec 11 is not exempt (RESET $p = 0.006$). Third, **low-order serial correlation is concentrated in the aggregated and disaggregated-no-CCI forms**: the Breusch–Godfrey AR(1) test rejects for Specs 1–5 but not for the baseline Spec 6, the faithful Spec 11, or the CCI-augmented Specs 6b–10/12, so the credit-spline and disaggregated dynamics absorb the serial dependence the aggregated specifications leave in the residual. Fourth, **the headline LIVES form (Spec 11) carries the best BIC among the $n = 146$ specifications bar Spec 8** (–945.0 vs –948.5): the form correction is not bought at the cost of fit, but neither does it dominate the BIC screen, which favours the over-parameterised Spec 8 — one strand of the selector divergence documented in §6.

1.8.5 7.3 Comparison with Williams (2010, 2012): the conventional baseline

The faithful comparison with Williams — where the LIVES structure recovers his error-correction speed and core wealth m.p.c.s — is in §7.0 and is taken up systematically in §9. Here we present the comparison for the *conventional baseline* Spec 6, because that comparison is what an earlier draft (and much of the literature) would have offered as the headline, and because it exhibits a numerical coincidence that the form-is-decisive thesis must explain rather than celebrate.

We compare Spec 6 to Williams' published Table 1 estimates from the BIS chapter (Muellbauer and Williams 2012). Williams reports **structural** long-run coefficients γ ; our OLS coefficients relate to those γ by the ECM identity $\text{OLS_coef} = \lambda \times \gamma$, so a difference in the implied γ can come from either the OLS coefficient or λ . Reporting both forms separates the two channels (`australia_williams_comparison.csv`):

Term	Williams γ	Williams implied OLS	Our OLS	Our γ	OLS gap	γ gap
λ	–0.2860	(same)	–0.1801	(same)	–37 %	(same)
Housing wealth ha_y	0.0488	0.0140	0.0088	0.0491	–37 %	+1 %
Illiquid eq_y + super_y	0.0220	0.0063	0.0054	0.0300	–14 %	+36 %
Net liquid nla_y	0.1590	0.0455	0.0354	0.1963	–22 %	+23 %
$\log(\text{HP}/y)$	–0.1300	–0.0372	–0.0170	–0.0943	—	—
ψ at CCI = 0	0.2000	0.0572	0.1999	1.1097	—	—

Mechanically, our OLS coefficients on the disaggregated wealth components are 14–37 per cent below Williams' implied OLS values, while our $|\lambda|$ is 37 per cent below his. Because the OLS gap and the λ gap have the same sign, the two deficits **partly cancel under the ECM identity** $\gamma = \text{OLS}/|\lambda|$: the implied structural γ on housing wealth comes out at 0.049, numerically all but equal to Williams' 0.049, and the implied γ on net liquid and illiquid financial wealth sit modestly *above* Williams in the same direction.

We treat this cancellation as a **property of the conventional baseline, not as evidence**, for two reasons. First, the closeness is fragile: it is a ratio of two estimates each measured impre-

cisely on $n = 86$, and §7.3.1 shows the confidence interval around γ_{HA} spans $[-0.05, 0.13]$ — wide enough to contain both Williams' value *and* zero. Second, and more fundamentally, Spec 6 is not the LIVES equation: its housing channel is a standalone level that the theory predicts to be ≈ 0 , so a coincidental match between an artefactual γ_{HA} and Williams' credit-unlocked m.p.c. carries no structural content. The faithful Spec 11 makes the same housing channel explicit ($\gamma_1 = 0.010$, $t = 1.03$) — right-signed, weakly identified, and *not* matching Williams numerically — which is the honest read on what the single-equation Australian data support. The §9 comparison is therefore built on Spec 11, not on this baseline coincidence.

A natural follow-up question is whether the simpler disaggregated no-CCI specification (Spec 4) — closer in form to Williams' Table 1 long-run cointegrating regression than Spec 6 is — would align with Williams' Table 1 on a longer sample. Refitting Spec 4 on the back-extended 1976Q3+ sample using the disaggregated wealth proxies (§3.13; `spec46_extended_comparison.csv`):

LR coefficient	1988+ baseline (n=146)	1976+ extended (n=190)	Williams Table 1
λ (<code>ecm_lag</code>)	−0.140	−0.193	−0.286
<code>nla_y</code>	+0.035	−0.002	+0.159
<code>eq_y</code>	−0.119	−0.104	+0.022‡
<code>super_y</code>	+0.040	+0.024	(incl.)‡
<code>ha_y</code>	+0.068	+0.040	+0.0488

‡ Williams reports a single illiquid-financial-asset MPC ($\gamma_{\text{IFA}} = 0.022$), shown here against the combined `eq_y` + `super_y`. Williams' structural MPCs ($\gamma_{\text{HA}} = 0.0488$, $\gamma_{\text{NLA}} = 0.159$, $\gamma_{\text{IFA}} = 0.022$) are the BIS-chapter values in `australia_williams_comparison.csv`, the same source used in §7.3's headline comparison and §11.4.

On Spec 4 the speed of adjustment moves 37 per cent closer to Williams ($-0.140 \rightarrow -0.193$, still 32 per cent short of -0.286), but the individual OLS wealth coefficients become *smaller*, not larger: γ_{NLA} collapses toward zero and γ_{EQ} retains a wrong sign. The back-extension therefore does not push the Spec 4 estimates closer to Williams' Table 1; the post-1988 sample window is not, in itself, what generates the divergence between Spec 4 and Williams' values. **Sample length is not the binding constraint.**

Reading the two exercises together points to the same conclusion as §7.0.1: the wedge between any single-equation OLS estimate and Williams' system FIML is the single-equation framing itself, not sample length, knot count, or sign-prior structure. The placebo evidence (§5.2) and the two-equation SUR result (§5.4) — in which joint estimation delivers a negligible cross-equation residual correlation and no efficiency gain — corroborate this reading.

We do *not* present Spec 8 (the free six-interaction form, $\lambda = -0.445$) as a CCI success here. Adding the full Williams CCI interaction set re-allocates the long-run identification across wealth components ($\gamma_{\text{HA}} = 0.022$, $\gamma_{\text{IFA}} = 0.053$, $\gamma_{\text{NLA}} = 0.098$) without moving systematically toward Williams' Table 1, and three of its six interaction coefficients carry the wrong sign against their priors (§5.5, §8.4). The magnitude of its λ reflects the collinear interaction set absorbing variation, not a sharpening of the credit channel.

1.8.6 7.3.1 How precisely is the structural profile identified?

The implied structural coefficients $\gamma_i = \beta_i/|\lambda|$ are ratios of two imprecisely estimated quantities on $n = 86$, so they inherit wide sampling uncertainty that the point-estimate comparison above conceals. Computing delta-method standard errors from the Newey–West covariance of (β_i, λ) — which carries the correlation between each numerator and the speed of adjustment — gives (`australia_gamma_inference.csv`):

Term	Implied γ	95% CI (delta)	Williams Table 1	Williams in CI?
Housing <code>ha_y</code>	0.042	[−0.050, 0.133]	0.0488	✓
Net liquid <code>nla_y</code>	0.150	[−0.317, 0.617]	0.159	✓
Equities <code>eq_y</code>	−0.044	[−0.506, 0.419]	0.011	✓
Super <code>super_y</code>	0.047	[−0.027, 0.121]	0.011	✓
<code>log(HP/y)</code>	−0.091	[−0.447, 0.265]	−0.130	✓
Permanent income	1.237	[−0.337, 2.811]	0.200	✓
Wealth aggregate (Σ)	0.194	[−0.768, 1.157]	0.230	✓

A moving-block residual bootstrap (block length 8, $B = 1000$) gives a near-identical picture, with the bootstrap median γ_{HA} at 0.038 and a 95 per cent interval [−0.45, 0.26] that includes zero (consistent with the standalone `ha_y` being statistically indistinguishable from zero in Spec 6).

Two things follow, and they discipline the §7.3 reading. First, **Williams' Table 1 value lies inside the 95 per cent confidence interval for every coefficient** — so the data are statistically consistent with his calibrations, the same conclusion the formal Wald test reaches (§9; companion $\chi^2 = 2.24$, $p = 0.90$). Second, and equally true, the intervals are so wide that they also contain zero (for the aggregate wealth effect, for equities, and for housing at the lower bound) and values far from Williams. The close numerical agreement of the baseline point estimates is therefore a **non-rejection driven by imprecision, not positive evidence** that the Australian structural coefficients *equal* Williams': on this sample the single-equation estimates cannot distinguish his values from a broad range of alternatives, including zero. This is the honest ceiling on the §7.3 comparison — consistency, not confirmation — and it is exactly why §7.0 leads with the faithful Spec 11 form (where the *significant* channels are NLA and IFA, not the credit-scaled housing m.p.c.) rather than with the baseline γ -profile coincidence.

For use in a calibrated model (MARTIN, §10.3), the policy-relevant summary is the aggregate long-run wealth elasticity $\gamma_W = 0.19$ (95% CI [−0.77, 1.16]); the point estimate is close to MARTIN's calibrated net-wealth elasticity of ≈ 0.17 , but the interval is too wide for the free estimate to discipline the calibration — it neither confirms nor rejects it. The disaggregated split into four components is even less precisely identified and should be treated as indicative.

(Caveat: both the delta method and the residual bootstrap hold the right-hand side fixed, so they propagate sampling uncertainty in the ECM coefficients but **not** the first-stage uncertainty in the generated permanent-income and CCI regressors; the true intervals are therefore

at least this wide. The real-time permanent-income sensitivity of §7.4 and §8.9 gives a partial read on the first-stage component.)

1.8.7 7.4 The Italy / AR comparison and the real-time check

The permanent-income measure matters materially for two coefficients — the speed of adjustment and the long-run permanent-income coefficient. Refitting Spec 6 ($n = 86$) under each measure on a common data flow (`australia_pi_realtime_robustness.csv`):

Term	AR (real-time)	Italy full-sample (head-line measure)	Italy real-time	Williams
<code>ecm_lag</code> (λ)	-0.051	-0.197	-0.118	-0.286
<code>ln_yp_over_y</code>	-0.222	+0.244	-0.105	(calib. 0.20)
<code>implied γ_{HA}</code>	0.019	0.0082	0.0118	0.049

(Common-refit values from `australia_pi_realtime_robustness.csv`; the canonical pipeline's Spec 6 λ under the full-sample Italy measure is -0.180 , marginally different from the -0.197 refit here because the diagnostic re-derives the data flow rather than reusing the cached pipeline outputs. The `ha_y` row reports the OLS coefficient, which equals the implied γ only after dividing by $|\lambda|$; we report the OLS coefficient here, consistent with the CSV.)

Two readings follow. First, the full-sample Italy measure roughly quadruples $|\lambda|$ relative to AR ($-0.05 \rightarrow -0.20$) and flips the long-run permanent-income coefficient from significantly negative (-0.22 , $t = -2.23$ — the "Australian permanent-income puzzle") to positive ($+0.24$, in sign agreement with theory and Williams' calibrated value, though insignificant at $t = 0.95$). Second — the operational caveat — **neither move fully survives a causal real-time projection**. The real-time Italy variant keeps about half the $|\lambda|$ gain (-0.118) but the permanent-income coefficient returns to modestly negative (-0.105 , $t = -1.53$), and the `ha_y` coefficient sits at $+0.012$ ($t = 1.86$) rather than at the full-sample $+0.008$. The positive-PI sign is therefore a property of the full-sample, two-sided measure, not of the real-time forecaster a model like MARTIN would use.

We retain the structural reasons the AR and Italy measures diverge — the rolling AR(8) forecaster lacks the labour-force-share predictor that captures Australia's slow-moving demographic effects on trend income, compounds short-run AR misspecification across 40 horizons, and over-estimates persistence after large income shocks, all of which the one-step direct projection avoids — but we read the puzzle's reversal as a feature of full-sample permanent-income *measurement* rather than a clean real-time resolution. We carry the full-sample measure as the headline and the real-time and AR variants as robustness columns (§8.9), and we flag throughout that the strong permanent-income response in the faithful Spec 11 ($t = 4.0$, §7.0) inherits this look-ahead caveat: it is conditioned on the full-sample permanent-income forecaster, and the real-time read is the operational lower bound.

1.9 8. Robustness

We run the Italian-style robustness suite of De Bonis, Liberati, Muellbauer and Rondinelli (2020) on the conventional constant-MPC baseline (Spec 6) and, where the committed batteries permit, on the automated-selector specification (Spec 2) and the faithful LIVES specification

(Spec 11). The suite is deliberately weighted toward exposing rather than concealing weakness: it spans estimator choice (OLS, IV, SUR), break testing, the credit-conditions construction, the permanent-income measure, the affordability adjustment, sample length, and out-of-sample forecast accuracy. Several of the batteries return honest negatives — the out-of-sample loss to a random walk, the near-zero cross-equation residual correlation, the placebo failure, and the partial reversal under a causal permanent-income measure. We report these prominently. They are substantive findings: a freely-estimated single-equation framework permits them to surface, whereas the imposed restrictions of a four-equation FIML system would hide them, and each points to why FIML and pre-1988 back-extension are the routes forward (§5, §9).

1.9.1 8.1 OLS vs IV on current income (Hall 1978 endogeneity)

Current income is instrumented by lagged income, lagged unemployment and the lagged mortgage rate (six instruments; `australia_iv_robustness.csv`, Spec 2, $n = 86$). The speed of adjustment is essentially unmoved: $\text{ecm_lag} = -0.0829$ (OLS) $\rightarrow -0.0852$ (IV), a change of -0.0023 (+2.82 per cent). The net-worth elasticity shifts by -4.81 per cent ($0.0615 \rightarrow 0.0585$) and the permanent-income coefficient by $+85$ per cent ($-0.0031 \rightarrow -0.0057$), but both are economically negligible in level terms. Current-income endogeneity is therefore not a material source of bias for the speed of adjustment on this sample.

1.9.2 8.2 Joint permanent-income + consumption SUR

Estimating the consumption equation jointly with the permanent-income equation by SUR (`australia_joint_pi_robustness.csv`, Spec 2, $n = 86$) leaves every coefficient within sampling noise of the single-equation OLS values. The largest level shift is in the speed of adjustment ($\text{ecm_lag} -0.0829 \rightarrow -0.1273$, $+53.6$ per cent in relative but a third of a percentage point in level), and the net-worth elasticity moves $+13.1$ per cent ($0.0615 \rightarrow 0.0695$). The cross-equation residual correlation is negligible; the SUR-versus-OLS coefficient stability is the load-bearing evidence (the literal $\hat{\rho} \approx -0.004$ figure cited in the system narrative is not a separately committed scalar — we report the coefficient comparison rather than the correlation). Single-equation OLS is therefore an acceptable estimator for the consumption block, and — consistent with the two-equation SUR result in §8.18 — the case for the multi-equation framework rests on cross-equation parameter restrictions, not on residual covariance or efficiency gain.

1.9.3 8.3 Chow battery and multiple-break tests

Chow tests (`australia_chow_battery.csv`, Spec 2) at 2008Q3 do not reject parameter stability (Chow stat 1.248, $p = 0.270$; $n_{\text{pre}} = 21$, $n_{\text{post}} = 65$), while the 2020Q1 break is strongly rejected (Chow stat 8.852, $p = 9.9 \times 10^{-10}$; $n_{\text{pre}} = 67$, $n_{\text{post}} = 19$) — the COVID structural break that the event dummies absorb. The 1995Q1 and 2000Q1 break dates fall too close to the sample edge to estimate. The Bai–Perron / CUSUM battery (`australia_breaks.csv`) returns a single dominant break: $\text{supF} = 164.20$ ($p = 0$) at 2020Q4, with a CUSUM p -value of 0.9963 (recursive residuals stable away from the COVID episode). The break structure is therefore concentrated entirely at COVID, not at the GFC.

1.9.4 8.4 Williams CCI interactions (Spec 8) — reallocation, not identification

Spec 8 enters all six Williams CCI interactions freely on the 1988Q4+ sample with the reduced-form `cci_williams`. Following Williams (Aust paper §5.1) the variables interacted with CCI are de-meant over the estimation sample before forming the interaction, so that each inter-

action term has a clean conditional interpretation rather than absorbing an implicit linear CCI level shift. The sign-prior verdicts (`australia_spec8_sign_prior_verdicts.csv`; full vector in `australia_all_results.csv`, $n = 146$) are:

Williams interaction	Sign prior	OLS coef	t	p	Verdict
$r \times \text{CCI}$	–	+0.00192	+1.85	0.066	wrong sign, marginal – FAIL
$\log(\text{HP}/y) \times (1 - 1.2 \cdot \text{CCI})$	–	+0.00765	+1.00	0.319	wrong sign on composite, insignificant – FAIL
$\log(y^p/y) \times \text{CCI}$	+	–0.6113	–2.12	0.036	wrong sign, significant – FAIL
$\log(y^p/y)$	+	+0.4677	+3.44	0.001	right sign – PASS
$\text{HA} \times \text{CCI} (\gamma_1)$	+	+0.00159	+0.32	0.748	right sign, insignificant

Three of the four sign-priced interactions fail. Of the six free interactions only the permanent-income level term ($\ln_yp_over_y$, +0.4677, $t = 3.44$) passes its prior cleanly; the housing-collateral interaction ha_x_cci carries its theoretically correct positive sign but is far from significant (+0.00159, $t = 0.32$). Against Williams' Table 1 the raw-coefficient gaps remain structural (`australia_williams_spec8_comparison.csv`): his α_{c1} ($r \times \text{CCI}$) = –0.871 against our structural +0.0043; his α_{c4} ($\text{HP}/y \times (1 - 1.2 \cdot \text{CCI})$) = –0.13 against our structural +0.017; his calibrated $\psi_1 = +0.93$ against our freely-estimated structural –1.374.

The substantive effect of Spec 8 is to *re-allocate* the long-run identification, not to recover Williams' channels. Standalone wealth strengthens (nla_y +0.0435***, struct +0.098; $super_y$ +0.0139**, struct +0.031; ha_y +0.0097**, struct +0.022) and the speed of adjustment shifts from –0.180 (Spec 6) to –0.445 ($t = -3.30$) on the full sample – past Williams' value in magnitude. Spec 8 also attains the highest adjusted R^2 among the $n = 146$ specifications (0.8215) and the best BIC overall (–948.5). We do **not** read any of this as a credit-conditions success. The pre-COVID estimates are revealing: there the wealth coefficients sign-correctly (ha_y +0.0139***) but the *de-meaned interaction* ha_x_cci turns significantly negative (–0.0114**, struct –0.046), the wrong sign for the LIVES collateral channel, and $hp_x_1_minus_cci$ is significantly negative (–0.0194***). Read together with the calibration collapse (§9, Spec 10/12) and the identification-vs-detrending decomposition (§5), the honest reading is that the six CCI-interacted regressors in a single equation act as flexible parameter time-variation rather than as the structurally identified common-factor channel that Williams' four-equation system delivers. Their mutual collinearity (each is approximately proportional to CCI) is the structural reason FIML is required; we discuss it as a first-class identification result in §5.

1.9.5 8.5 Net-liquid-assets restriction $\gamma_{LA} + \gamma_{LOANS} = 0$

We refit each disaggregated specification with deposits/ y and debt/ y entered separately and conduct a Wald test of $H_0 : \gamma_{LA} + \gamma_{LOANS} = 0$ using `car::linearHypothesis` with the Newey–West variance estimator. The restriction is **accepted at the 5 per cent level in every specification $\times$ sample combination** (`australia_nla_restriction_test.csv`):

Spec	Sample	$\gamma_{LA} + \gamma_{LOANS}$	NW SE	t	p	Restriction
4	full	+0.0800	0.0559	1.430	0.153	accepted
5	full	+0.1377	0.1134	1.214	0.225	accepted
6	full	+0.1045	0.1028	1.017	0.309	accepted
4	pre-COVID	+0.0398	0.0269	1.480	0.139	accepted
5	pre-COVID	-0.0356	0.0494	-0.721	0.471	accepted
6	pre-COVID	-0.0201	0.0532	-0.378	0.706	accepted

The data cannot distinguish separate liquid-asset and debt propensities. We read this as non-rejection-by-imprecision rather than as positive confirmation of exact netting, but it validates the Italian convention of netting deposits against debt and supports the use of the constructed nla_y series — the net-liquid channel that carries a significant, correctly-signed marginal propensity in the faithful LIVES specification (Spec 11, nla_y +0.037***, struct 0.077; §7).

1.9.6 8.6 Drehmann amortising-mortgage adjusted real rate

De Bonis et al. (2020) apply the BIS Drehmann, Juselius and Korinek (2017) amortisation-adjusted rate $adjR = R / (1 - (1+R)^{-N})$. For Australia we set $N = 25$ years (100 quarters), consistent with the longer Australian average mortgage maturity (`australia_drehmann_robustness.csv`, Spec 2, $n = 86$). The maturity choice materially changes the estimated speed of adjustment: ecm_lag -0.1928 (base) $\rightarrow$ -0.0866 (Drehmann), with the net-worth elasticity moving 0.0013 $\rightarrow$ 0.0612 and the permanent-income coefficient 0.2404 $\rightarrow$ -0.0032. Unlike the Italian result, the consumption equation is *not* invariant to the amortisation adjustment on the Australian sample; we flag the CCI/rate maturity convention as a genuine source of sensitivity, consistent with the more general finding that the credit-channel calibrations are weakly pinned down off a single equation.

1.9.7 8.7 Scaled-income robustness

De Bonis et al. (2020) average disposable income with labour-plus-transfer income to down-weight property-income mismeasurement. Re-running this construction (`australia_scaled_income_robustness.csv`, Spec 2, $n = 86$) shifts the speed of adjustment from -0.1928 to -0.1117 (SE 0.0466) and lifts the net-worth elasticity from 0.0013 to 0.0569. The income-measure choice moves $|\lambda|$ by roughly 0.08 — not negligible — but does not change the substantive ranking of the wealth coefficients.

1.9.8 8.8 Williams non-property income (NPY) robustness

Replacing the disposable-income series with `npv_real_pc` constructed per Williams (2009) §4.2.1 (`australia_williams_income_robustness.csv`, Spec 2, $n = 86$) provides the closest methodological match to Williams' income concept (property income stripped, but not symmetrically averaged with labour-plus-transfer income). The substitution shifts ecm_lag from -0.1928 to -0.0938 (SE 0.0431) and lifts the net-worth elasticity from 0.0013 to 0.0572. This moves $|\lambda|$ by roughly 0.10 under the Williams-NPY measure relative to the baseline net-worth specification, in the same direction as — and somewhat larger than — the scaled-income result.

1.9.9 8.9 Permanent-income method comparison (AR, full-sample Italy, real-time Italy)

§7.4 reports the headline three-way comparison; the committed columns (`australia_pi_method_comparison.csv` and `australia_pi_realtime_robustness.csv`, $n = 86$) refit under all three permanent-income measures:

PI measure	λ (ecm_lag)	t	$\log(y^p/y)$	t	adj-R ²
AR forecaster (full)	-0.0829	—	-0.0031 (puzzle)	—	0.7578
AR (real-time)	-0.0511	-0.89	-0.2224	-2.23	—
Italy LP (full-sample)	-0.1974	-1.95	+0.2442	+0.95	0.7690
Italy LP (real-time)	-0.1182	-1.86	-0.1051	-1.53	—

(adjusted R² from `australia_pi_method_meta.csv`; the Italy LP forecaster fits marginally better, 0.7690 vs 0.7578.)

The full-sample Italy LP measure is the headline measure (framed as a *measurement*, §4.3), and the AR forecaster delivers the significantly negative "Australian permanent-income puzzle" coefficient (-0.0031 full, -0.2224 under the real-time AR variant). The real-time Italy LP variant is the operationally honest benchmark: it is causal, re-fitting the projection at each t on data whose full k -quarter horizon is realised by t , so it is usable at forecast time. It shows that about half the speed-of-adjustment gain from AR to the full-sample Italy measure is genuine (λ falls from -0.197 to -0.118 once look-ahead is removed), but that the positive permanent-income sign is **not** — it reverses to -0.105 under the causal measure. We carry the full-sample measure as the headline and disclose explicitly that its positive-PI sign and λ magnitude are full-sample, non-causal properties. The same look-ahead caveat attaches to the strong permanent-income coefficient in the faithful LIVES specification (Spec 11, $\ln_{yp_over_y} +0.504$, $t = 4.0$); the real-time column is the operational robustness check.

1.9.10 8.10 Permanent-income filter sensitivity

A nine-cell grid over discount factor $\delta \in \{0.90, 0.95, 0.97\}$ and horizon $k \in \{20, 40, 60\}$ quarters, with the GFC learning-weight on and off (`australia_permanent_income_sensitivity.csv`), shows the speed of adjustment to be extremely stable within the AR method: λ ranges only from -0.0824 to -0.0830 across the eighteen $\delta \times k \times$ GFC combinations (baseline $\delta = 0.95$, $k = 40$, the $\eta = 0.05$ mapping of §4.3: $\lambda = -0.0829$, structural PI weight -0.037, structural main-wealth +0.742). The GFC learning weight has no effect on these summaries. Switching to an HP-filter permanent income ($\lambda = 1600$) moves λ to -0.2006 and flips the structural PI weight to +1.397. The within-AR-method PI tuning is therefore not what drives the $|\lambda|$ gap with Williams; the dominant factor is the AR-vs-Italy-LP method choice itself (§8.9).

1.9.11 8.11 COVID-period robustness

Across four sample variants (full, pre-COVID, COVID-dropped, COVID-rich) the speed of adjustment is sign-stable for every estimable specification except the two Williams-calibration-imposed specifications (`australia_lambda_robustness.csv`). For the conventional baseline (Spec 6) λ runs -0.180 (full) / -0.123 (pre-COVID) / -0.100 (COVID-dropped) / -0.149 (COVID-rich) — all correctly signed. The faithful LIVES specification (Spec 11) gives -0.480 (full) / -0.245 (pre-COVID) / -0.256 (COVID-dropped) — also sign-stable, with the full-sample

value inflated by the COVID quarters and the pre-COVID -0.245 treated as the identified value (close to Williams' -0.286). Only Spec 10 (Williams-prior calibrated; sign-flip to $+0.005$ COVID-dropped) and Spec 12 (Williams calibrations imposed; sign-flip to $+0.030$ pre-COVID) are not sign-stable across samples — an artefact of the calibration collapse documented in §9, not of the COVID episode itself.

1.9.12 8.12 Rolling-window estimation

A 60-quarter rolling estimation of the baseline specification shows the wealth coefficients trending mildly downward post-2014 (consistent with the macroprudential era flattening the wealth-consumption transmission) and λ becoming slightly less negative in the most recent windows. We do not interpret this as model instability but as a symptom of the limited identifying variation in the post-deregulation portion of the sample (§5): the financial-liberalisation episode that would identify the credit channels largely predates the 1988Q3 start of ABS sectoral balance-sheet data.

1.9.13 8.13 Out-of-sample forecast validation

We run a rolling out-of-sample validation (`australia_oos_rmse.csv`) on five structural specifications (Spec 4 disagg-no-CCI, Spec 6 conventional baseline, Spec 7 cohort-burden, Spec 8 Williams-CCI interactions, Spec 9 Kalman-CCI) over 36 expanding-window cuts at horizons $h \in \{1, 4, 8\}$ quarters ($n = 36$ at $h = 1, 4$; $n = 32$ at $h = 8$), against random-walk-with-drift and AR(1) benchmarks:

Specification	$h = 1$ RMSE	$h = 4$ RMSE	$h = 8$ RMSE
Benchmark RW drift	0.03094	0.03094	0.03282
Benchmark AR(1)	0.03703	0.03102	0.03283
Spec 4 (disagg, no CCI)	0.03191	0.03252	0.05025
Spec 6 (conventional baseline)	0.03218	0.03323	0.04163
Spec 7 (cohort-burden)	0.03080	0.03461	0.04428
Spec 8 (Williams CCI)	0.03242	0.03151	0.03657
Spec 9 (Kalman CCI)	0.03203	0.03222	0.03854

At $h = 1$ the structural specifications are competitive with the random-walk benchmark and Spec 7 narrowly beats RW-drift (0.03080 vs 0.03094); no other structural spec does. At $h = 4$ and $h = 8$ the random walk with drift dominates **every** structural specification (the best structural performer at $h = 8$ is Spec 8 at 0.03657 vs RW-drift 0.03282). This is the standard macro-forecasting puzzle: the LIVES framework's identification advantage is in interpreting historical co-movement, not in beating naive benchmarks at multi-step prediction. We record it honestly rather than overstating forecast performance.

Two construction caveats attach to the CCI specifications (Spec 8, Spec 9). The credit-conditions series and its de-mean constants are full-sample objects, so the Spec 8/9 columns are conditional on a full-sample-constructed CCI and are an upper bound on what a fully real-time credit-conditions forecaster would deliver; the permanent-income input to the validator is the leak-free rolling AR forecaster.

1.9.14 8.14 Back-extension robustness — Spec 1 on the 1976Q3+ sample

Refitting Spec 1 (aggregate net worth) on the back-extended sample using `ln_networth_y_proxy` (`spec1_extended_comparison.csv`):

LR coefficient	1988+ baseline (n = 146)	1976+ extended (n = 190)	% change
λ (<code>ecm_lag</code>)	-0.1772	-0.2021	+14.0
<code>ln_networth_y</code>	+0.1119	+0.1062	-5.1
<code>ln_hp_over_y</code>	-0.0151	-0.0033	-78.1
<code>real_rate</code>	-0.00137	+0.00086	-163
<code>ln_yp_over_y</code>	+0.9612	+0.9732	+1.2

The wealth elasticity is essentially stable across samples ($0.112 \rightarrow 0.106$, a 5 per cent change) — a positive validation of the aggregate-net-worth proxy: doubling the sample length and adding the deregulation-era regime does not shift the structural wealth-to-consumption coefficient. The permanent-income elasticity is also stable, and λ moves slightly more negative on the longer sample. The house-price-to-income coefficient collapses to near zero on the longer sample — a real signal reflecting lower `hp_over_y` variation in the pre-deregulation 1970s — and the real-rate coefficient sign-flips, though both estimates are economically negligible. The aggregate net-worth proxy correctly includes the household M3 liquid component (the earlier \$bn/\$m unit defect that had made the M3 term numerically inert is fixed and propagated through a reproducible cold rebuild; see Appendix D).

1.9.15 8.15 Spec 4 on the back-extended sample

The disaggregated-wealth proxies of §3 allow Spec 4 to fit on the back-extended sample (`spec46_extended_comparison.csv`): λ moves 37 per cent closer to Williams ($-0.1404 \rightarrow -0.1927$, still about a third short of -0.286), but the individual OLS wealth coefficients become *smaller* rather than larger — `nla_y` collapses ($+0.035 \rightarrow -0.002$, a sign-flip), `ha_y` falls 41.6 per cent ($0.068 \rightarrow 0.040$) and `super_y` falls 41.7 per cent ($0.040 \rightarrow 0.024$). The exercise establishes that sample length is **not** the binding constraint on whether the disaggregated single-equation form reproduces Williams' Table 1: the longer sample sharpens the speed of adjustment but blunts, rather than sharpens, the individual wealth channels.

1.9.16 8.15.1 Spec 6b — conventional baseline on the back-extended sample

Spec 6 binds at 2002Q3+ on the baseline sample because `cci_ratio` (ABS Cat 5601.0 housing-loan flow) begins there. Spec 6b retains the Spec 6 long-run and short-run structure but replaces the short-run CCI regressor with the second difference of log RBA D02 total credit (available from 1976Q3) and switches the wealth components to their back-extended proxies. This lets the conventional baseline fit on the full back-extended sample (n = 190 in the cointegration screen; n = 180 full / 160 pre-COVID in the diagnostics) (`australia_all_results.csv`; `australia_all_diagnostics.csv`):

LR coefficient	Spec 6 (n = 86)	Spec 6b (n = 180)	Williams Table 1
λ (ecm_lag)	-0.180 (t = -1.76)	-0.229 (t = -4.18)	-0.286
ha_y / ha_y_proxy γ	0.049	0.038	0.049
nla_y / nla_y_proxy γ	0.196	0.013	0.159
eq_y / eq_y_proxy γ	-0.035	-0.081	(calibrated 0.011)
super_y / super_y_proxy γ	0.065	0.029	(calibrated 0.011)
ln_hp_over_y γ	-0.094	-0.036	-0.130
ln_yp_over_y (CCI = 0)	+0.200	+1.234	+0.20 (calibrated)
BIC	-493.8	-1 116.3	n/a

Two patterns recur. First, the speed of adjustment moves substantially closer to Williams' published value — $\lambda = -0.229$ on the back-extended sample (80 per cent of Williams' -0.286 , vs 63 per cent on Spec 6), with the t-statistic improving sharply from -1.76 to -4.18 , and the pre-COVID estimate similar at -0.240 ($t = -4.75$). Second, the wealth γ profile shifts toward smaller individual elasticities — γ_{NLA} collapses from 0.196 to 0.013 and γ_{EQ} becomes more negative, mirroring the Spec 4 back-extension finding. γ_{HA} stays positive (0.038 vs Williams' 0.049) but sits about 22 per cent below Williams. This is consistent with the substantive reading throughout the paper: on the back-extended sample the disaggregated wealth proxies do not separately identify with Williams-like precision, even when the canonical short-run dynamics, the post-2008 permanent-income break, and a long-history credit proxy are all available. Sample length sharpens $|\lambda|$ but not the individual wealth γ profile — the residual gap is structural to the single-equation framing, not a sample-length artefact.

1.9.17 8.16 Maximal-GETS placebo on the back-extended sample

We place the deployed institutional CCI against 200 random 15-knot, 15-prior draws under the same sign-survival protocol on the back-extended 1976Q3+ sample. The committed verdict files report three nested placebo variants (`australia_williams_knot_placebo_verdict.csv`, `australia_williams_knot_placebo_extended_summary.csv`, `australia_williams_knot_placebo_maximal_extended_summary.csv`):

Placebo variant	Williams/ canonical adj-R ²	adj-R ² percentile	$ \lambda $ percentile	Verdict
Standard (Williams literal 4-knot)	0.7209	34th	58th	"detrending critique vindicated — below median"
Extended (Williams-canonical)	0.6789	18th	10th	"detrending critique persists — below median"
Maximal-GETS (deployed institutional)	0.6852	64th	37th	"weak support — above median but not far"

The deployed/institutional CCI sits in the **18th–34th adjusted-R² percentile** band across the literal and Williams-canonical variants, and the maximal-GETS protocol lifts this to the 64th adjusted-R² percentile but only the 37th $|\lambda|$ percentile (placebo medians: adj-R² 0.6832, $|\lambda|$ 0.2226, surviving knots 8; canonical surviving knots 7). Random combinations of 15 candidate knots and priors produce faster mean reversion than the canonical institutional choice in 64 per cent of draws. The honest reading is that the maximal-GETS protocol does some identification work, but most of the lift comes from the adaptiveness of the drop-on-violation reduction (15 candidate knots is a great deal of flexibility) rather than from Williams' specific knot or prior choice. We carry this placebo failure as a core negative result, not as an embarrassment: it is the empirical expression of why single-equation CCI is weakly identified and why the joint system is needed for regime classification (§10).

1.9.18 8.17 Sectional sign-prior CCI

Williams (Aust paper §5.1) imposes sign priors over periods rather than knot by knot. We construct a sectional CCI basis with one knot per period (1982 / 1990 / 1993 / 2007, plus 2014 / 2017 / 2020 / 2021 extensions) and re-run the placebo. The sectional canonical sits at the 36th adjusted-R² percentile and 40th $|\lambda|$ percentile — worse than the maximal-GETS canonical (64th/37th). Williams' specific period dating does not outperform random period placements on the back-extended sample.

1.9.19 8.18 Two-equation SUR (consumption + house prices)

Joint SUR estimation of the consumption equation and a Williams-style house-price ECM (Aust paper eq. 11) on the back-extended 1976Q3+ sample yields a residual correlation $\hat{\rho}(\varepsilon_C, \varepsilon_H) \approx -0.004$, with SUR coefficients within 0.1 per cent of equation-by-equation OLS for nearly every term. The finding is robust across specification variants (no CCI: $\hat{\rho} \approx -0.083$; no event dummies: $\hat{\rho} \approx +0.043$; minimal LR + SR: $\hat{\rho} \approx -0.025$). Joint estimation gives no efficiency gain at the quarterly frequency. The case for the multi-equation framework therefore rests on cross-equation parameter restrictions, not on residual covariance — the same conclusion as the consumption + PI SUR of §8.2.

1.9.20 8.19 Three-equation joint cross-equation CCI identification

We extend the maximal-GETS protocol to require sign-prior survival across **three** equations simultaneously (consumption + house prices

- mortgage stock). Of the seven knots that survive consumption-only fitting, only **two** pass the joint test:

Survival regime	Surviving knots
Consumption only	1979, 1986, 1992, 2007, 2009, 2017, 2020
Joint (C $\cap$ H $\cap$ M)	1986, 2017

The joint-identified `cci_williams_joint` flips the house-price equation's CCI loading from significantly negative (-0.024 under consumption-only CCI) to significantly positive ($+0.024$ under joint CCI), consistent with Williams' cross-equation sign restrictions working as intended. The mortgage-stock equation's CCI loading remains negative (joint sign-survival is a sign restriction, not a parameter-equality restriction; full FIML would be required to re-sign the mortgage equation's loading). The wealth-coefficient profile against Williams' Table

1 is barely affected by joint identification, confirming that the residual gap is structural to the single-equation framing, not a CCI-construction artefact — and that the route to sharpening the credit channels runs through the four-equation FIML system, not through further single-equation tuning.

1.10 9. Comparison with Williams (2010, 2012)

This section consolidates the comparison with Williams' published Table 1 and BIS chapter estimates. The organising thesis is the one that runs through the paper: **the LIVES structure transfers to Australia, but Williams' Australian calibrations do not**. The faithful single-equation LIVES form (Spec 11) recovers Williams' error-correction speed and the correctly signed wealth-m.p.c. structure; imposing his permanent-income gearing (Spec 12, Spec 10) collapses the equilibrium; and the six CCI interactions through which his framework identifies the credit channels are too mutually collinear to be separated off a single equation. We deliberately retire the earlier draft's "the preferred specification reproduces Williams' profile almost exactly" framing: that reading rested on the conventional constant-m.p.c. baseline (Spec 6), on an offsetting-deficit coincidence, and on confidence intervals so wide that they contain both Williams' values and zero. The honest ceiling is consistency, not confirmation.

1.10.1 9.1 Where the structure transfers — the faithful LIVES form recovers Williams' speed and wealth structure

The decisive agreement is on the error-correction speed and the *sign and significance* of the core wealth channels, and it appears only once the equation is written in the faithful LIVES form.

On the pre-COVID sample the faithful LIVES specification (Spec 11) estimates a speed of adjustment of $\lambda = -0.245$ ($t = -4.80$) against Williams' FIML estimate of -0.286 (SE 0.083 , $t = -3.45$) — close to his value (about 86 per cent of it), and tightly identified (`australia_lambda_robustness.csv`; `australia_all_results.csv`). The full-sample estimate is $\lambda = -0.480$ ($t = -3.59$); it is inflated by the COVID quarters and fails the $|\lambda|$ upper-bound screen, so we read the pre-COVID figure as the identified one (§7.0). Either way the contrast with the conventional baseline is sharp: Spec 6 returns an insignificant $\lambda = -0.180$ ($t = -1.76$), 37 per cent below Williams.

Within the faithful form the wealth structure is correctly signed and, for two of the three components, individually significant on the full sample (`australia_all_results.csv`):

- net liquid assets (γ_{NLA}): OLS $+0.037$ ($t = 3.40$, ***), implied structural m.p.c. **0.077**, against Williams' calibrated 0.159 — same sign, same order of magnitude, roughly half his value;
- illiquid financial assets (γ_{IFA} , equities + superannuation combined): OLS $+0.018$ ($t = 3.03$, ***), implied structural m.p.c. **0.037**, against Williams' calibrated **0.022** — same sign, within 0.015;
- housing-collateral (γ_1 , the CCI · (HA/4y) interaction): OLS $+0.0049$ ($t = 1.03$), implied structural m.p.c. **0.010**, against Williams' peak housing m.p.c. of 0.0488 — correctly signed but insignificant.

The permanent-income response is strong and correctly signed ($\ln_{\text{yp_over_y}}$ OLS $+0.504$, $t = 4.0$). And critically, because the CCI-spline interactions replace Spec 6's 2002Q3-binding `cci_ratio` short-run term, the faithful form estimates on $n = 146$ rather than $n = 86$ — a near-

doubling of the identification window relative to the baseline against which the previous draft compared Williams.

The single most important interpretive point for the comparison is that **the housing channel only appears when the form is faithful**. In the LIVES theory there is no classical housing wealth effect: the housing m.p.c. is zero at CCI = 0 and is unlocked as credit conditions loosen. Reading Spec 6's insignificant standalone ha_y (OLS +0.0088, $t = 1.52$) as a *failed* housing wealth effect was therefore a category error — the theory predicts that coefficient to be ≈ 0 , and the housing effect lives in the credit interaction ha_x_cci , where it is correctly signed.

1.10.2 9.2 Where the calibrations do not transfer — imposing Williams' gearing collapses the equilibrium

The opposite result holds for Williams' Australian *calibrations*. The natural single-equation response to weakly identified interactions is Williams' own — calibrate the credit channels and estimate only what the data support — but the data reject that route decisively.

Spec 12 imposes Williams' scale-robust calibrations ($\gamma_IFA = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$) via an iterative fixed-point offset and frees only the housing-collateral m.p.c., the net-liquid m.p.c. and λ . The result is a near-total collapse of the error-correction mechanism: $\lambda = -0.029$ ($t = -0.68$) on the full sample, and it flips to the wrong sign (+0.030) pre-COVID, so it is not even sign-stable across samples (*australia_lambda_robustness.csv*). This is independently reproduced by the pre-existing Williams-prior specification (Spec 10), which keeps the rate and affordability channels free and still returns $\lambda = -0.048$ ($t = -0.76$) full-sample and a still-insignificant -0.027 ($t = -0.62$) pre-COVID, flipping to the wrong sign once the COVID quarters are dropped (*australia_lambda_robustness.csv*). Two separate calibration routes therefore reach the same conclusion.

The mechanism is straightforward: the Australian data freely estimates the permanent-income weight at $\psi_0 \approx 0.50$ — roughly two-and-a-half times Williams' 0.20 — so forcing his value injects a large, mis-signed contribution that destroys the long-run equilibrium. (The " $\approx 2.5\times$ " framing is a comparison of Spec 11's level coefficient $\ln_yp_over_y = +0.504$ with Williams' calibrated $\psi_0 = 0.20$; the exact ψ_0/ψ_1 split is not separately estimable on a single equation because yp_x_cci is insignificant, so we report it as an order-of-magnitude contrast rather than a precise ratio.)

Williams' rate, affordability and autonomous-consumption loadings cannot be imposed at all at their published magnitudes. His raw real-rate loading $\alpha_r = -0.871$ enters his FIML system on differently scaled regressors; on this repository's *percent*-scaled real rate against a *unit*-normalised CCI it is roughly thirty times too large, and the iterative fixed point diverges. This is itself a finding: the LIVES structure is portable, but its numerical calibrations are specific to Williams' scaling conventions and sample, and cannot be transplanted mechanically.

1.10.3 9.3 Reconciling the companion paper's Wald non-rejection

The companion paper reports that a joint Wald test does **not reject** Williams' calibration ($\chi^2 = 2.24$, $p = 0.90$). Read alongside §9.2 this looks paradoxical: how can a calibration that *collapses* the equilibrium when imposed also be *not rejected* by a test? The resolution is power. The freely estimated single-equation coefficients are so imprecise that they cannot statistically reject Williams' values — but the same imprecision means the data also cannot reject zero, or any number of other points. **Low power is not the same as good fit**. A calibration

can be non-rejected by an underpowered test and still wreck the model when imposed; both facts hold here simultaneously, and the gamma-inference confidence intervals (§9.4) make the imprecision explicit.

(The $\chi^2 = 2.24$ figure is reported in the companion paper; it is not a value in this repository's committed results CSVs. The load-bearing committed evidence for the non-rejection-by-imprecision reading is the gamma-inference interval table `australia_gamma_inference.csv`, in which every Williams value lies inside our 95 per cent CI *and* every CI also contains zero.)

1.10.4 9.4 The structural-gamma comparison: consistency, not confirmation

The earlier draft's headline — that the implied structural γ profile "matches Williams almost exactly" — was a property of the conventional baseline (Spec 6), and it survives only as a much-weakened, hedged statement. Two features demote it from confirmation to consistency.

First, the apparent agreement rests on offsetting deficits. Under the ECM identity $OLS = \lambda \times \gamma$, Spec 6's OLS wealth coefficients run 14–37 per cent below Williams' implied OLS values while its $|\lambda|$ runs 37 per cent below Williams'; the two deficits roughly cancel, leaving the implied γ close to Williams (`australia_williams_comparison.csv`). That cancellation is a numerical coincidence of the baseline, not evidence that the structure has been identified. We therefore no longer present it as a positive result.

Second, and decisively, every component's confidence interval is wide enough to contain *both* Williams' value and zero (`australia_gamma_inference.csv`):

Term	OLS	Implied γ	95% CI	Bootstrap 95% CI	Williams	In CI?
ha_y (housing)	+0.0082	0.042	[−0.050, 0.133]	[−0.451, 0.255]	0.0488	✓
nla_y (net liquid)	+0.0295	0.150	[−0.317, 0.617]	[−0.940, 0.656]	0.159	✓
eq_y (equities)	−0.0086	−0.044	[−0.506, 0.419]	[−0.821, 0.578]	0.011	✓
super_y (super)	+0.0092	0.047	[−0.027, 0.121]	[−0.062, 0.210]	0.011	✓
ln_hp_over_y	−0.0179	−0.091	[−0.447, 0.265]	[−0.718, 1.220]	−0.130	✓
ln_yp_over_y	+0.2442	1.237	[−0.337, 2.811]	[−2.065, 4.813]	0.20	✓
WEALTH_AGG	+0.0384	0.194	[−0.768, 1.157]	—	0.230	✓

The honest summary is therefore: **every Williams value lies inside our 95 per cent confidence interval — but so does zero.** This is non-rejection driven by imprecision, the same low-power phenomenon that produces the companion Wald result. The $n = 86$ baseline sample is small and the intervals are correspondingly wide; point-estimate proximity to Williams is consistency, not confirmation.

1.10.5 9.5 Adding the CCI interactions does not close the gap (Spec 8)

Estimating the six CCI interactions freely (Spec 8) raises the speed of adjustment to $\lambda = -0.445$ ($t = -3.30$) full-sample on $n = 146$, in magnitude above Williams' -0.286 . We do not present this as a credit-channel success. The wealth coefficients shift relative to the baseline ($\gamma_{NLA} \rightarrow 0.098$, $\gamma_{IFA} \rightarrow 0.053$, housing-collateral $\rightarrow 0.004$) without moving systematically toward Williams' Table 1, and three of the four interaction sign priors fail outright (`australia_spec8_sign_prior_verdicts.csv`):

- `r_x_cci` (Williams $\alpha_{c1} = -0.871$ at CCI = 1): prior negative, OLS $+0.0019$, $p = 0.066 \rightarrow$ **FAIL** (wrong sign);
- `hp_x_1_minus_cci` (Williams $\alpha_{c4} = -0.13$ at CCI = 0): prior negative, OLS $+0.0077$, $p = 0.319 \rightarrow$ **FAIL**;
- `yp_x_cci` (Williams $\psi_1 = +0.93$): prior positive, OLS -0.611 , $p = 0.036 \rightarrow$ **FAIL** (wrong sign);
- `ln_yp_over_y` (Williams $\psi_0 \approx 0.20$): prior small positive, OLS $+0.468$, $p = 0.0008 \rightarrow$ **PASS**.

The raw-coefficient comparison confirms that the interactions do not land where Williams' framework predicts (`australia_williams_spec8_comparison.csv`): his $\alpha_{c1} = -0.871$ against our structural $+0.0043$ ($t = 1.85$), his $\alpha_{c4} = -0.13$ against our $+0.017$ ($t = 1.00$), his calibrated $\psi_1 = +0.93$ against our freely estimated -1.37 ($t = -2.12$). Adding the interactions therefore **re-allocates** the long-run identification across components — and raises λ — but does not close the residual gap with the joint FIML estimates. The earlier draft's framing that Spec 8 "exceeds Williams" is withdrawn.

1.10.6 9.6 The interactions are collinear: the structural reason Williams uses FIML

The reason single-equation estimation cannot deliver Williams' result is, at root, an identification problem. The six CCI-interacted regressors are 0.74–0.97 mutually correlated on this sample because each is approximately proportional to the latent CCI; they cannot be separately identified off one equation. (The 0.74–0.97 range is a property of the interaction-regressor matrix reported in §5 and the analysis narrative; it is not itself a standalone committed CSV. The load-bearing committed evidence is threefold: the wrong-signed, insignificant Spec 8 interaction coefficients of §9.5; the identification-versus-detrending shifts of §5; and the Spec 10/Spec 12 collapse of §9.2.)

This is precisely the structural reason Williams (2010) uses a four-equation FIML system rather than single-equation OLS. The same CCI enters all four of his equations with sign constraints; the affordability multiplier ω in the wealth $\times (1 - \omega \text{CCI})$ interaction is shared across equations; and $\zeta_h = 1$ normalises the house-price equation. Those cross-equation restrictions supply the identifying variation that lets the credit channels be separated — variation that no single-equation specification has. The joint-survival result of §5 corroborates this from the other direction: when the CCI knots are required to satisfy sign priors *jointly* across the consumption, house-price and mortgage equations, only two of the seven single-equation survivors pass, so the maximal-GETS single-equation identification was partly equation-specific. This aligns with both Williams' framework (CCI as a common factor under parameter restrictions) and the Duca, Muellbauer and Murphy (2013) state-space implementation in which the latent factor is identified jointly across equations.

1.10.7 9.7 Sample length is not the binding constraint

A natural conjecture is that the divergence from Williams reflects sample period — his 1978Q1–2008Q2 window spans the financial-liberalisation episode that identifies the credit channels, whereas our balance-sheet data begin only in 1988Q3, largely after deregulation. The back-extension exercise tests this directly and finds it is not the binding constraint.

Refitting the simpler disaggregated no-CCI specification (Spec 4) on the back-extended 1976Q3+ sample moves λ 37 per cent closer to Williams ($-0.140 \rightarrow -0.193$, still 32 per cent short of -0.286) — but the individual wealth coefficients *shrink* rather than grow on the longer window: γ_{NLA} collapses toward zero and γ_{EQ} retains a wrong sign (§8). Lengthening the sample sharpens the speed of adjustment but not the wealth structure, so the post-1988 window is not, in itself, what generates the divergence between any single-equation OLS estimate and Williams' jointly identified Table 1. The back-extension is retained as a contribution (it builds a 1976Q3-anchored master dataset and permits the direct sample-length test), but it is not the route to closing the gap with Williams.

1.10.8 9.8 The permanent-income measure and the Australian PI puzzle

A final point of comparison concerns the permanent-income channel and the measure on which it depends. Under the AR-method forecaster we replicate the often-noted "Australian permanent-income puzzle": the long-run coefficient on $\log(y^p/y)$ is negative — near-zero on the full-sample (look-ahead) AR variant (-0.003) and significantly negative once the AR forecaster is run in real time (-0.22 , $t = -2.23$) (*australia_pi_method_comparison.csv*, *australia_pi_realtime_robustness.csv*). Under the full-sample Italy Jordà local-projection *measure* it is positive ($+0.24$ on the net-worth forecaster regression; $+0.504$ in the faithful Spec 11). The structural reasons the measures diverge are real — the rolling-AR forecaster lacks the labour-force-share predictor that captures slow demographic effects, compounds short-run AR misspecification across 40 horizons, and over-estimates persistence after large income shocks, all of which the one-step direct projection avoids. But the positive sign is partly a property of the full-sample, non-causal construction of the measure: under a causal real-time projection the coefficient returns to modestly negative (-0.11 , §8). The strong positive permanent-income response that the faithful LIVES form recovers therefore rests on the full-sample (look-ahead) permanent-income measure; we flag this explicitly and direct readers to the real-time robustness column rather than treating the puzzle as resolved.

1.10.9 9.9 Summary

The comparison with Williams resolves into a clean two-part statement. The LIVES *structure* transfers: the faithful single-equation form recovers his error-correction speed (pre-COVID $\lambda = -0.245$ against -0.286) and the correctly signed wealth-m.p.c. structure (net-liquid and illiquid-financial m.p.c.s individually significant; housing-collateral correctly signed). His Australian *calibrations* do not transfer: imposing his permanent-income gearing collapses λ to ≈ 0 (Spec 12, Spec 10), because Australia freely estimates $\psi_0 \approx 0.50$, roughly $2.5\times$ his 0.20 , and his rate loading cannot even be imposed at its published scale. The companion Wald non-rejection is reconciled as low-power non-rejection — every Williams value sits inside a wide CI that also contains zero, so the structural- γ "agreement" is consistency, not confirmation. The six CCI interactions are 0.74 – 0.97 collinear and so weakly identified off a single equation, which is exactly why Williams' identification comes from cross-equation FIML restrictions; the back-extension shows sample length is not the binding constraint. The path to a tighter

reconciliation with Williams' published values therefore runs through a full four-equation FIML build, not through any single-equation OLS refinement.

1.11 10. Decomposition and policy implications

This section reads the estimated long run and the policy counterfactuals through the lens of the faithful LIVES specification (Spec 11) of §7.0, while being explicit about which committed decompositions are available on which long-run parameterisation. Two caveats frame everything below. First, the long-run contributions decomposition committed to the repository (`australia_longrun_contributions.csv`) is built on the *net-worth* (Spec 2-style) long run, not the disaggregated LIVES bracket of Spec 11; we therefore use it to characterise the shape of the fitted equilibrium path rather than to attribute consumption to the credit-scaled housing-collateral channel. Second, the counterfactuals (`australia_counterfactuals_summary.csv`) were generated on the conventional baseline (Spec 6 dummies) and the interaction spec (Spec 8); we report them as committed but read the CCI counterfactual through the de-meaning convention rather than as a structural housing-collateral experiment. The policy reading in §10.3, by contrast, is anchored on the Spec 11 channels.

1.11.1 10.1 Long-run contributions decomposition

The long-run decomposition (an Australian counterpart to Williams (2010) Charts 2–8) splits fitted de-meaned $\log(c/y)$ into the partial contribution of each long-run regressor over the 86 quarters from 2003Q3 to 2024Q4, the window over which the underlying series (including the Italy-LP permanent-income measure) are jointly available. Because the committed decomposition rests on the net-worth long run, the regressors are aggregate net worth ($\ln_networth_y$), house-price affordability ($\ln_hp_over_y$), the real mortgage rate ($real_rate$) and the permanent-income ratio ($\ln_yp_over_y$); each term is de-meaned and so sums to approximately zero by construction over the window, and the actual de-meaned path and the residual are reported alongside (`australia_longrun_contributions.csv`).

At the last observation (2024Q4) the partial contributions are: net worth -0.002 , house-price affordability -0.041 , real rate $+0.002$, and permanent income $+0.041$, against an actual de-meaned $\log(c/y)$ of -0.009 and a residual of -0.009 . The two large, near-offsetting wedges at end-of-sample are the negative affordability term (elevated house prices relative to income depressing the equilibrium consumption ratio) and the positive permanent-income term, with aggregate net worth close to its sample mean. The residual is not negligible: it carries a sizeable, slowly-evolving level – around $+0.10$ to $+0.18$ in the mid-2000s, drifting through zero and turning negative later in the sample – with the sharpest excursions around the GFC and, especially, early COVID (-0.24 in 2020Q2). By 2024Q4 it has returned close to zero (-0.009).

The honest qualifier is that this is a net-worth decomposition, not the disaggregated LIVES bracket. Under the faithful form (Spec 11) the housing contribution would enter *only* through the credit-scaled collateral channel $\gamma_1 \cdot CCI \cdot (HA/4y)$, whose implied structural MPC is 0.010 (right-signed but insignificant, $t = 1.03$; §7.0), so a LIVES-consistent decomposition would attribute far less of the equilibrium path to housing than a standalone-wealth reading would suggest. We do not present a separate Spec 11 path decomposition because the credit interaction is weakly identified off a single equation (§5, §10.3); attributing the fitted path to a γ_1 that is statistically indistinguishable from zero would overstate the precision of the channel. The committed net-worth decomposition is therefore best read as descriptive of the equilibrium

shape — affordability and permanent income as the dominant moving wedges — rather than as a structural attribution to the LIVES collateral channel.

Source: australia_longrun_contributions.csv, australia_longrun_decomposition.png.

1.11.2 10.2 Counterfactuals

We report three policy counterfactuals, holding all non-counterfactual regressors at their observed values and integrating the implied $\Delta \log c$ paths back to log-consumption levels (australia_counterfactuals_summary.csv). The APRA and COVID scenarios are computed on the conventional baseline event dummies; the CCI scenario is computed on the interaction spec. We retain them as committed evidence, with the hedges noted below.

Scenario	Event date	Basis	h = 4 q gap	h = 8 q gap	End-of-sample gap
No 2014/2017 APRA macro-prudential	2014Q4	Spec 6 dummies	+0.8 %	+2.3 %	+28.3 %
No COVID income support	2020Q1	Spec 6 dummies	−9.6 %	−9.6 %	−9.6 %
CCI at peak vs CCI = 0	1988Q4	Spec 8 interactions	n/a	n/a	≈ 0

(Gaps are cumulative deviations in $\log(c)$ from the baseline path, expressed in per cent; h-quarter values are measured from the relevant event date.)

Counterfactual 1 — no 2014/2017 APRA macroprudential. Zeroing the smoothed-step d_{apra_2014} and d_{apra_2017} dummies implies that consumption would have been about 0.8 per cent higher four quarters after the 2014 round (+0.0078 in log points) and about 2.3 per cent higher after eight quarters (+0.0234). The end-of-sample gap of +28.3 per cent (+0.2828 by 2024Q4) compounds the persistent post-event ogive shift over a decade and must be read as a diagnostic upper bound, not a forecast: it assumes nothing else in the macroeconomic environment would have adjusted over the same window, which is not how the economy operates. The four- and eight-quarter figures are the policy-relevant range and are consistent with the APRA programme accounting for roughly 1–3 per cent of consumption over its first two years.

Counterfactual 2 — no COVID income support. Zeroing the $d_{jobkeeper_2020}$, d_{2020_covid} and $d_{2020_rebound}$ dummies implies that consumption would have been about 9.6 per cent *lower* in the COVID period (−0.0962 in log points). The cumulative gap is constant across horizons because the COVID event dummies are bounded in time (zero before and after the 2020–21 window), so unlike the persistent APRA ogive the deviation does not continue to compound past the event window; the scenario is modelled as a one-off.

Counterfactual 3 — CCI at peak vs zero. Evaluating the interaction spec with the CCI-interacted regressors at CCI = 1 (historical peak) versus CCI = 0 (no liberalisation) implies an essentially zero cumulative consumption gap across the back-extended sample (-4.9×10^{-15} , i.e. zero to numerical precision). This is a mechanical consequence of the de-meaning convention: under de-meaned interactions the average contribution of CCI variation to fitted

$\Delta \log c$ is zero by construction, so the *cumulative* effect of a permanent CCI shift integrates to zero across the sample. The CCI interactions therefore operate as pure timing/distribution effects — they reallocate where consumption growth lands across the cycle without shifting its unconditional level. This is the LIVES-theoretic reading: credit conditions matter for *when* households extract housing equity and respond to permanent-income news, not for the long-run level of the consumption-to-income ratio. It is not a structural housing-collateral experiment, and it should not be read as evidence on the magnitude of γ_1 (for which see §7.0, where the credit-scaled collateral MPC is right-signed but insignificant).

Source: `australia_counterfactuals.csv`, `australia_counterfactuals_summary.csv`,
`australia_counterfactual_paths.png`.

1.11.3 10.3 Policy implications

We read the policy implications off the faithful LIVES specification (Spec 11), distinguishing the channels that come through clearly from those that remain weakly identified.

Wealth channel of monetary policy. Under Spec 11 the two plain-MPC wealth channels are significant and correctly signed: the net-liquid-asset MPC is $\gamma_3 = 0.077$ (OLS coefficient +0.037, $t = 3.40$) and the illiquid-financial-asset MPC is $\gamma_2 = 0.037$ (OLS +0.018, $t = 3.03$), both as structural propensities recovered as $OLS/|\lambda|$ on the full sample ($|\lambda| = 0.480$). Housing wealth, by contrast, enters *only* through the credit-scaled collateral channel: the implied structural MPC is $\gamma_1 = 0.010$ — right-signed but statistically insignificant (OLS +0.0049, $t = 1.03$), against Williams' calibrated peak of 0.0488. The policy reading is therefore asymmetric across asset classes: liquid and illiquid financial wealth transmit to consumption with well-identified marginal propensities, whereas the housing-collateral channel that LIVES theory makes conditional on credit conditions is, on Australian single-equation post-deregulation data, of the predicted sign but too imprecise to quantify. Movements in mortgage rates that change housing values propagate to consumption with a speed of adjustment we identify on the pre-COVID sample at $\lambda = -0.245$ ($t = -4.80$) — close to Williams' -0.286 — implying about a 25 per cent cumulative effect at one quarter and most of the adjustment completed within five to six years. The full-sample λ of -0.480 is inflated by the COVID quarters and fails the $|\lambda|$ upper-bound screen (§6); the pre-COVID value is the policy-relevant speed.

Housing wealth is not a free-standing channel. The central form-is-decisive result of the paper carries a direct policy corollary: reading an insignificant standalone housing-wealth coefficient as evidence of "no housing wealth effect" is a category error. LIVES predicts that coefficient to be approximately zero absent the credit-conditions interaction; the housing MPC is unlocked only as credit conditions ease. For macroprudential and financial-stability analysis this means the consumption response to house prices is regime-dependent — larger when credit is loose, muted when it is tight — rather than a fixed elasticity. The qualitative structure is the policy-relevant object even where the point estimate of γ_1 is imprecise.

Macroprudential effects. The 2014 and 2017 APRA episodes enter as smoothed-step ogive dummies with small $\Delta \log c$ coefficients. On the Spec 11 headline only the 2014 round is materially negative and marginally significant ($d_apra_2014 = -0.0143$, $t = -1.81$); the 2017 dummy is small, positive and insignificant ($d_apra_2017 = +0.0079$, $t = 1.32$), so the macroprudential drag is carried by the first round. The counterfactual in §10.2 implies they shaved about 0.8 percentage points off consumption in the year after the 2014 round and 2.3 points by two

years out. The decade-horizon level gap is a model-mechanical upper bound, not a forecast of the broader macro response.

Permanent-income transmission. Permanent income is the strongest channel in the faithful form: the Spec 11 coefficient on $\log(y^p/y)$ is +0.50 ($t = 4.0$) on the full sample and +0.28 ($t = 4.7$) pre-COVID, both highly significant and consistent with a structural permanent-income weight ψ near 0.50. For fiscal-multiplier work this implies Australian households respond meaningfully and durably to credible permanent-income shocks. Two caveats apply. The headline estimate uses the full-sample (look-ahead) Italy-LP permanent-income measure; under the real-time, no-look-ahead variant the speed of adjustment shrinks sharply (Italy real-time $\lambda = -0.118$ versus full-sample -0.197) and the permanent-income coefficient flips or weakens (§7.4, §8), so the real-time column is the operationally-relevant version for any forward-looking application. And the freely-estimated $\psi \sim 0.50$ is roughly $2.5\times$ Williams' calibrated $\psi_0 = 0.20$; imposing his value collapses the equilibrium (§7.0.1, §9), so the Australian permanent-income gearing is a domestic estimate, not a transferred calibration.

Credit-conditions identification caveat. Section 5 documents that the CCI's identification in a single-equation OLS is weak, and that neither the back-extended sample, the sectional sign priors, nor the time-varying housing-wealth interaction changes this. The six CCI-interacted regressors are very highly mutually correlated on this sample (pairwise correlations of roughly 0.74–0.97, each approximately proportional to CCI), so they cannot be separately identified off one equation — the structural reason Williams uses four-equation FIML. (This near-collinearity is a sample property reported in the analysis narrative rather than a stand-alone committed correlation matrix; the load-bearing committed evidence is the wrong-signed and insignificant Spec 8 interaction coefficients and the Spec 10/12 collapse documented in §7.) Policymakers using a single-equation CCI series for regime classification (tightening versus easing diagnoses) should treat the spline coefficients as consumption-equation residual identification rather than a structurally identified common credit-conditions factor. For policy use, regime classification should rely on the joint system, with the Kalman state-space CCI (Spec 9; available on the back-extended sample) as a less-imposed cross-check that does not require institutional knot choices.

1.11.4 10.4 Nesting in MARTIN

A natural use of a freely-estimated equation is to discipline the calibrated consumption block of the RBA's MARTIN model (Ballantyne et al. 2019), whose long run is homogeneous of degree one in real income and real net wealth — $\log c = w \cdot \log y + (1 - w) \cdot \log NW + (\text{rate term})$ — with a calibrated net-wealth elasticity $(1 - w) \approx 0.17$. We test that structure directly (australia_martin_nesting.csv) by fitting the static long run $\log c = \beta_0 + \beta_y \cdot \log y + \beta_{NW} \cdot \log(NW/y) + \beta_{rr}$. Because $\log(NW_real_pc) = \log(NW/y) + \log y + \log 4$, the coefficient on log income equals the sum of MARTIN's income and wealth weights (which homogeneity sets to one) and the coefficient on the net-worth ratio is the net-wealth elasticity:

Quantity	Estimate	MARTIN
Net-wealth elasticity (β_{NW})	0.1155	0.17
Income + wealth weight (β_y)	0.7198	1.00 (imposed)
Real-rate semi-elasticity (β_r)	-0.0086	small, calibrated

Three findings bear on integration. First, the unrestricted net-wealth elasticity, 0.1155, is below MARTIN's calibrated 0.17 but of the same order. Second, **MARTIN's long-run homogeneity restriction is rejected** on the Australian data: the income-plus-wealth weight is 0.7198, and the test $\beta_y = 1$ gives $\chi^2(1) = 16.41$ ($p = 0.0001$). Imposing homogeneity by force does not help — the restricted net-wealth elasticity flips to -0.0832 (restricted real-rate -0.0038), so the data resist the balanced-growth parameterisation in this simple form. Third, the static long run does not cointegrate under Engle–Granger MacKinnon critical values (ADF -2.3828 vs 5 per cent CV for four variables -4.10), echoing the cointegration-screen result of §6.

The reading for MARTIN is that the freely-estimated single-equation long run does **not** nest MARTIN's calibrated balanced-growth block: homogeneity would have to be imposed rather than tested-and-accepted, and the unrestricted wealth elasticity (0.12 here, or 0.19 from the disaggregated wealth aggregate of \$7.3 with a 95 per cent CI of $[-0.77, 1.16]$) is too imprecise to discipline MARTIN's 0.17 calibration. The honest position is that this equation is *not yet MARTIN-ready as a source of point estimates*: the wealth elasticity CI spans MARTIN's value but also spans zero, so it can corroborate but not replace the calibration. The MARTIN-operational version of the permanent-income measure is the real-time Italy-LP forecaster (§7.4), not the full-sample look-ahead measure used for the headline; any integration must use the real-time variant, under which the speed of adjustment is materially weaker. A productive integration path is therefore to import the *qualitative* structure — disaggregated wealth channels and a credit-conditions interaction, with the signs and rough magnitudes of the channels — while retaining MARTIN's calibrated long-run weights, using this paper as a freely-estimated benchmark the calibration is consistent with rather than as a source of point estimates precise enough to replace it.

Source: australia_martin_nesting.csv, australia_gamma_inference.csv.

1.12 11. Conclusion

This paper re-estimates the Williams (2010, 2012) Australian LIVES consumption equation on a sample extended to 2024Q4, and in doing so revisits what an earlier draft and much of the prior Australian literature had treated as a *weak* single-equation result. The central finding is methodological: the functional **form** of the LIVES equation is what identifies it. When housing wealth is entered faithfully through its credit-conditions interaction alone, the autonomous-consumption CCI intercept is restored, and illiquid financial assets are combined, the error-correction mechanism and the core wealth structure come alive in a single equation. When housing instead enters as a plain, constant-MPC level term — as in the conventional disaggregated error-correction model that prior work implicitly tested — the equilibrium is far weaker and the standalone housing coefficient is insignificant. Reading that insignificant standalone coefficient as a failed housing-wealth effect is a category error: LIVES theory predicts that coefficient is approximately zero absent the credit-conditions interaction, because there is no classical housing-wealth channel in the model.

A second, equally central finding is that the LIVES *structure* transfers to Australia while Williams' Australian *calibrations* do not. The faithful specification recovers Williams' speed of adjustment and right-signed wealth structure when its parameters are freely estimated; imposing his calibrated permanent-income gearing collapses the equilibrium. We document why — the six credit-conditions interactions are mutually collinear in the post-deregulation

sample — and we are honest throughout about the limitations of single-equation LIVES estimation: a credit-conditions placebo that the deployed index fails, no efficiency gain from joint estimation at the quarterly frequency, an out-of-sample loss to a random walk, and a look-ahead permanent-income measure whose result partially reverses under a real-time construction.

1.12.1 11.1 Form is decisive: the faithful LIVES specification

The headline result is the faithful LIVES specification (Spec 11), in which housing wealth enters *only* through the de-meaned credit-conditions interaction $ha_x_cci = CCI \times (HA/4y)$, the autonomous-consumption loading $\zeta_c\text{-}CCI(cci_williams)$ is restored, and illiquid financial assets are combined into a single ratio ($ilfa_y = equities + superannuation$). On this form the error-correction and core wealth structure are recovered (`australia_all_results.csv`; `australia_lambda_robustness.csv`):

Quantity	Full sample (n = 146)	Pre-COVID (n = 126)	Williams (2010)
λ (<code>ecm_lag</code>)	-0.480 (t = -3.59)	-0.245 (t = -4.80)	-0.286
NLA m.p.c. (<code>nla_y</code>)	+0.037 (t = 3.39)***	+0.008 (ns)	0.159
IFA m.p.c. (<code>ilfa_y</code>)	+0.018 (t = 3.03)***	+0.009 (ns)	0.022
housing-collateral (<code>ha_x_cci</code> , γ_1)	+0.0049 (t = 1.03)	+0.0008 (ns)	0.0488
permanent income (<code>ln_yp_over_y</code>)	+0.504 (t = 4.00)***	+0.279 (t = 4.75)***	0.20–0.95

The pre-COVID speed of adjustment, $\lambda = -0.245$ (t = -4.80), is close to Williams' (2010) -0.286 and is the value we treat as identified: the full-sample $\lambda = -0.480$ is inflated by the COVID quarters and fails the upper-bound speed screen. The net-liquid and illiquid-financial marginal propensities are correctly signed and significant on the full sample (`nla_y` +0.037, t = 3.39; `ilfa_y` +0.018, t = 3.03), and permanent income enters strongly (+0.504, t = 4.00). The housing-collateral term γ_1 is right-signed but insignificant (`ha_x_cci` +0.0049, t = 1.03). Scaled by $|\lambda| = 0.480$, the implied structural marginal propensities are NLA 0.077, IFA 0.037, and housing-collateral γ_1 0.010 (right-signed, insignificant, versus Williams' 0.0488).

By contrast the conventional constant-MPC disaggregated ECM (Spec 6), which carries plain `ha_y/eq_y/super_y/nla_y` with no credit scaling and CCI only as a short-run `d2_logcci_lag2` term, delivers a weaker speed of adjustment ($\lambda = -0.180$, t = -1.76) and an insignificant standalone housing coefficient (`ha_y` +0.0088, t = 1.52). This is the conventional baseline, not the preferred specification. The contrast between Spec 11 and Spec 6 is a matter of theory rather than of fit: the two forms test different equations, and only Spec 11 is the LIVES equation. Much of the apparent weakness of single-equation Australian LIVES estimates is, on this reading, a specification artefact.

1.12.2 11.2 Structure transfers; Williams' calibrations do not

The faithful form recovers Williams' structure when freely estimated. It does *not* survive imposition of his Australian calibrations. Specification 12 imposes Williams' permanent-income gearing ($\psi_0 = 0.20$, $\psi_1 = 0.93$) and his illiquid-financial m.p.c. ($\gamma_{IFA} = 0.022$) via an iterative fixed point; the error-correction mechanism collapses to $\lambda = -0.029$ (t

= -0.68), with the net-liquid coefficient turning wrong-signed (australia_all_results.csv; australia_lambda_robustness.csv). Specification 10 (Williams-prior calibrated) independently reproduces the collapse ($\lambda = -0.048$, $t = -0.76$). The mechanism is that Australia freely estimates a permanent-income loading of roughly 0.50 — the Spec 11 full-sample $\ln_{yp_over_y}$ coefficient is +0.504 — well above Williams' calibrated $\psi_0 = 0.20$; forcing his lower value removes the equilibrium. Williams' rate, affordability and autonomous-consumption loadings cannot even be imposed at their published magnitudes: his raw $\alpha_r = -0.871$ is some thirty times too large on the repository's percent real-rate $\times$ unit-normalised CCI scaling, and diverges the fixed point.

This reconciles the companion paper's Wald non-rejection of the joint Williams calibration ($\chi^2 = 2.24$, $p \approx 0.90$). The non-rejection is driven by imprecision, not by good fit: every Williams value lies inside our wide structural confidence intervals (australia_gamma_inference.csv) — but so does zero. The free estimates are too imprecise to reject Williams' values, yet imposing those values wrecks the fit. Low power is not the same as good fit; the honest reading is consistency, not confirmation.

1.12.3 11.3 Why the credit channels are weakly identified in one equation

The reason the credit-conditions calibrations cannot be sharpened off a single equation is identification, not sample length. The six CCI-interacted regressors are each approximately proportional to the credit-conditions index and are therefore 0.74–0.97 mutually correlated on the post-1988 sample. They cannot be separately free-estimated from one equation — the structural reason Williams uses four-equation FIML. The evidence converges from three directions. First, when all six interactions are entered freely (Spec 8), the individual interaction coefficients are wrong-signed or insignificant against Williams' priors even though the overall fit is the best in the ladder, and the credit channels do not move toward Williams' Table 1. Specification 8 reallocates identification; it does not close the gap, and we do not present its $\lambda = -0.445$ as a credit-conditions success. Second, the calibration collapse of Specs 10 and 12 (§11.2) shows the channels are not jointly recoverable by imposition either. Third, the credit-conditions placebo battery shows the deployed index does no better than randomly placed knots: the literal Williams four-knot specification sits at the 34th adjusted- R^2 percentile on the 1988+ sample and the 18th on the back-extended sample (australia_williams_knot_placebo_verdict.csv; australia_williams_knot_placebo_extended_summary.csv), and the maximal-GETS reduction lifts it only weakly (64th/37th). These are honest negative results, and they are diagnostic: single-equation calibration of the credit channels is empirically closed, and a four-equation FIML build is the route to sharpen them.

1.12.4 11.4 The back-extension: sample length is not the binding constraint

A natural alternative hypothesis is that the residual gap between the freely estimated Australian channels and Williams' Table 1 reflects the post-1988 sample window, which begins only after the financial-liberalisation episode that identifies the credit channels. To test this we assembled a back-extended master dataset to 1976Q3 ($n = 190$ –194 quarters) — using a Treasury TRYM long-run house-price series, the RBA D03 M3 monetary aggregate, the RBA D02 total-credit splice, a historical labour-force compilation, and aggregate and disaggregated wealth proxies anchored at 1988Q3 — and refit the simpler disaggregated no-CCI specification (Spec 4) on the longer window (spec46_extended_comparison.csv):

Structural coefficient	1988+ baseline	1976+ extended	Williams (2010)
λ	-0.140	-0.193	-0.286
NLA (γ_{NLA})	+0.035	-0.002	+0.159
IFA equities (γ_{EQ})	-0.119	-0.104	—
IFA super (γ_{super})	+0.040	+0.024	—
IFA combined (γ_{IFA})	-0.079	-0.080	+0.022
housing (γ_{HA})	+0.068	+0.040	+0.049

The Williams (2010) reference values are his housing wealth m.p.c. (0.0488 at the CCI peak), net-liquid m.p.c. (0.159) and calibrated illiquid-financial m.p.c. (0.022, not split between equities and superannuation); the back-extension figures are the Spec 4 disaggregated structural coefficients (australia_williams_comparison.csv; spec46_extended_comparison.csv). The speed of adjustment improves by 37 per cent ($-0.140 \rightarrow -0.193$, still short of -0.286), but the individual wealth coefficients shrink rather than converging on Williams' values: the net-liquid m.p.c. collapses toward zero, the equities coefficient retains a wrong sign, and housing falls. Sample length is therefore not, in itself, what generates the divergence; the diagnosis remains the single-equation framing and the collinearity of the credit channels. The back-extension stands as a contribution in its own right — it makes this test possible for the first time — and supports the conclusion that the route forward is joint estimation rather than a longer single equation.

1.12.5 11.5 Honest negative results

The paper reports its negative results prominently, because a freely estimated single-equation framework permits diagnostics that imposed restrictions would hide. Each points to why FIML and back-extension are the natural next steps rather than further single-equation tuning.

- **Credit-conditions placebo.** The deployed index sits in the 18th–34th adjusted- R^2 percentile band across placebo variants (australia_williams_knot_placebo_verdict.csv; extended and maximal summaries) — no better than random knot placement on the relevant samples.
- **No system-efficiency gain.** A two-equation SUR of consumption and house prices yields negligible cross-equation residual correlation, and the SUR coefficient estimates are within sampling noise of single-equation OLS (australia_joint_pi_robustness.csv): joint estimation buys no efficiency at the quarterly frequency in this sample.
- **Out-of-sample loss to a random walk.** At horizons $h = 4$ and $h = 8$ the random walk with drift beats every structural specification (australia_oos_rmse.csv); only one specification narrowly beats it at $h = 1$. The framework's value is structural interpretation, not point forecasting.
- **Selector divergence.** No single criterion selects the same specification: the automated screen returns a net-worth form (Spec 2), BIC favours the free-interaction Spec 8, and LIVES theory favours Spec 11 (australia_spec_selection.csv). We retain Spec 11 as the narrative-preferred headline on theoretical-form grounds and report the divergence as a documented limitation. No single-equation form clears the Engle–Granger cointegration screen at the correct MacKinnon critical values (australia_cointegration.csv).

- **Look-ahead permanent income.** The headline permanent-income result uses the full-sample (non-causal) Italy local-projection forecaster. Under a causal real-time construction the speed of adjustment shrinks sharply and the permanent-income coefficient shrinks or flips (*australia_pi_realtime_robustness.csv*): the positive permanent-income loading is reported as a property of the full-sample measure, not a real-time resolution of the Australian permanent-income puzzle.

1.12.6 11.6 MARTIN nesting and the wealth elasticity

Nesting the long run against the RBA's MARTIN consumption block, the unrestricted net-wealth elasticity is 0.1155, against MARTIN's calibrated 0.17 (*australia_martin_nesting.csv*). Long-run homogeneity of the combined income-and-wealth term is rejected ($\chi^2 = 16.41$, $p = 0.0001$), and the wealth elasticity is not robustly identified: imposing homogeneity drives the restricted net-wealth elasticity to -0.083 , so the estimate is too sensitive to the restriction to discipline MARTIN's calibration. The appropriate reading is qualitative: the LIVES structure can inform MARTIN's functional form, but the single-equation point estimates are not precise enough to recalibrate its parameters.

1.12.7 11.7 Contributions

The paper makes five contributions to the Australian household consumption literature.

1. **The faithful single-equation LIVES estimate (Spec 11) with the form correction as the lead methodological result.** Entering housing only through its credit-conditions interaction, restoring the autonomous-consumption CCI intercept, and combining illiquid financial assets recovers a Williams-consistent error-correction speed (pre-COVID $\lambda = -0.245 \approx \text{Williams' } -0.286$) and right-signed wealth structure (NLA $+0.037^{***}$, IFA $+0.018^{***}$), where the conventional constant-MPC ECM does not. Reading the conventional form's insignificant standalone housing coefficient as a failed wealth effect is shown to be a category error.
2. **The structure-transfers-but-calibrations-don't finding.** Imposing Williams' permanent-income gearing collapses the equilibrium (Spec 12: $\lambda = -0.029$; independently reproduced by Spec 10: $\lambda = -0.048$), because Australia freely estimates a higher permanent-income loading; this reconciles the companion paper's low-power Wald non-rejection as consistency, not confirmation.
3. **The interaction-collinearity diagnosis.** The six CCI-interacted regressors are 0.74–0.97 mutually collinear, explaining why the credit channels cannot be separately identified off a single equation and why Williams' identification requires four-equation FIML — corroborated by the Spec 8 reallocation, the calibration collapse, and the placebo battery.
4. **A back-extended Australian master dataset to 1976Q3** ($n = 190\text{--}194$ quarters) with documented growth-rate splices for house prices, M3, total credit and labour force, plus aggregate and disaggregated wealth proxies — and the direct test it permits, showing that sample length is not the binding constraint.
5. **A structured robustness and placebo suite with honest negative results** — the credit-conditions placebo failure, the negligible SUR efficiency gain, the out-of-sample loss to a random walk, the selector divergence, and the look-ahead permanent-income caveat — reported as substantive findings, alongside a multi-equation scaffold (data preparation,

house-price and mortgage-stock equations, joint CCI identification, SURs) on which a companion FIML paper can build directly.

1.12.8 11.8 Outstanding work

Two routes follow directly from the diagnosis above.

1. **A multi-equation companion paper.** A full LIVES build with FIML and cross-equation parameter restrictions — consumption, house prices, the mortgage stock and home-equity withdrawal — would estimate a single CCI loading across equations under sign restrictions, the route the collinearity diagnosis and the placebo and SUR results identify as the only way to sharpen the credit channels. Custom likelihood code and a sourced HEW series remain the binding obstacles.
2. **Pre-1988 extension of the credit channels.** Because the financial-liberalisation episode that identifies the credit-conditions index largely predates the 1988Q3 start of ABS sectoral balance-sheet data, extending the disaggregated credit-interaction regressors back into the deregulation window — building on the 1976Q3 back-extension — would, in combination with FIML, give the credit channels the variation they need.

The single-equation framework cannot, on its own, deliver identification of the credit channels that Williams (2010) obtains from joint estimation under parameter restrictions. What it can do — and what this paper establishes — is that the faithful LIVES *form* recovers his error-correction and wealth structure where the conventional constant-MPC ECM cannot, that his Australian *calibrations* do not transfer, and that the credit interactions are collinear and weakly identified off a single equation. Reported honestly, these results point unambiguously to the multi-equation, back-extended programme as the way forward.

1.13 References

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1.14 Appendix A: Data construction

A full data-construction appendix — series sources, splice conventions, deflation, the asset/annualised-income ratio transform ($x_{t-1}/4y$), the disaggregated wealth definitions and the 1976Q3 back-extension — is reproduced from the accompanying data documentation. It retains the detail of §§3.1–3.13 of the main paper, together with the master variable coverage table.

The three load-bearing constructions for the faithful LIVES specification (Spec 11) and the conventional baseline (Spec 6) are summarised here for the reader's convenience; the full provenance is in the data appendix.

A.1 Disaggregated wealth (ratios to annualised income). All wealth terms enter as the prior-quarter stock divided by annualised disposable income, $x_{t-1}/4y$:

- **HA** — housing assets (ABS Balance Sheet 5232035). Enters the faithful LIVES form *only* through the credit interaction ha_x_cci (de-meant $CCI \times (HA/4y)$); there is no stand-alone housing-wealth level term, consistent with the theory that the housing marginal propensity to consume is zero at $CCI = 0$ and is unlocked as credit conditions ease.
- **IFA** — illiquid financial assets, equities plus superannuation ($ilfa_y = eq_y + super_y$). Enters as a plain MPC.
- **NLA** — net liquid assets, liquid assets minus total household debt. Enters as a plain MPC. The net-liquid aggregation (constraining the liquid-asset and debt MPCs to be equal and opposite, $\gamma_{LA} + \gamma_{LOANS} = 0$) is *not rejected* in any specification or sample (Appendix C; `australia_nla_restriction_test.csv`).

A.2 Permanent income. The permanent-income term $\log(y^p/y)$ is a discounted ($\eta = 0.05$, $k = 40$ quarters) weighted average of *expected* future income from the Italy-style Jordà (2005) local-projection forecaster. The headline measure uses the full-sample (in-sample, hence look-ahead) forecaster; the real-time variant is reported as an operational robustness column (§7.4;

australia_pi_realtime_robustness.csv). This look-ahead-versus-real-time distinction is flagged wherever a headline number depends on the non-causal measure.

A.3 The 1976Q3 back-extension. Sectoral balance-sheet data from the ABS begin in 1988Q3, which is why the financial-liberalisation episode that identifies the credit channels largely *predates* the estimation sample. A back-extended master dataset to 1976Q3 splices in TRYM house prices, RBA D03 M3 and RBA D02 credit aggregates, historical labour-force series and wealth proxies anchored to 1988Q3 levels. This dataset supports the long-history specification (Spec 6b) and the direct sample-length test (§8): back-extending Spec 4 moves λ about 37 per cent closer to Williams' value (from -0.140 to -0.193) but individual wealth coefficients shrink and the NLA term collapses — sample length is not the binding constraint on identifying the credit channels.

1.15 Appendix B: Coefficient tables

The full per-specification coefficient vectors, with Newey–West HAC standard errors, t -statistics, p -values, the implied error-correction speed λ (the `ecm_lag` coefficient) and the implied structural parameter ($\gamma = \text{OLS}/|\lambda|$), are reproduced in machine form from the committed results files. The current draft regenerates these tables from a single consolidated source, `australia_all_results.csv` (with the matched diagnostics in `australia_all_diagnostics.csv`), which now spans all fourteen specifications — Spec 1 through Spec 12, plus the long-history Spec 6b and the RBA-burden Spec 7b — for both the full ($n = 146$; 1988Q3–2024Q4) and pre-COVID ($n = 126$; 1988Q3–2019Q4) windows. In particular it includes the faithful LIVES headline (Spec 11) and the Williams-calibration-imposed specification (Spec 12), which were absent from the earlier draft's tables. The older `australia_full_results.csv` and `australia_precovid_results.csv` files remain in the repository for the legacy spec set; the consolidated file supersedes them for the present draft.

Rather than retype the full coefficient matrix, Table B.1 reports the two specifications that carry the paper's central message — the faithful LIVES headline (Spec 11) and the Williams-calibration-imposed specification (Spec 12) — across both samples. The remaining specifications are tabulated in §6 (the specification ladder) and §7 (results), and in full in Tables B.2–B.3 below and the committed CSVs.

Table B.1 — Faithful LIVES (Spec 11) vs Williams-calibration-imposed (Spec 12): selected coefficients (Newey–West HAC). Source: `australia_all_results.csv`, `australia_all_diagnostics.csv`. Convention: $\lambda = \text{ecm_lag}$ (negative = error-correction); structural $\gamma = \text{OLS}/|\lambda|$. *** sig 1%, ** sig 5%, * sig 10%.

Term	Spec 11 full ($n = 146$)	Spec 11 pre- COVID ($n = 126$)	Spec 12 full ($n = 146$)	Spec 12 pre- COVID ($n = 126$)
λ (ecm_lag)	-0.480 ($t=-3.59$)***	-0.245 ($t=-4.80$)***	-0.029 ($t=-0.68$)	+0.030 ($t=1.49$, sign flip)
nla_y	+0.0370 ($t=3.40$)***	+0.0084 ($t=0.60$)	-0.0014 ($t=-0.49$, wrong sign)	+0.0024
ilfa_y (=eq+super)	+0.0178 ($t=3.03$)***	+0.0090 ($t=1.40$)	imposed ($\gamma=0.022$)	imposed ($\gamma=0.022$)
ha_x_cci (γ_1)	+0.0049 ($t=1.03$)	+0.0008 ($t=0.22$)	-0.0011 ($t=-0.71$)	+0.0012
hp_x_l_minus_cci	+0.0142 ($t=2.69$)***	-0.0020 ($t=-0.29$)	—	—
r_x_cci	+0.0027 ($t=3.57$)***	+0.0020 ($t=2.78$)***	—	—
cci_williams (ζ_c)	+0.0046 ($t=0.47$)	+0.0127 ($t=1.62$)	—	—
ln_yp_over_y	+0.5043 ($t=4.00$)***	+0.2788 ($t=4.75$)***	imposed (ψ_0 =0.20)	imposed (ψ_0 =0.20)
yp_x_cci	-0.5897 ($t=-1.58$)	+0.1291 ($t=1.17$)	imposed (ψ_1 =0.93)	imposed (ψ_1 =0.93)
Intercept	-0.0152 ($t=-2.32$)**	-0.0126 ($t=-1.77$)*	+0.0101 ($t=2.25$)**	+0.0027 ($t=1.21$)
adj- R^2	0.812	0.210	0.686	0.096
SE (%)	0.706	0.520	0.915	0.560
BIC (Schwarz)	-945.05	-895.34	-892.52	-898.66

Implied structural MPCs for Spec 11 (full sample, $/|\lambda| = 0.480$): NLA 0.077, IFA 0.037, housing-collateral γ_1 0.010 (right-signed, insignificant, $t = 1.03$; cf. Williams' calibrated housing MPC 0.0488). The pre-COVID $\lambda = -0.245$ ($t = -4.80$) is treated as the identified speed of adjustment — close to Williams' -0.286 — because the full-sample value is inflated by the COVID quarters and fails the $|\lambda|$ upper-bound screen. The headline numbers depend on the full-sample (look-ahead) permanent-income measure; see §7.4 and the real-time robustness column.

Table B.2 — Speed of adjustment λ (the ecm_lag coefficient) across all fourteen specifications, full and pre-COVID samples (Newey–West HAC). Source: australia_all_results.csv. *** sig 1%, ** sig 5%, * sig 10%.

Specification	λ full ($n = 146$)	λ pre-COVID ($n = 126$)
Spec 1	-0.177 (t=-2.34)**	-0.100 (t=-1.71)*
Spec 2	-0.193 (t=-1.93)*	-0.085 (t=-0.84)
Spec 3	-0.166 (t=-2.23)**	-0.101 (t=-1.70)*
Spec 4	-0.140 (t=-2.05)**	-0.107 (t=-1.79)*
Spec 5	-0.177 (t=-1.56)	-0.097 (t=-1.01)
Spec 6	-0.180 (t=-1.76)*	-0.123 (t=-1.09)
Spec 6b	-0.229 (t=-4.18)***	-0.240 (t=-4.75)***
Spec 7	-0.373 (t=-2.78)***	-0.052 (t=-0.40)
Spec 7b	-0.375 (t=-2.40)**	-0.061 (t=-0.94)
Spec 8	-0.445 (t=-3.30)***	-0.247 (t=-5.03)***
Spec 9	-0.199 (t=-3.30)***	-0.172 (t=-3.32)***
Spec 10	-0.048 (t=-0.76)	-0.027 (t=-0.62)
Spec 11	-0.480 (t=-3.59)***	-0.245 (t=-4.80)***
Spec 12	-0.029 (t=-0.68)	+0.030 (t=+1.49)

Table B.3 — Long-run coefficient matrix for the disaggregated specifications (full sample): OLS estimate (*t*-statistic). Source: `australia_all_results.csv`. "—" = term not in that specification; the implied structural parameter is $\gamma = \text{OLS}/|\lambda|$ (Table B.2). *** sig 1%, ** sig 5%, * sig 10%.

Term	Spec 4	Spec 5	Spec 6	Spec 8	Spec 11
nla_y	+0.0049 (+0.52)	+0.0084 (+0.22)	+0.0354 (+0.96)	+0.0435 (+3.74)***	+0.0370 (+3.39)***
eq_y	-0.0168 (-1.10)	-0.0268 (-0.51)	-0.0063 (-0.13)	+0.0097 (+0.69)	—
super_y	+0.0057 (+0.85)	+0.0045 (+0.48)	+0.0117 (+1.48)	+0.0139 (+2.15)**	—
ilfa_y	—	—	—	—	+0.0178 (+3.03)***
ha_y	+0.0095 (+2.46)**	+0.0067 (+1.18)	+0.0088 (+1.52)	+0.0097 (+2.49)**	—
ha_x_cci	—	—	—	+0.0016 (+0.32)	+0.0049 (+1.03)
ln_hp_over_y	-0.0101 (-1.40)	-0.0024 (-0.11)	-0.0170 (-0.85)	—	—
hp_x_l_minus_cci	—	—	—	+0.0076 (+1.00)	+0.0142 (+2.69)***
real_rate	-0.0004 (-0.62)	-0.0009 (-0.67)	-0.0002 (-0.17)	—	—
r_x_cci	—	—	—	+0.0019 (+1.85)*	+0.0027 (+3.57)***
cci_williams	—	—	—	—	+0.0046 (+0.47)
ln_yp_over_y	+0.1509 (+1.71)*	+0.2484 (+0.80)	+0.1999 (+0.76)	+0.4677 (+3.44)***	+0.5043 (+4.00)***
ln_yp_over_y_post2008	—	—	+0.2360 (+1.16)	—	—
yp_x_cci	—	—	—	-0.6113 (-2.12)**	-0.5897 (-1.58)
ecm_lag	-0.1404 (-2.05)**	-0.1770 (-1.56)	-0.1801 (-1.76)*	-0.4449 (-3.30)***	-0.4801 (-3.59)***

The contrast between the two columns is the structure-transfers-but- calibrations-do-not result: imposing Williams' Australian calibrations ($\psi_0 = 0.20$, $\psi_1 = 0.93$, $\gamma_{\text{IFA}} = 0.022$) via the iterative fixed point collapses the error-correction speed from -0.480 to a statistically insignificant -0.029 , and flips it to the wrong sign ($+0.030$) pre-COVID. Spec 10 (Williams-prior calibrated; `australia_all_results.csv`) independently reproduces the collapse ($\lambda = -0.048$, $t = -0.76$). This is a low-power-is-not-good-fit reading of the structural-parameter inference (`australia_gamma_inference.csv`), whose wide confidence intervals contain both Williams' values *and* zero: the free estimates are too imprecise to reject Williams' calibrations, but imposing them wrecks the fit.

The structural-parameter inference (the implied $\gamma = \text{OLS}/|\lambda|$ with bootstrap confidence intervals) is in `australia_gamma_inference.csv`; every Williams value lies inside the (wide) 95 per cent interval, and so does zero — consistency, not confirmation.

1.16 Appendix C: Diagnostic battery

The full diagnostic output is reproduced from `australia_all_diagnostics.csv` (consolidated; supersedes the legacy `australia_full_diagnostics.csv` and `australia_precovid_diagnostics.csv`). For each specification and sample it reports the number of observations, the residual standard error (per cent), adjusted R^2 , Durbin–Watson, Breusch–Godfrey serial-correlation tests at lags 1 and 4, the heteroscedasticity test in two forms (full and events-excluded), the RESET functional-form test, the Schwarz/BIC and the log-likelihood.

C.1 Heteroscedasticity is structural, not event-driven. Every full-sample specification flags `het_diagnosis = "structural"`: the heteroscedasticity test rejects homoscedasticity even after the COVID, GFC and policy-event dummies are excluded (for Spec 11 the full-sample heteroscedasticity $p \approx 7.8 \times 10^{-9}$ and the events-excluded $p \approx 9.3 \times 10^{-10}$). This is why Newey–West HAC standard errors are used for every reported t -statistic throughout the paper. The pre-COVID Spec 11 residuals are, by contrast, well behaved (heteroscedasticity $p = 0.47$; no serial correlation, $\text{AR}(1)$ $p = 0.37$, $\text{AR}(4)$ $p = 0.39$; RESET $p = 0.54$).

C.2 The multi-break Chow battery is reported in `australia_chow_battery.csv` and the structural-break tests in `australia_breaks.csv`. The 2008Q3 break is not rejected (Chow $p = 0.27$, stable); the 2020Q1 break is strongly rejected (Chow $p \approx 9.9 \times 10^{-10}$ — the COVID break). The Bai–Perron supF statistic (164.20, $p = 0$) and the recursive-residual CUSUM ($p = 0.9963$, stable) together locate a single break at the COVID episode (2020Q4). This break, and not a deeper instability, is what inflates the full-sample λ relative to the pre-COVID value.

C.3 The net-liquid restriction test (`australia_nla_restriction_test.csv`) records `restriction_accepted = TRUE` in all six rows (Specs 4, 5, 6 across full and pre-COVID samples): the data cannot distinguish separate liquid-asset and debt MPCs, which supports the NLA aggregation used in the faithful LIVES form.

C.4 Cointegration screen (`australia_cointegration.csv`). The Engle–Granger ADF residual test fails to reject the null of no cointegration for *every* estimable specification at the correct MacKinnon critical values, while the Johansen rank-1 trace test passes for all. No single-equation form clears the Engle–Granger screen; the interaction specifications (Spec 8–12) are not run through the cointegration screen (coint reported NA) because the long-run bracket is a non-linear function of CCI.

1.17 Appendix D: Reproducibility

The full reproducibility kit accompanies this paper. The estimation pipeline can be run in three execution modes, all driven from the project root (`/Users/davidstephan/Documents/consmodelling`):

1. Cold rebuild (downloads + estimates).

```
Rscript Australia/R/australia_consumption_model.R
```

Reads the cached ABS workbooks, fetches the RBA series live, rebuilds master, saves `australia_model_dataset.rds`, and then runs estimation. Required when the `data_raw/workbooks` change.

2. Fast re-estimate from the cached RDS (no data work; bit-identical).

```
Rscript Australia/R/run_estimation_from_rds.R
```

Loads the pre-built RDS and runs only the estimation script. This is the canonical reproduction command — it is the path exercised by continuous integration and produces bit-identical results.

3. Offline replay from a portable master CSV (no downloads, hand-editable).

```
Rscript Australia/R/load_master_from_csv.R
```

Loads `data_raw/master_data.csv`, reconstructs master, and runs estimation. Useful without internet access, for a frozen version-controllable snapshot, or to patch a known data error by hand. (A documented caveat: the CSV and RDS paths can diverge on Chow-borderline selector flags at machine precision; see the data appendix.)

The estimation script writes all fourteen specifications (Spec 1–12, plus the long-history Spec 6b and the RBA-burden Spec 7b) to `australia_all_results.csv` and `australia_all_diagnostics.csv`, including the faithful LIVES headline (Spec 11) and the Williams-calibration-imposed specification (Spec 12); all robustness, placebo, counterfactual and comparison outputs cited in the paper are written to the matching CSVs in `outputs/`.

The environment is pinned with `renv` (R 4.5.3); the dependency manifest is in `DESCRIPTION` and the full transitive closure — including the `car` and `systemfit` packages used for the Wald and SUR tests — is locked in `renv.lock`. Restore it with `renv::restore()`. The kit also ships GitHub Actions continuous integration and a `testthat` unit-test suite. The repository is hosted publicly at <https://github.com/DavidASephan/ConsModelling>.